Online Supplement: Modelling with Sensitive Variables

Felix Chan[a], László Mátyás[b], Ágoston Reguly[c,d,*]

[a]*School of Accounting, Economics and Finance, Curtin University, Kent Street, Bentley, 6102, Australia, felix.chan@cbs.curtin.edu.au*
[b]*Department of Economics, Central European University, Quellenstraße 51, Vienna, 1100, Vienna, matyas@ceu.edu*
[c]*Institute of Economics, Corvinus University of Budapest, Fovam ter 8, Budapest, 1093, Hungary, agoston.reguly@uni-corvinus.hu*
[d]*Scheller College of Business, Georgia Institute of Technology, 800 W Peachtree St NW, Atlanta, 30308, Georgia, United States*

A1. Further Supplements for Shifting Method

A1.1. Proof of Proposition 2

Let $\mathbf{z}$ be a realisation of $\mathbf{Z}$ and the probability of $\mathbf{z}$ falls into the grid $\mathcal{C}_{\mathbf{b}}^{\mathbf{WS}}$ of the working sample is

$$\Pr\left(\mathbf{z} \in \mathcal{C}_{\mathbf{m}}^{\mathbf{WS}}\right) = \sum_{s=1}^{S} \Pr(\mathbf{z} \in \mathcal{S}_s) \sum_{m=1}^{M} \Pr\left(\mathbf{z} \in \mathcal{C}_{\mathbf{b}}^{\mathbf{WS}} | \mathbf{z} \in \mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\right) \int_{\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}} f_{\mathbf{Z}}(\mathbf{z}) d\mathbf{z}. \quad (1)$$

Let $\mathbf{z}_i = (z_{1i}, \ldots, z_{Pi})'$ be the i^{th} realisation of $\mathbf{z}$ for $i \in \mathbb{Z}^+$ and define $\mathbf{z}_1 \preceq \mathbf{z}_2$ if $z_{1i} \leq z_{2i}$ $\forall i$. In other words, $\mathbf{z}_1$ is 'less than or equal to' $\mathbf{z}_2$ element-wise. The definition extends natural to other related notations namely $\prec$, $\succ$ and $\succeq$.

Similar to the univariate case, $\Pr(\mathbf{a} \preceq \mathbf{c}_1, \mathbf{a} \preceq \mathbf{c}_2) = \Pr(\mathbf{a} \preceq \mathbf{c}_1)$ if $\mathbf{c}_1 \preceq \mathbf{c}_2$. Under the assumption $\Pr(\mathbf{z} \in \mathcal{S}_s) = 1/S$, then the proof follows the same argument as the proof for Proposition 1. This completes the proof. ■

*Corresponding author; *email*: agoston.reguly@uni-corvinus.hu.

A1.2. Speed of Convergence for the Shifting Method

The probabilities of the synthetic variable $Z^\dagger$ to be fall into $\mathcal{C}_b^{WS}$ bin is given by,

$$\Pr\left(Z^\dagger \in \mathcal{C}_b^{WS}\right) = \begin{cases} 0, & \text{if } s = 1 \text{ and } m = 1, \\ \frac{1}{S}\sum_{s=2}^{S} \frac{1}{s-1} \int_{\mathcal{C}_1^{(s)} | \mathcal{C}_b^{WS} \subset \mathcal{C}_1^{(s)}} f_Z(z)\mathrm{d}z, & \text{if } s \neq 1 \text{ and } m = 1, \\ \frac{1}{S^2}\sum_{s=1}^{S} \int_{\mathcal{C}_m^{(s)} | \mathcal{C}_b^{WS} \subset \mathcal{C}_m^{(s)}} f_Z(z)\mathrm{d}z, & \text{if } 1 < m < M, \\ \frac{1}{S}\sum_{s=1}^{S} \frac{1}{S-s+1} \int_{\mathcal{C}_M^{(s)} | \mathcal{C}_b^{WS} \subset \mathcal{C}_M^{(s)}} f_Z(z)\mathrm{d}z, & \text{if } m = M\,. \end{cases} \tag{2}$$

For each of the conditions in Equation (2), the corresponding expression is $o(1)$. To see this, note that $f_Z(\cdot)$ is a density, so the integral is less than 1. First, consider the case of $s \neq 1$ and $m = 1$,

$$\begin{aligned} \frac{1}{S}\sum_{s=2}^{S} \frac{1}{s-1} \int_{\mathcal{C}_1^{(s)} | \mathcal{C}_b^{WS} \subset \mathcal{C}_1^{(s)}} f_Z(z)\mathrm{d}z, &\leq \frac{1}{S}\sum_{s=2}^{S} \frac{1}{s-1} \\ &= \frac{1}{S}\sum_{s=1}^{S} \frac{1}{s} \\ &= \frac{1}{S}\int_1^S \frac{1}{s} ds \\ &= \frac{\log S}{S}. \end{aligned} \tag{3}$$

As $S \to \infty$, the ratio in the last line goes to 0. This is expected if the widths of the classes in the working sample go to zero. This is straightforward, while the probability that an observation belongs to a point is 0. The same derivations apply to the case when $m = M$. Now, consider the case of $1 < m < M$,

$$\begin{aligned} \frac{1}{S^2}\sum_{s=1}^{S} \int_{\mathcal{C}_m^{(s)} | \mathcal{C}_b^{WS} \subset \mathcal{C}_m^{(s)}} f_Z(z)\mathrm{d}z &\leq \frac{1}{S^2}\sum_{s=1}^{S} 1 \\ &= \frac{1}{S}, \end{aligned} \tag{4}$$

which also converges to 0 as $S \to \infty$, but at a faster rate than in the previous cases.

A1.3. One-by-one discretization

A seemingly competitive alternative is to use discretization that discretizes each variable in $\mathbf{Z}$ one by one. However, this method will not ensure convergence in distribution for the joint $\mathbf{f}_{\mathbf{Z}}(\cdot)$ that is needed for point identification. To illustrate our argument let us use a simple illustrative example with $P = 2$, hence only two variables are discretized, $M = 3$ and $\mathbf{a}_l = [0, 0]$, $\mathbf{a}_u = [6, 6]$. For $s = 1$, $\mathcal{C}_{\mathbf{m}}^{(\mathbf{1})} = \{[(\mathbf{0}, \mathbf{0}), (\mathbf{2}, \mathbf{2})], [(\mathbf{2}, \mathbf{2}), (\mathbf{4}, \mathbf{4})], [(\mathbf{4}, \mathbf{4}), (\mathbf{6}, \mathbf{6})]\}$. Let us use $S = 6$, thus the shift size $\mathbf{h} = [0.5, 0.5]$. Discretizing the variables independently from each other will result in fixed M intervals along all other dimensions while learning more and more along one dimension. This will result in gaps in the domain of $\mathbf{f}_{\mathbf{Z}}(\cdot)$. Figure A1 shows this case, where grey blocks show the mapped/learned parts of the distribution of $\mathbf{f}_{\mathbf{Z}}(\cdot)$.

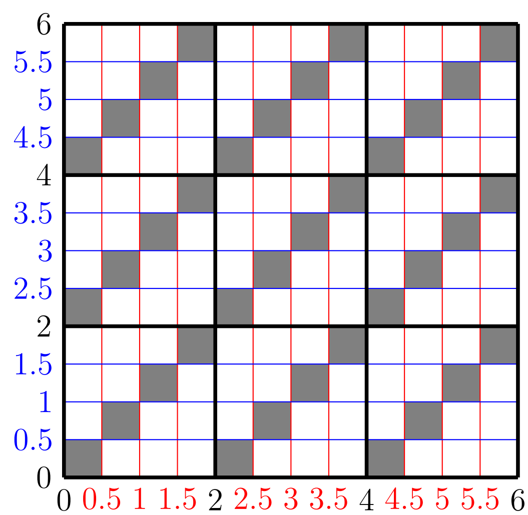

Figure A1: One-by-one variable discretization

A1.4. Algorithms for Shifting method

Algorithm A1 The shifting method - creation of split samples

1: For any given S and M, set $s = 1$

$$B = S(M-1)\ , \qquad h = \frac{a_u - a_l}{B}\ , \qquad \Delta = \frac{a_u - a_l}{M-1}\ . \tag{5}$$

2: Set $c_0^{(s)} = a_l$ and $c_M^{(s)} = a_u$.

3: If $s = 1$, set

$$c_m^{(s)} = c_{m-1}^{(s)} + \Delta, \qquad m = 2, \ldots, M-1 \tag{6}$$

else

$$c_m^{(s)} = c_m^{(s-1)} + h, \qquad m = 1, \ldots, M-1. \tag{7}$$

Note: $c_1^{(1)}$ does not exist.

4: If $s < S$ then $s := s + 1$ and goto Step 2.

Algorithm A2 The shifting method – creation of synthetic variable ($Z_i^{\dagger}$)

1: Set $s := 1, m := 1, Z_i^{\dagger} = \emptyset$.

2: Create $V(s,m)$, the set of possible working sample choice values,

$$\mathcal{C}_m^{(s)} = \begin{cases} \{\emptyset\}, & \text{if, } s = 1 \text{ and } m = 1, \\ \bigcup_{b=1}^{s-1}\{\mathcal{C}_b^{WS}\}, & \text{if, } s \neq 1 \text{ and } m = 1, \\ \bigcup_{b=s-1+(m-2)(M-1)}^{s-1+(m-1)(M-1)}\{\mathcal{C}_b^{WS}\}, & \text{if, } 1 < m < M, \\ \bigcup_{b=B-S+s-1}^{B}\{\mathcal{C}_b^{WS}\}, & \text{if } m = M. \end{cases} \tag{8}$$

3: Create the set of observations from the defined split sample class:

$$\mathcal{A}_m^{(s)} := \{Z_i^{(s)} \in \mathcal{C}_m^{(s)}\}\ \forall i,$$

where $\mathcal{A}_m^{(s)}$ has $N_m^{(s)}$ number of observations.

4: Draw observations from $\mathcal{A}_m^{(s)}$ and assign a new value (e.g. mid-value) to $\mathcal{V}_j$ based on a randomly selected interval of $\mathcal{C}_b^{WS} \in \mathcal{C}_m^{(s)}$, with probabilities given by,

$$\Pr\left(Z^{\dagger} \in \mathcal{C}_b^{WS} | Z^{(s)} \in \mathcal{C}_m^{(s)}\right) = \begin{cases} 1, & \text{if } s = 1 \text{ and } m = 1, \\ 1/(s-1), & \text{if } s \neq 1 \text{ and } m = 1, \\ 1/S, & \text{if } 1 < m < M, \text{ or} \\ 1/(S-s+1), & \text{if } m = M. \end{cases} \tag{9}$$

Example: Let $\mathcal{C}_3^{(2)} = [2.5, 4.5]$, $\mathcal{A}_m^{(s)} = \{3.5, 3.5, 3.5\}, N_m^{(s)} = 3$. The different intervals in the working sample is $C_{b_1}^{WS} = [2.5, 3]$, $C_{b_2}^{WS} = [3, 3.5]$, $C_{b_3}^{WS} = [3.5, 4]$ and $C_{b_4}^{WS} = [4, 4.5]$. Now we assign each member of $\mathcal{A}_m^{(s)}$ to one of the mid-value of C_b^{WS} with $1/4$ probabilities which results in the values of $\mathcal{V}_j = \{2.75, 3.25, 3.75, 4.25\}$.

5: Add these new values to $Z_i^{\dagger}$,

$$Z_i^{\dagger} := \left\{ Z_i^{\dagger}, \bigcup_{j=1}^{N_m^{(s)}} \mathcal{V}_j \right\} \tag{10}$$

6: If $s < S$, then $s := s + 1$ and go to Step 3.

7: If $s = S$, then $s := 1$ and set $m = m + 1$ and go to Step 3.

A2. Asymptotic Properties of Conditional Mean Estimators for the Shifting Method

We investigate the asymptotic properties of the estimator for the conditional means while using the shifting method to map $\mathbf{f_Z}(\mathbf{z};\mathbf{a}_l,\mathbf{a}_u)$. We restrict our attention to multiple linear regressions defined in Section 4 in the main manuscript. First, we discuss the estimator $\hat{\boldsymbol{\kappa}}$ for $\mathbb{E}\left[\mathbf{X}|\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\right]$, used to estimate $\hat{\boldsymbol{\beta}}$, when one or more explanatory variables $\mathbf{X}$ are discretized. Secondly, we investigate the properties of the conditional mean estimator $\hat{\boldsymbol{\pi}}$ that is used when $\mathbf{y}$ is discretized on the right-hand side.

A2.1. $\hat{\boldsymbol{\kappa}}$ *estimator for* $\mathbb{E}\left[\mathbf{X}|\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\right]$

The conditional expectation function for a given grid in $\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}$is defined as $\kappa(s,\mathbf{m}):=$
$\mathbb{E}\left[\mathbf{X}|\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\right]$. $\boldsymbol{\kappa}=(\kappa(1,\mathbf{1}),\ldots,\kappa(S,\mathbf{1}),\ldots,\kappa(S,\mathbf{m}),\ldots,\kappa(S,\mathbf{M}))'$ is the parameter vector of interest containing each conditional means,
$\mathbb{E}\left[\mathbf{X}|\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\right]$, where we condition on the used discretization intervals (or grid) $\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}$. Let us define the $\hat{\boldsymbol{\kappa}}$, the estimator of $\boldsymbol{\kappa}$ via OLS as,

$$\hat{\boldsymbol{\kappa}}=\left(\mathbf{1}'_{\{\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}\mathbf{1}_{\{\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}\right)^{-1}\mathbf{1}'_{\{\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}\mathbf{X}^\dagger\ , \tag{11}$$

where $\mathbf{1}_{\{\mathbf{X}\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}$ is a matrix with dimensions $(N\times SM^K)$ and takes the value of one if observations fall into class $\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}$. We construct $\mathbf{1}_{\{\mathbf{X}\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}$, such that first, we iterate along s then along each grid points $\mathbf{m}=(m_1,\ldots,m_k,\ldots,m_K)$. The regression equation is given by,

$$\mathbf{X}^\dagger=\mathbf{1}_{\{\mathbf{X}^*\in\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}\}}\boldsymbol{\kappa}+\boldsymbol{\eta}_\kappa, \tag{12}$$

where $\boldsymbol{\eta}_\kappa=(\eta_{\kappa,1},\ldots,\eta_{\kappa,N})'$ the idiosyncratic term.[1]

By the weak law of large numbers, $\hat{\boldsymbol{\kappa}}$ is converging to the true underlying distribution's conditional expectations, $\hat{\boldsymbol{\kappa}}\to\boldsymbol{\kappa}$ as $\lim_{S\to\infty}F_{\mathbf{X}^\dagger}(\cdot)=F_{\mathbf{X}}(\cdot)$ under Assumptions 2.a), 2.b) and $\Pr(z\in\mathcal{S}_s)=1/S$ as shown in Section 3.2

[1] We use $\boldsymbol{\eta}$ as the idiosyncratic term for the different cases, hence the subscript to differentiate among the cases.

from the main manuscript. The asymptotic properties of $\hat{\boldsymbol{\kappa}}$ can be derived by using Lindeberg–Lévy central limit theorem with standard OLS assumptions,

$$\sqrt{N}\left(\hat{\boldsymbol{\kappa}}-\boldsymbol{\kappa}\right) \overset{\text{a}}{\sim} \mathcal{N}\left(\mathbf{0},\boldsymbol{\Omega}_{\boldsymbol{\kappa}}\right) . \tag{13}$$

The asymptotic variance of the OLS estimator is given by

$$\boldsymbol{\Omega}_{\boldsymbol{\kappa}} = V\left(\boldsymbol{\eta}_{\kappa}\right)\left(\mathbf{1}'_{\{\mathbf{X}^*\in\mathcal{C}^{(\mathbf{s})}_{\mathbf{m}}\}}\mathbf{1}_{\{\mathbf{X}^*\in\mathcal{C}^{(\mathbf{s})}_{\mathbf{m}}\}}\right)^{-1}, \tag{14}$$

where $V\left(\boldsymbol{\eta}_{\kappa}\right)$ is the variance of the corresponding idiosyncratic term. Algorithm A3 describes the process for creating the working sample, when discretization happens with one or more right-hand side variables.

Algorithm A3 Discretization of explanatory variables – creation of working sample

1: Estimate $\hat{\boldsymbol{\kappa}}$ as defined in Equation (11).
2: Set $c := 1, s := 1, \mathbf{m} := (1,\dots,1), k := 0,\ \{\mathbf{y}^{WS},\mathbf{X}^{WS},\mathbf{W}^{WS}\} = \emptyset$.
3: Assign the conditional mean for $\mathbf{X}^*\in\mathcal{C}^{(\mathbf{s})}_{\mathbf{m}}$ from the c'th element of $\hat{\boldsymbol{\kappa}}$ to all discretized observations $\mathbf{X}^*\in\mathcal{C}^{(\mathbf{s})}_{\mathbf{m}}$. and the observed values $\mathbf{y}^{(s)}_j, \mathbf{W}^{(s)}_j$ to the working sample,

$$\left\{\mathbf{y}^{WS}_i,\mathbf{X}^{WS}_i,\mathbf{W}^{WS}_i\right\} := \left\{\mathbf{y}^{WS}_i,\mathbf{X}^{WS}_i,\mathbf{W}^{WS}_i,\bigcup_{j=1}^{N}\left(\mathbf{y}^{(s)}_j,\hat{\boldsymbol{\kappa}}(c),\mathbf{w}^{(s)}_j \mid \mathbf{X}^*_j\in\mathcal{C}^{(\mathbf{s})}_{\mathbf{m}}\right)\right\} . \tag{15}$$

4: Set $c := c+1$.
5: If $s < S$, then $s := s+1$ and go to Step 3.
6: If $s = S$, then $s := 1, k := k+1$ and set $\mathbf{m} = \mathbf{m}+\mathbf{1}_{\{k\}}$, where $\mathbf{1}_{\{k\}}$ is a $(K\times 1)$ indicator function with value of 1 at element k, otherwise 0. Go to Step 3.

A2.2. Bias and speed of convergence for $\hat{\boldsymbol{\kappa}}$

For a class $C^{(s)}_m = [c^{(s)}_{m-1}, c^{(s)}_m)$ in split sample s, define

$$w^{(s)}_m := c^{(s)}_m - c^{(s)}_{m-1}, \qquad d^{(s)}_m := \frac{w^{(s)}_m}{2}, \qquad v^{(s)}_m := \frac{c^{(s)}_{m-1}+c^{(s)}_m}{2}.$$

For the pdf $f_Z(z)$ let us define $f_k := f_Z^{(k)}\left(v_m^{(s)}\right)$ for its derivatives, $k = 1, 2, 3$ at the midpoint. Furthermore let us recall that under the shifting method, the working step is

$$h = \frac{a_u - a_\ell}{S(M-1)} \qquad \text{so that (for interior bins)} \qquad w_m^{(s)} = S\,h.$$

We have shown that $\sqrt{N}(\hat{\boldsymbol{\kappa}} - \boldsymbol{\kappa}) \overset{a}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_{\boldsymbol{\kappa}})$ under $N, S \to \infty$. A natural question is: with a *fixed* S, how large is the approximation bias of $\hat{\boldsymbol{\kappa}}$. To quantify the magnitude of the bias, we use the midpoint $v_m^{(s)} = (c_m^{(s)} + c_{m-1}^{(s)})/2$ as substituted values. First, we focus on the case $S = 1$, then extend to any fixed S.

Bias with $S = 1$. The true within-bin conditional mean is

$$\kappa\left(C_m^{(s)}\right) := \mathbb{E}\left[X \mid X^* \in C_m^{(s)}\right] = \frac{\int_{c_{m-1}^{(s)}}^{c_m^{(s)}} z\, f_Z(z)\, dz}{\int_{c_{m-1}^{(s)}}^{c_m^{(s)}} f_Z(z)\, dz}.$$

Let us fix an interior bin $C = [v_m^{(s)} - d_m^{(s)},\, v_m^{(s)} + d_m^{(s)}]$, $z = v_m^{(s)} + t$, with $t \in [-d_m^{(s)}, d_m^{(s)}]$ and define,

$$I_0 = \int_C f_Z(z)\, dz, \qquad I_1 = \int_C z\, f_Z(z)\, dz,$$

for simplicity. Next, we use local Taylor approximation, $f_Z\left(v_m^{(s)} + t\right) = f_0 + f_1 t + \frac{f_2}{2} t^2 + \frac{f_3}{6} t^3 + O(t^4)$. Then

$$I_0 = 2d_m^{(s)} f_0 + \frac{f_2}{3}\left(d_m^{(s)}\right)^3 + O\left((d_m^{(s)})^5\right),$$

and

$$\begin{aligned}
I_1 &= \int_{-d_m^{(s)}}^{d_m^{(s)}} (v_m^{(s)} + t) f_Z(v_m^{(s)} + t) \mathrm{d}t \\
&= v_m^{(s)} I_0 + \int_{-d_m^{(s)}}^{d_m^{(s)}} t f_Z(v_m^{(s)} + t) \mathrm{d}t \\
&= v_m^{(s)} \left[2d_m^{(s)} f_0 + \frac{f_2}{3}\left(d_m^{(s)}\right)^3 + O\left(\left(d_m^{(s)}\right)^5\right)\right] \\
&\quad + \frac{2}{3} f_1 \left(d_m^{(s)}\right)^3 + \frac{1}{15} f_3 \left(d_m^{(s)}\right)^5 + O\left(\left(d_m^{(s)}\right)^7\right)
\end{aligned}$$

Hence,

$$\begin{aligned}
\kappa\left(C_m^{(s)}\right) &= v_m^{(s)} + \frac{\frac{2}{3} f_1 \big(d_m^{(s)}\big)^3 + O\big((d_m^{(s)})^5\big)}{2 d_m^{(s)} f_0 + O\big((d_m^{(s)})^3\big)} \\
&= v_m^{(s)} + \frac{(d_m^{(s)})^2}{3} \frac{f_1}{f_0} + O\big((d_m^{(s)})^4\big) \\
&= v_m^{(s)} + \frac{(d_m^{(s)})^2}{3} (\log f_Z)'\big(v_m^{(s)}\big) + O\big((d_m^{(s)})^4\big).
\end{aligned}$$

Thus, where the density is increasing at $v_m^{(s)}$, the within-bin mean lies above the midpoint (and below if decreasing). The *midpoint substitution error* is therefore

$$\kappa\left(C_m^{(s)}\right) - v_m^{(s)} = \frac{(d_m^{(s)})^2}{3} (\log f_Z)'\big(v_m^{(s)}\big) + O\big((d_m^{(s)})^4\big),$$

which vanishes for (locally) uniform densities where $(\log f_Z)' = 0$, providing the same result as in Section A6.

Exact bound for any fixed S. Let $I_{0,h}, I_{1,h}$ denote the composite-midpoint approximations on the step-h grid. Set

$$g_0(z) := f_Z(z), \qquad g_1(z) := z\, f_Z(z), \qquad \underline{f}_C := \inf_{z \in C} f_Z(z) > 0.$$

The composite midpoint rule yields

$$\big|I_{0,h} - I_0\big| \;\le\; \frac{w_m^{(s)}}{24}\, h^2 \sup_{z \in C} \big|g_0''(z)\big|, \qquad \big|I_{1,h} - I_1\big| \;\le\; \frac{w_m^{(s)}}{24}\, h^2 \sup_{z \in C} \big|g_1''(z)\big|. \tag{16}$$

A first-order ratio expansion gives

$$\big|\kappa_h(C) - \kappa(C)\big| \;\le\; \frac{1}{I_0} \big|I_{1,h} - I_1\big| \;+\; \frac{|I_1|}{I_0^2} \big|I_{0,h} - I_0\big| \;+\; o(h^2). \tag{17}$$

Using $I_0 \ge w_m^{(s)}\, \underline{f}_C$ and $|I_1| = |\kappa(C)|\, I_0$ together with (16) in (17),

$$\big|\kappa_h(C) - \kappa(C)\big| \;\le\; \frac{h^2}{24} \cdot \frac{\sup_{z \in C} \big|g_1''(z)\big| \;+\; |\kappa(C)| \sup_{z \in C} \big|g_0''(z)\big|}{\underline{f}_C} \;+\; o(h^2). \tag{18}$$

Since $h = \frac{w_m^{(s)}}{S}$, we obtain the curvature–driven rate

$$\kappa_h(C) - \kappa(C) \;=\; O\Big(\frac{w_m^{(s)\,2}}{S^2}\Big), \tag{19}$$

with a bin–local constant determined by the second derivatives of f_Z (through g_0'' and g_1'') and the local mass floor $\underline{f}_C$.

Bias reduction with S. Now, relative to the $S = 1$ the *remaining fraction* is approximately $1/S^2$: for $S = 2$ the reduction is about 75%, for $S = 5$ about 96%, and for $S = 10$ about 99%.

Boundary bins. For the raw working-sample probabilities, the CDF mapping converges at rate $1/S$ in the interior and $\log S/S$ near the support boundaries; however, for the conditional mean on a fixed bin, the composite-midpoint error remains $O(h^2)$ provided f_Z is smooth on that bin.
Remark. The class half-width $d_m^{(s)} = w_m^{(s)}/2$ is set by M and does not change with S. Increasing S *shrinks the working step* $h = w_m^{(s)}/S$, which drives the $O(h^2)$ improvement.

A2.3. Bias reduction

Based on Equation (18), we get an exact upper boundary for the bias. Next we propose plug-in estimators for the quantities of $\sup_{z\in C}\big|g_1''(z)\big|$, $\sup_{z\in C}\big|g_0''(z)\big|$ and $\underline{f}_C$ to quantify the bias.

Within the class C, we observe the S working class $\{(u_j, h)\}_{j=1}^S$ with midpoints u_j and working probabilities $\hat{p}_j$ (counts divided by total sample size). Estimate the ingredients as follows:

1. *Density at class midpoints:* $\hat{f}(u_j) = \hat{p}_j/h$.
2. *Local quadratic fit* on $\{(u_j, \hat{f}(u_j))\}_{j=1}^S$ to obtain $\widehat{f'}(u_j), \widehat{f''}(u_j)$. Then

$$\overline{\widehat{g_0''}}(C) \;\approx\; \frac{1}{S}\sum_{j=1}^{S}\widehat{f''}(u_j), \qquad \overline{\widehat{g_1''}}(C) \;\approx\; \frac{1}{S}\sum_{j=1}^{S}\Big(2\,\widehat{f'}(u_j) + u_j\,\widehat{f''}(u_j)\Big),$$

   using $g_1''(z) = 2f'(z) + zf''(z)$.
3. *Bin-average density:* $\widehat{\overline{f}}(C) = \frac{1}{w}\sum_{j=1}^S \hat{p}_j$.

This implies a plug-in bias estimator

$$\hat{b}(C) \;=\; \frac{h^2}{24}\,\frac{\widehat{\overline{g_1''}}(C) \;-\; \hat{\kappa}(C)\,\widehat{\overline{g_0''}}(C)}{\widehat{\overline{f}}(C)}\,, \tag{20}$$

hence, one can reduce the bias in the conditional estimator,

$$\tilde{\kappa}_h(C) \;=\; \hat{\kappa}_h(C) + \hat{b}(C).$$

Remarks. Using only interior classes in the local fit (dropping the first/last classes) stabilizes the curvature estimates. Note that this bias-correction does not eliminate the bias entirely, but change the rate from $O(h^2)$ to $O(h^4)$ coming from the local Taylor approximation.

A2.4. Bias of $\hat{\boldsymbol{\beta}}$ under fixed S (with W observed)

This section studies the OLS estimator of $\boldsymbol{\beta}$ when the number of split samples S is *fixed*. Recall the DGP,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\boldsymbol{\gamma} + \mathbf{u}, \qquad \mathbb{E}[\mathbf{u} \mid \mathbf{X}, \mathbf{W}] = 0, \tag{21}$$

and $\mathbf{M}_W := I - \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$ be the residual-maker matrix with observed $\mathbf{W}$. Let $\ddot{\mathbf{X}}$ denote the imputed regressor. With fixed S let us note the LS estimator,

$$\hat{\boldsymbol{\beta}}_S := \left(\ddot{\mathbf{X}}'\mathbf{M}_W\ddot{\mathbf{X}}\right)^{-1}\ddot{\mathbf{X}}'\mathbf{M}_W\mathbf{y}. \tag{22}$$

The true conditional mean function is $\kappa(\mathcal{C}_m^{(s)}) := \mathbb{E}[X \mid X^* \in \mathcal{C}_m^{(s)}]$ and define the (fixed-S) shifting target $\kappa_h(\mathcal{C}_m^{(s)})$, i.e. the limit (as $N \to \infty$) of the shifting-based approximation on working step h (with interior interval width $w = Sh$). Let

$$\delta_h(\mathcal{C}_m^{(s)}) := \kappa_h(\mathcal{C}_m^{(s)}) - \kappa(\mathcal{C}_m^{(s)}).$$

For each observation, write $\mu_i := \kappa(X_i^*)$ and $\mu_{S,i} := \kappa_h(X_i^*) = \mu_i + \delta_i$, with $\delta_i := \delta_h(X_i^*)$. Let $\tilde{\mathbf{R}} := \mathbf{M}_W\mathbf{R}$ denote residualized variables. Define the population matrices

$$A := \mathbb{E}[\tilde{\mu}\,\tilde{\mu}'], \qquad A_S := \mathbb{E}[\tilde{\mu}_S\,\tilde{\mu}_S'], \qquad B := \mathbb{E}[\tilde{\mu}\,\tilde{X}'], \qquad B_S := \mathbb{E}[\tilde{\mu}_S\,\tilde{X}'].$$

Assume A has non-degenerate variation after partialling out W. The LS estimator with fixed S is defined as

$$\boldsymbol{\beta}_S := A_S^{-1}B_S\,\boldsymbol{\beta}. \tag{23}$$

Note that when $S \to \infty$ (so $h \to 0$), $\delta_h(\cdot) \to 0$ and $\boldsymbol{\beta}_S \to \boldsymbol{\beta}$.

Let us assume that for interior bins, there exists $K < \infty$ such that

$$\sup_{\mathcal{C}_m^{(s)}} \|\delta_h(\mathcal{C}_m^{(s)})\| \leq Kh^2, \tag{24}$$

where $h = w/S$ for interior bins and $\mathcal{C}_m^{(s)}$ denotes the set of interior classes. The under usual LS assumptions with fixed S,

$$\hat{\boldsymbol{\beta}}_S \xrightarrow{p} \boldsymbol{\beta}_S,$$

where $\boldsymbol{\beta}_S$ is defined in (23). Moreover,

$$\|\boldsymbol{\beta}_S - \boldsymbol{\beta}\| = O(h^2) = O\Big(\frac{w^2}{S^2}\Big). \tag{25}$$

Consequently,

$$\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta} = O_p(N^{-1/2}) + O\Big(\frac{w^2}{S^2}\Big). \tag{26}$$

Proof. Write $\tilde{\mathbf{y}} = \mathbf{M}_W \mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{u}}$. Then (22) can be written as

$$\hat{\boldsymbol{\beta}}_S = \left(\frac{1}{N}\ddot{\mathbf{X}}'\mathbf{M}_W\ddot{\mathbf{X}}\right)^{-1}\left(\frac{1}{N}\ddot{\mathbf{X}}'\tilde{\mathbf{y}}\right).$$

For fixed S, $\ddot{\mathbf{X}}_i \xrightarrow{p} \mu_{S,i}$ cell-wise as $N \to \infty$, hence by LLN,

$$\frac{1}{N}\ddot{\mathbf{X}}'\mathbf{M}_W\ddot{\mathbf{X}} \xrightarrow{p} A_S, \qquad \frac{1}{N}\ddot{\mathbf{X}}'\tilde{\mathbf{y}} \xrightarrow{p} \mathbb{E}[\tilde{\mu}_S\tilde{\mathbf{y}}] = \mathbb{E}[\tilde{\mu}_S\tilde{\mathbf{X}}']\boldsymbol{\beta} + \mathbb{E}[\tilde{\mu}_S\tilde{\mathbf{u}}].$$

By exogeneity and measurability of μ_S w.r.t. $(\mathbf{X}^*, \mathbf{W})$, $\mathbb{E}[\tilde{\mu}_S\tilde{\mathbf{u}}] = 0$, giving $\hat{\boldsymbol{\beta}}_S \xrightarrow{p} A_S^{-1}B_S\boldsymbol{\beta} = \boldsymbol{\beta}_S$.

For the bias rate, write $\mu_S = \mu + \delta$ with $\|\delta\| \leq Kh^2$ uniformly on interior bins by (24) and use the expansions

$$A_S = A + \mathbb{E}[\tilde{\mu}\tilde{\delta}'] + \mathbb{E}[\tilde{\delta}\tilde{\mu}'] + \mathbb{E}[\tilde{\delta}\tilde{\delta}'], \qquad B_S = B + \mathbb{E}[\tilde{\delta}\tilde{\mathbf{X}}'].$$

Under finite moments, $\|A_S - A\| = O(h^2)$ and $\|B_S - B\| = O(h^2)$. A first-order matrix inverse expansion yields

$$\boldsymbol{\beta}_S - \boldsymbol{\beta} = A^{-1}(B_S - B)\boldsymbol{\beta} - A^{-1}(A_S - A)\boldsymbol{\beta} + O(h^4),$$

so $\|\boldsymbol{\beta}_S - \boldsymbol{\beta}\| = O(h^2)$. Finally, $\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta} = (\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta}_S) + (\boldsymbol{\beta}_S - \boldsymbol{\beta})$ with $\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta}_S = O_p(N^{-1/2})$ by standard LS arguments.

Effect of de-biasing κ_h. If a de-biasing scheme is applied at the κ-stage so that the imputation error satisfies $\sup_{C\in\mathcal{C}_{\mathrm{int}}}\|\delta_h(C)\| \le Kh^4$, then the same argument yields $\|\boldsymbol{\beta}_S - \boldsymbol{\beta}\| = O(h^4)$, and

$$\hat{\boldsymbol{\beta}}_S - \boldsymbol{\beta} = O_p(N^{-1/2}) + O(h^4).$$

Uniformity. If the bound (24) holds uniformly over a class of DGPs (e.g. uniformly bounded curvature and bin probabilities bounded away from zero on interior bins, plus a uniform eigenvalue bound for A), then (25) holds uniformly over that class:

$$\sup_{P\in\mathcal{P}}\|\boldsymbol{\beta}_S(P) - \boldsymbol{\beta}(P)\| \le K_\beta h^2.$$

Exterior boundary bins. The fixed-S bias analysis and rate apply to exterior (i.e., boundary) intervals bins as well. The key points are:
(i) Same quadrature order at the boundary. Let $C = [a, a+w]$ be a boundary bin (say the first bin at the lower support $a = a_\ell$) and let $g_0(z) = f_Z(z)$, $g_1(z) = zf_Z(z)$ be twice continuously differentiable on the *closed* interval $[a, a+w]$, with $\inf_{z\in[a,a+w]} f_Z(z) > 0$. The composite midpoint rule on a uniform working grid of step $h = w/S$ over $[a, a+w]$ still admits the $O(h^2)$ error for each integral $\int_a^{a+w} g_r(z)\,dz$ $(r = 0, 1)$, because the integrands are smooth up to the endpoint and the subinterval midpoints remain interior to $[a, a+w]$. Consequently, the ratio expansion that yields $\kappa_h(C)-\kappa(C) = O(h^2)$ on interior bins remains valid for boundary bins under the same smoothness and positive-mass conditions on $[a, a+w]$
(ii) Uniformity and the $\hat{\boldsymbol{\beta}}$ bias. If the boundary-bin versions of the regularity conditions (bounded g_r'' on $[a, a+w]$ and bin probability bounded below on boundary bins) hold uniformly, then the bound $\sup_{C\in\mathcal{C}}\|\delta_h(C)\| \le Kh^2$ extends to include boundary bins. The proof of then goes through unchanged, yielding the same $O(h^2) = O(w^2/S^2)$ coefficient bias rate with fixed S, uniformly over interior *and* boundary bins.

A2.5. $\hat{\boldsymbol{\pi}}$ *estimator for* $\mathbb{E}\left[\mathbf{y}|\mathbf{y}^* \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_{\mathbf{l}}\right]$

Let $\pi(s, m, \mathbf{l})$ be the parameter of interest for each conditional mean: $\mathbb{E}\left[\mathbf{y}|\mathbf{y}^* \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_{\mathbf{l}}\right]$. We condition the used discretization intervals for $\mathbf{y}$ with $\mathcal{C}_m^{(s)}$ and mutually the exclusive partitions of $\mathbf{X}$ denoted by $\mathcal{D}_{\mathbf{l}}$. Note $\mathcal{D}_{\mathbf{l}}$ elements are vectors denoting each grid point for partition $\mathbf{l}$. Overall, we have $S \times M \times L^K$ different conditional expectations defined by $\pi(s, m, \mathbf{l})$. Note that L is a fixed number by the researcher, K is also a fixed number of explanatory variables, asymptotically $\frac{L^K}{N} \to 0$ is always true, thus the number of observations in these partitions are asymptotically increasing as well.[2]

Let us define the $\boldsymbol{\pi} = (\pi(1, 1, \mathbf{1}, \ldots, \pi(S, 1, \mathbf{1}), \ldots, \pi(S, M, \mathbf{1}), \ldots, \pi(S, M, \mathbf{l}), \ldots, \pi(S, M, \mathbf{L}))'$ the vectorized version of $\pi(s, m, \mathbf{l})$ iterating along all s, m and all partitions $\mathbf{l}$. We propose $\hat{\boldsymbol{\pi}}$, an estimator of $\boldsymbol{\pi}$ via OLS estimator,

$$\hat{\boldsymbol{\pi}} = \left(\mathbf{1}'_{\{\mathbf{y}^\dagger \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \mathbf{1}_{\{\mathbf{y}^\dagger \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \right)^{-1} \mathbf{1}'_{\{\mathbf{y}^\dagger \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \mathbf{y}^\dagger . \tag{27}$$

Under the same assumptions as for $\hat{\boldsymbol{\kappa}}$ in Section A2.1, $\hat{\boldsymbol{\pi}} \to \boldsymbol{\pi}$. The asymptotic distribution of the estimator can be derived, similarly. Let us write,

$$\mathbf{y}^\dagger = \mathbf{1}_{\{\mathbf{y}^\dagger \in \mathcal{C}_\mathbf{m}^{(\mathbf{s})}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \boldsymbol{\pi} + \boldsymbol{\eta}_\pi, \tag{28}$$

where $\boldsymbol{\eta}_\pi = (\eta_{\pi,1}, \ldots, \eta_{\pi,N})'$ is the idiosyncratic term. Under standard OLS assumption, we have

$$\sqrt{N} \left(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi} \right) \overset{\text{a}}{\sim} \mathcal{N} \left(\mathbf{0}, \boldsymbol{\Omega}_{\boldsymbol{\pi}} \right) . \tag{29}$$

The variance of the OLS estimator is given by

$$\boldsymbol{\Omega}_{\boldsymbol{\pi}} = V \left(\boldsymbol{\eta}_\pi \right) \left(\mathbf{1}'_{\{\mathbf{y}^\dagger \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \mathbf{1}_{\{\mathbf{y}^\dagger \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathcal{D}_\mathbf{l}\}} \right)^{-1}, \tag{30}$$

where $V \left(\boldsymbol{\eta}_\pi \right)$ is the variance of the idiosyncratic term. Algorithm A4 describes how to create in practice the working sample that can be used for estimation.

[2] In finite samples L should be chosen such that there are enough observations in each conditional set.

Algorithm A4 Discretization of the outcome variable – creation of working sample

1: Partition the data into L mutually exclusive sets based on the values of $\mathbf{X}$.
2: Estimate $\hat{\boldsymbol{\pi}}$ as defined in Equation (27).
3: Set $c := 1, s := 1, m := 1, k := 1, \mathbf{l} := (1, \dots, 1)$, $\{\mathbf{y}^{WS}, \mathbf{X}^{WS}\} = \emptyset$.
4: Calculate the sample conditional mean: $\hat{\mathbb{E}}\left[\mathbf{X}|\mathbf{X} \in \mathcal{D}_{\mathbf{l}}\right]$ defined by the partition $\mathbf{l}$.
5: Add the c'th element of the conditional mean estimator $\hat{\boldsymbol{\pi}}$ and the calculated sample means of $\mathbf{X}$ to the working sample,

$$\left\{\mathbf{y}_i^{WS}, \mathbf{X}_i^{WS}\right\} := \left\{\mathbf{y}_i^{WS}, \mathbf{X}_i^{WS}, \bigcup_{j=1}^{N}\left(\hat{\boldsymbol{\pi}}(c), \hat{\mathbb{E}}\left[\mathbf{X}|\mathbf{X} \in \mathcal{D}_{\mathbf{l}}\right] \mid \mathbf{X}_j \in \mathcal{D}_{\mathbf{l}}\right)\right\} . \tag{31}$$

6: If $s < S$, then $s := s + 1$ and go to Step 4.
7: If $s = S$, then set $s := 1$, $m := m + 1$ and go to Step 4.
8: If $s = S$, then set $s := 1$, $m := 1$, $k := k + 1$,and set $\mathbf{l} = \mathbf{l} + \mathbf{1}_{\{k\}}$, where $\mathbf{1}_{\{k\}}$ is a $(K \times 1)$ indicator function with value of 1 at element k, otherwise 0. and go to Step 4.

A2.6. Bias, speed of convergence and bias correction for $\hat{\boldsymbol{\pi}}$

Let $g_l(y) = f_{Y|X\in D_l}(y)$ and consider an interior outcome bin $C_m^{(s)}$ with $w_Y = c_m^{(s)} - c_{m-1}^{(s)}$, $d_Y = w_Y/2$, $v_m^{(s)} = (c_{m-1}^{(s)} + c_m^{(s)})/2$, and working step $h_Y = (a_u^Y - a_\ell^Y)/[S(M-1)]$ so that $w_Y = S\,h_Y$. Then, for each (s, m, l),

$$\pi(s, m, l) \;=\; v_m^{(s)} + \frac{w_Y^2}{12}\,(\log g_l)'\big(v_m^{(s)}\big) + O(w_Y^4),$$

wheres for fixed S,

$$\hat{\pi}_S(s, m, l) \;=\; \pi(s, m, l) + O(h_Y^2) \;=\; \pi(s, m, l) + O\Big(\frac{w_Y^2}{S^2}\Big).$$

Hence shifting reduces the midpoint's $O(w_Y^2)$ miss by a factor of $1/S^2$.

Finally, let us remark that the same bias-correction applies to $\pi(s, m, l)$ by replacing f with the conditional density $g_l = f_{Y|X\in D_l}$ and working along the y-axis with step $h_Y = w_Y/S$.

A2.7. Left-hand-side discretization

When $\mathbf{y}$ is discretized, the method constructs cell means $\boldsymbol{\pi}(s,m,l) = \mathbb{E}[\mathbf{y} \mid \mathbf{y}^* \in \mathcal{C}_m^{(s)}, \mathbf{X} \in \mathbf{D}_l]$ and an estimator $\hat{\boldsymbol{\pi}}_S(s,m,l)$. Define the imputed outcome $\ddot{\mathbf{y}}$ by replacing $\mathbf{y}_i^*$ with $\hat{\boldsymbol{\pi}}_S(s(i),m(i),l(i))$. With $\mathbf{W}$ observed, the estimator becomes

$$\hat{\boldsymbol{\beta}}_S^{(y)} := (\mathbf{X}'\mathbf{M}_W\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_W\ddot{\mathbf{y}}.$$

For fixed S, $\ddot{\mathbf{y}}_i \xrightarrow{p} \mathbf{y}_i + \Delta_{S,i}$ where the deterministic approximation error satisfies $\sup \|\Delta_{S,i}\| = O(h_Y^2) = O(w_Y^2/S^2)$ on interior outcome bins. Then

$$\operatorname*{plim}_{N\to\infty} \hat{\boldsymbol{\beta}}_S^{(\mathbf{y})} = \boldsymbol{\beta} + \left(\mathbb{E}[\tilde{\mathbf{X}}\tilde{\mathbf{X}}']\right)^{-1} \mathbb{E}[\tilde{\mathbf{X}}\,\Delta_S], \qquad \|\operatorname{plim} \hat{\boldsymbol{\beta}}_S^{(\mathbf{y})} - \boldsymbol{\beta}\| = O(h_Y^2).$$

Thus the coefficient bias is again of order $O(w_Y^2/S^2)$ (or $O(h_Y^4)$ under debiasing of $\boldsymbol{\pi}$), and the overall error decomposes as $O_p(N^{-1/2}) + O(w_Y^2/S^2)$.

A3. Discretization on Both Sides

Let us investigate the case when discretization happens with both outcome $\mathbf{y}^*$, and with one or more explanatory variables $\mathbf{X}^*$. In this case, we do not need to partition the domain of $\mathbf{X}$, but can use the discretization grids $\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}$ for the explanatory variable. Note as $\mathbf{y}$ is discretized, we need to partition $\mathbf{W}$ with $\mathcal{D}_{\mathbf{l}}$, similarly to the case discussed in Section 4.2 from the main text. To simplify our derivations, let us assume the number of split samples and the number of classes are the same for both $\mathbf{y}$ and for $\mathbf{X}$, thus $S = S_Y = S_X$ and $M = M_Y = M_X$.[3]

Step 1: Identification with $\mathbf{y}^*$ and $\mathbf{X}^*$

Identification, when discretization happens on both sides, Equation (30) holds from the main text, but instead of $\mathbf{X} \in \mathcal{D}_{\mathbf{l}}$, one need to condition on $\mathbf{X} \in \mathcal{C}_{\mathbf{m_X}}$, correct conditioning for $\mathbf{y}$ to $\mathbf{y}^* \in \mathcal{C}_{m_Y}$ and add the conditioning $\mathbf{W} \in \mathcal{D}_{\mathbf{l}}$. This leads to

$$\sum_l \sum_{m_X} \mathbb{E}[\mathbf{y}|\mathbf{y}^* \in \mathcal{C}_{m_Y}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}] \Pr[\mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}] =$$
$$\mathbb{E}\left[\mathbf{X}|\mathbf{y}^* \in \mathcal{C}_{m_Y}^{(s)}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right]\boldsymbol{\beta} + \mathbb{E}\left[\mathbf{W}|\mathbf{y}^* \in \mathcal{C}_{m_Y}^{(s)}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right]\boldsymbol{\gamma} \tag{32}$$

[3] One can extend the analysis, all results are the same.

Note that $\Pr[\mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}]$ and $\mathbb{E}\left[\mathbf{W}|\mathbf{y}^* \in \mathcal{C}^{(s)}_{m_Y}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right]$ are identified as usual. The conditional expectations affected by discretization are given by

$$
\begin{aligned}
\psi_Y(s,\mathbf{m},\mathbf{l}) :&= \mathbb{E}\left(\mathbf{y}|\mathbf{y}^* \in \mathcal{C}^{(s)}_{m_Y}, \mathbf{X}^* \in \mathcal{C}^{(\mathbf{s})}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right) \\
&= \lim_{N,S\to\infty} \mathbb{E}\left(\mathbf{y}^\dagger|\mathbf{y}^\dagger \in \mathcal{C}^{(s)}_{m_Y}, \mathbf{X}^\dagger \in \mathcal{C}^{(\mathbf{s})}_{\mathbf{m},\mathbf{X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right)\ , \\
\psi_X(s,\mathbf{m},\mathbf{l}) :&= \mathbb{E}\left(\mathbf{X}|\mathbf{y}^* \in \mathcal{C}^{(s)}_{m_Y}, \mathbf{X}^* \in \mathcal{C}^{(\mathbf{s})}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right) \\
&= \lim_{N,S\to\infty} \mathbb{E}\left(\mathbf{X}^\dagger|\mathbf{y}^\dagger \in \mathcal{C}^{(s)}_{m_Y}, \mathbf{X}^\dagger \in \mathcal{C}^{(\mathbf{s})}_{\mathbf{m_X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right)\ ,
\end{aligned}
\tag{33}
$$

where $\mathcal{C}^{(s)}_{m_Y}$ is the discretization intervals for $\mathbf{y}$, $\mathcal{C}^{(\mathbf{s})}_{\mathbf{m_X}}$ is the discretization grid for $\mathbf{X}$. Note that, $s = \{s_Y, s_X\}$ and $m = \{m_Y, \mathbf{m}_X\}$ in this case, for $\mathcal{C}^{(s)}_{m_Y}$ and for $\mathcal{C}^{(\mathbf{s})}_{\mathbf{m_X}}$.

Step 2: OLS estimators for conditional expectations

We propose conditional mean estimators, $\hat{\boldsymbol{\psi}}_Y$ and $\hat{\boldsymbol{\psi}}_X$ via OLS in the same spirit as for the other cases. To show the properties of the estimators, let us make our notation more tractable. First, let us define $\mathbf{s} = (s_Y, s_X)$ the split samples used for $\mathbf{y}$ and $\mathbf{X}$. The number of split samples S_Y and S_X can be the same or different. Let $\mathbf{m} = (m_Y, \mathbf{m}_X)$, vector for the intervals for $\mathbf{y}$ (scalar) and grids for $\mathbf{X}$ (vector with $(K \times 1)$ elements). The number of used intervals and grids may be the same or different. Finally, $\mathbf{l}$ represents the partition vector for the J variables from $\mathbf{W}$.

Overall for $\psi_Y(s,\mathbf{m},\mathbf{l})$ and $\psi_X(s,\mathbf{m},\mathbf{l})$, we have $S_Y \times S_X \times M_Y \times M_X^K \times L^J$ different cases. We use vectorized versions for both denoted by $\boldsymbol{\psi}_Y$ and $\boldsymbol{\psi}_X$, that contains these elements iterating in the order of $s_y, s_m, m_y, \mathbf{m}_X, \mathbf{l}$. We propose estimators $\widehat{\boldsymbol{\psi}}_Y$ and $\widehat{\boldsymbol{\psi}}_X$ via OLS. To simplify our notation, let $\mathbf{1}'_{\{\psi\}} = \mathbf{1}'_{\{\mathbf{y}^\dagger \in \mathcal{C}^{(s)}_{m,Y}, \mathbf{X}^\dagger \in \mathcal{C}^{(\mathbf{s})}_{\mathbf{m},\mathbf{X}}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\}}$.

The estimators are:

$$
\begin{aligned}
\widehat{\boldsymbol{\psi}}_Y &= \left(\mathbf{1}'_{\{\psi\}}\mathbf{1}_{\{\psi\}}\right)^{-1}\mathbf{1}'_{\{\psi\}}\mathbf{y}^\dagger\,, \\
\widehat{\boldsymbol{\psi}}_X &= \left(\mathbf{1}'_{\{\psi\}}\mathbf{1}_{\{\psi\}}\right)^{-1}\mathbf{1}'_{\{\psi\}}\mathbf{X}^\dagger\,.
\end{aligned}
\tag{34}
$$

Under same assumptions as in Appendix B.1. and B.2. from the main text $\widehat{\boldsymbol{\psi}}_Y \to \boldsymbol{\psi}_Y$ and $\widehat{\boldsymbol{\psi}}_X \to \boldsymbol{\psi}_X$. To be specific, we utilize the weak law of large numbers in both cases and $\lim_{S_Y\to\infty} F_{\mathbf{y}^\dagger}(\cdot) = F(\cdot), \lim_{S_X\to\infty} F_{\mathbf{X}^\dagger}(\cdot) = F_{\mathbf{X}}(\cdot)$ under Assumptions 2.a), 2.b) and $\Pr(\mathbf{y} \in \mathcal{S}_s) = 1/S_Y, \Pr(\mathbf{X} \in \mathcal{S}_s) = 1/S_X$

as shown in Section 3.2. Note that K, J are the number of (discretized) regressors and fixed, as well as L the number of partitions. The asymptotic distribution of the estimator can be derived, similarly. Let us write,

$$\mathbf{y}^{\dagger} = \mathbf{1}_{\{\psi\}}\boldsymbol{\psi}_Y + \boldsymbol{\eta}_{\psi_Y} \,, \qquad \mathbf{X}^{\dagger} = \mathbf{1}_{\{\psi\}}\boldsymbol{\psi}_X + \boldsymbol{\eta}_{\psi_X}, \tag{35}$$

where $\boldsymbol{\eta}_{\psi_Y} = (\eta_{\psi_Y,1}, \dots, \eta_{\psi_Y,N})'$ and $\boldsymbol{\eta}_{\psi_X} = (\eta_{\psi_X,1}, \dots, \eta_{\psi_X,N})'$ are the corresponding idiosyncratic terms. Under standard OLS assumption, we have

$$\sqrt{N}\left(\widehat{\boldsymbol{\psi}}_Y - \boldsymbol{\psi}_Y\right) \overset{\text{a}}{\sim} \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Omega}_{\boldsymbol{\psi}_Y}\right) \,, \qquad \sqrt{N}\left(\widehat{\boldsymbol{\psi}}_X - \boldsymbol{\psi}_X\right) \overset{\text{a}}{\sim} \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Omega}_{\boldsymbol{\psi}_X}\right) \,. \tag{36}$$

The variance of the OLS estimators is given by

$$\boldsymbol{\Omega}_{\boldsymbol{\psi}_Y} = V\left(\boldsymbol{\eta}_{\psi_Y}\right)\left(\mathbf{1}_{\{\psi\}}'\mathbf{1}_{\{\psi\}}\right)^{-1} \,, \qquad \boldsymbol{\Omega}_{\boldsymbol{\psi}_X} = V\left(\boldsymbol{\eta}_{\psi_X}\right)\left(\mathbf{1}_{\{\psi\}}'\mathbf{1}_{\{\psi\}}\right)^{-1} \,, \tag{37}$$

where $V\left(\boldsymbol{\eta}_{\psi_Y}\right), V\left(\boldsymbol{\eta}_{\psi_X}\right)$ are the variances of the idiosyncratic term. Algorithm A5 describes how to create in practice a working sample when discretization happens on both sides of the regression equation.

Algorithm A5 Discretization on both sides – creation of working sample

1: Partition the $\mathbf{W}$ into L mutually exclusive sets.
2: Estimate $\hat{\boldsymbol{\psi}}_Y$ and $\hat{\boldsymbol{\psi}}_X$ as defined in Equation (34).
3: Set $c := 1, s_Y := 1, s_X := 1, m_Y := 1, \mathbf{m}_X := (1,\ldots,1), k := 0, \mathbf{l} := (1,\ldots,1),\ \ j := 0\,,\ \{\mathbf{y}^{WS}, \mathbf{X}^{WS}, \mathbf{W}^{WS}\} = \emptyset$.
4: Calculate the sample conditional mean:
$\bar{\mathbf{w}} := \hat{\mathbb{E}}\left[\mathbf{W}|\mathbf{y}^* \in \mathcal{C}_{m_y}^{(s_y)}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}^{(\mathbf{s_x})}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right]$.
5: Add the c'th element of the conditional mean estimators $\widehat{\boldsymbol{\psi}}_Y, \widehat{\boldsymbol{\psi}}_X$ and the calculated sample means of $\mathbf{W}$ to the working sample,

$$\left\{\mathbf{y}_i^{WS}, \mathbf{X}_i^{WS}, \mathbf{W}_i^{WS}\right\} := \left\{\mathbf{y}_i^{WS}, \mathbf{X}_i^{WS}, \mathbf{W}_i^{WS}\right. \tag{38}$$

$$\left.\bigcup_{j=1}^{N}\left(\widehat{\boldsymbol{\psi}}_Y(c), \widehat{\boldsymbol{\psi}}_X(c), \bar{\mathbf{w}} \mid \mathbf{y}_j^* \in \mathcal{C}_{m_y}^{(s_y)}, \mathbf{X}_j^* \in \mathcal{C}_{\mathbf{m_X}}^{(\mathbf{s_x})}, \mathbf{W_j} \in \mathcal{D}_{\mathbf{l}}\right)\right\}. \tag{39}$$

6: If $s_Y < S_Y$, then $s_Y := s_Y + 1$ and go to Step 4.
7: If $s_Y = S_Y$, then set $s_Y := 1$, $s_X := s_X + 1$ and go to Step 4.
8: If $s_X = S_X$, then set $s_Y := 1, s_X := 1$, $m_Y := m_Y + 1$ and go to Step 4.
9: If $m_y = M_Y$, then set $s_Y := 1, s_X := 1, m_Y := 1, k := k+1$,
$\mathbf{m}_X := \mathbf{m}_X + \mathbf{1}_{\{k\}}$ and go to Step 4.
10: If $k = K - 1$, then set $s_Y := 1, s_X := 1, m_Y := 1, k := 0$,
$\mathbf{m}_X := (1,\ldots,1), j := j+1$, $\mathbf{l} := \mathbf{l} + \mathbf{1}_{\{j\}}$ and go to Step 4.

Step 3: OLS estimator for $\boldsymbol{\beta}$

To get a consistent estimator for $\boldsymbol{\beta}$, let us define $\breve{\mathbf{y}}$ and $\breve{\mathbf{X}}$ that takes the corresponding values from $\hat{\boldsymbol{\psi}}_Y$, and $\hat{\boldsymbol{\psi}}_X$ based on the discretized values of $\mathbf{y}^*$ and $\mathbf{X}^*$. Note that if there are additional controls $\mathbf{W}$, one needs to replace it with $\breve{\mathbf{W}}$ that takes the sample conditional means of
$\mathbb{E}\left(\mathbf{W}|\mathbf{y}^* \in \mathcal{C}_{m_Y}^{(s)}, \mathbf{X}^* \in \mathcal{C}_{\mathbf{m_X}}^{(\mathbf{s})}, \mathbf{W} \in \mathcal{D}_{\mathbf{l}}\right)$.

$$\breve{\mathbf{y}} = \breve{\mathbf{X}}\boldsymbol{\beta} + \breve{\mathbf{W}}\boldsymbol{\gamma} + \breve{\boldsymbol{\nu}}\,, \tag{40}$$

where $\breve{\boldsymbol{\nu}} = (\nu_1, \ldots, \nu_N)$. The OLS estimator for $\boldsymbol{\beta}$ is,

$$\hat{\boldsymbol{\beta}} = \left(\breve{\mathbf{X}}'\mathbf{M}_{\breve{\mathbf{W}}}\breve{\mathbf{X}}\right)^{-1}\breve{\mathbf{X}}'\mathbf{M}_{\breve{\mathbf{W}}}\breve{\mathbf{y}}\,, \tag{41}$$

where $\mathbf{M}_{\breve{\mathbf{W}}}$ is the usual residual maker, using $\breve{\mathbf{W}}$. Note that $\mathbb{E}[\breve{\nu}_i] = 0$ for all $i = 1, \dots, N$ since $\breve{\mathbf{y}}$, $\breve{\mathbf{X}}$ and $\breve{\mathbf{W}}$ are consistent estimates of the corresponding conditional expectations. Moreover, $\mathbb{E}[\breve{\nu}_i \breve{\nu}_j] = 0$ for $i \neq j$ due to $\mathcal{D}_{\mathbf{l}}$, and $\mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}$ in m and k are mutually exclusive. $\mathbb{E}[\breve{\nu}_i | \breve{\mathbf{X}}, \breve{\mathbf{W}}] = 0$ since the discretization does not affect the sampling error. Furthermore,if the discretization is mean independent from the discretization scheme, then, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = o_p(1)$. Under the these assumptions, $\mathbb{E}\left[\breve{\boldsymbol{\nu}}\breve{\boldsymbol{\nu}}' | \breve{\mathbf{X}}, \breve{\mathbf{W}}\right] = \sigma_{\nu}^2 \mathbf{I}$, the asymptotic distribution of $\hat{\boldsymbol{\beta}}$ is given by $\sqrt{N}\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \overset{\text{a}}{\sim} \mathcal{N}\left(\mathbf{0}, \sigma_{\boldsymbol{\nu}}^2 \left(\breve{\mathbf{X}}' \mathbf{M}_{\breve{\mathbf{W}}} \breve{\mathbf{X}}\right)^{-1}\right)$.

A4. Set Identification with Sharp Boundaries

In this section, we analize how set identification relates to our approach. We derive sharp boundaries for β in the discussed cases of linear regression based on where discretization happens. Then, we extend these results to split sampling and show how the identified region behaves as $S \to \infty$. To do so, first, let us define the bin endpoints as

$$\ell_m := c_{m-1}, \qquad u_m := c_m, \qquad m = 1, \dots, M.$$

Here, we consider three canonical regression problems where only the discretized Z^* is observed:

1. Linear regression with an interval regressor. The model is

   $$Y = \alpha + \beta Z + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid Z] = 0,$$

   but the analyst observes (Y, Z^*) instead of (Y, Z).
2. Linear regression with interval outcome. The latent variable satisfies

   $$Z = \alpha + \beta X + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0,$$

   but only the discretized value Z^* (hence the bin containing Z) is observed together with X.
3. Linear regression with interval regressor and interval outcome. The latent variable satisfies

   $$Y = Z'\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0,$$

   but both the outcome Y^* and the regressor variable Z^* is discretized and observed.

In all cases, discretization yields *interval information* about Z, which leads to the *set identification* of the parameters. Below we derive sharp identified sets and their boundaries. To simplify our analysis we restrict our attention to simple linear regression cases without further covariates.

A4.1. Case 1: $Y = \alpha + \beta Z + \varepsilon$ *with* $\mathbb{E}[\varepsilon \mid Z] = 0$*; observe* (Y, Z^*)

Define the bin-specific conditional means for the outcome (identified from the data):

$$\mu_m := \mathbb{E}\big[\, Y \mid Z^* = v_m \,\big], \qquad m = 1, \ldots, M.$$

When $Z^* = v_m$, we know $Z \in [\ell_m, u_m)$. The linear model implies the conditional mean restrictions

$$\mu_m \in \big[\alpha + \beta\, \ell_m,\ \alpha + \beta\, u_m\big], \qquad m = 1, \ldots, M. \tag{42}$$

Sharp identified set for (α, β)

Define the strip for bin m:

$$\mathcal{S}_m := \big\{(\alpha, \beta) \in \mathbb{R}^2 :\ \mu_m \in [\alpha + \beta \ell_m,\ \alpha + \beta u_m]\big\}\,.$$

Then the *sharp* identified set is the intersection of strips:

$$\Theta \;=\; \bigcap_{m=1}^{M} \mathcal{S}_m. \tag{43}$$

Projection to β *and feasibility condition*

For a fixed β, inequalities (42) become

$$\mu_m - \beta u_m \;\le\; \alpha \;\le\; \mu_m - \beta \ell_m, \qquad m = 1, \ldots, M.$$

Thus a given β is feasible iff the implied intervals for α overlap:

$$L(\beta) \;\le\; U(\beta),$$

where

$$L(\beta) := \max_m \big(\mu_m - \beta u_m\big), \qquad U(\beta) := \min_m \big(\mu_m - \beta \ell_m\big).$$

Hence the *projected* identified set for β is

$$\mathcal{B} \;=\; \Big\{\beta \in \mathbb{R} :\ L(\beta) \le U(\beta)\Big\}. \tag{44}$$

Boundary characterization for β

The boundaries of $\mathcal{B}$ occur at values of β where the overlap becomes tight, i.e.,

$$\mu_j - \beta u_j \;=\; \mu_k - \beta \ell_k \quad \text{for some } j, k \in \{1, \ldots, M\}.$$

Solving yields candidate boundary values

$$\beta_{jk} \;=\; \frac{\mu_j - \mu_k}{u_j - \ell_k}, \qquad j, k = 1, \ldots, M,\ u_j \neq \ell_k. \tag{45}$$

Sort the β_{jk}'s; on each interval between adjacent candidates (and on the tails), test feasibility $L(\beta) \leq U(\beta)$. The union of feasible sub-intervals is $\mathcal{B}$; its smallest and largest points are the identified-set boundaries.

Now, let us express the feasibility condition, for which we define,

$$\mathcal{S}_+ := \{(j,k) : u_j > l_k\}, \quad \mathcal{S}_- := \{(j,k) : u_j < l_k\}, \quad \mathcal{S}_0 := \{(j,k) : u_j = l_k\}\,.$$

If there is no violation among equal-endpoint pairs, we get the feasibility condition:

$$\forall (j,k) \in \mathcal{S}_0 : \mu_j \leq \mu_k\,.$$

The lower and upper boundaries for β are:

$$\beta_L \;=\; \sup_{(j,k)\in\mathcal{S}_+} \frac{\mu_j - \mu_k}{u_j - \ell_k}, \qquad \beta_U \;=\; \inf_{(j,k)\in\mathcal{S}_-} \frac{\mu_j - \mu_k}{u_j - \ell_k}. \tag{46}$$

With the conventions $\sup \varnothing = -\infty$ and $\inf \varnothing = +\infty$, thus Equation (46) also covers cases where the identified set is one-sided unbounded. This expression is the same as Equation (2.21) in Molinari (2020).[4]

Using split samples: shifting method

Next, we extend our set-identification with multiple split samples, generated by the shifting method. We employ set-identification for each split sample and then use the fact that there is indeed one true β parameter. Let us write the lower and upper boundaries for β for split sample s:

$$\beta_L^{(s)} \;=\; \sup_{(j,k)\in\mathcal{S}_+^{(s)}} \frac{\mu_j^{(s)} - \mu_k^{(s)}}{u_j^{(s)} - \ell_k^{(s)}}, \qquad \beta_U^{(s)} \;=\; \inf_{(j,k)\in\mathcal{S}_-^{(s)}} \frac{\mu_j^{(s)} - \mu_k^{(s)}}{u_j^{(s)} - \ell_k^{(s)}}, \tag{47}$$

[4] This is a simplified version of the linear regression model outlined before and Molinari (2020) provides a more general theorem.

where each conditional means $(\mu_j^{(s)}, \mu_k^{(s)})$, bin endpoints $(u_j^{(s)} - \ell_k^{(s)})$ and feasibility conditions $(\mathcal{S}_+^{(s)}, \mathcal{S}_-^{(s)}, \mathcal{S}_0^{(s)})$ depends on the split sample. We utilize the fact that β is the same, therefore using the set-identified regions from each split sample we can use the largest lower-boundary value as well as the smallest upper-boundary:

$$\begin{aligned} \beta_L = \max_s \beta_L^{(s)} &= \max_s \sup_{(j,k)\in\mathcal{S}_+^{(s)}} \frac{\mu_j^{(s)} - \mu_k^{(s)}}{u_j^{(s)} - \ell_k^{(s)}}, \\ \beta_U = \min_s \beta_U^{(s)} &= \min_s \inf_{(j,k)\in\mathcal{S}_-^{(s)}} \frac{\mu_j^{(s)} - \mu_k^{(s)}}{u_j^{(s)} - \ell_k^{(s)}}. \end{aligned} \tag{48}$$

If we use the shifting method where we systematically map the domain of Z, then as $S \to \infty$, the lower and upper boundaries get closer to each other.

A4.2. Case 2: $Z = X'\beta + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$; observe Z^*

Let us define the (random) endpoints associated with the observation Z^*:

$$\ell(Z^*) := \sum_{m=1}^{M} \ell_m \, \mathbf{1}\{Z^* = v_m\}, \qquad u(Z^*) := \sum_{m=1}^{M} u_m \, \mathbf{1}\{Z^* = v_m\}.$$

Then almost surely

$$\ell(Z^*) \;\le\; Z \;<\; u(Z^*).$$

Assuming linear model, taking conditional expectations given $X = x$ yields

$$\mathbb{E}\big[\ell(Z^*) \mid X = x\big] \;\le\; \mathbb{E}[Z \mid X = x] \;=\; \alpha + x'\beta \;\le\; \mathbb{E}\big[u(Z^*) \mid X = x\big].$$

Define the *identified conditional bounds*

$$L(x) := \mathbb{E}\big[\ell(Z^*) \mid X = x\big], \qquad U(x) := \mathbb{E}\big[u(Z^*) \mid X = x\big].$$

These are point-identified because $\ell(Z^*)$ and $u(Z^*)$ are deterministic functions of the observed Z^*.

Sharp identified set for β

The *sharp* identified set is the collection of β satisfying the aforementioned linear inequality family. For a fixed β, the admissible interval for α:

$$\alpha \in \Big[\Gamma_L(\beta),\, \Gamma_U(\beta)\Big], \quad \Gamma_L(\beta) := \sup_x\{\, L(x) - \beta x \,\}, \quad \Gamma_U(\beta) := \inf_x\{\, U(x) - \beta x \,\}.$$

Hence the projected identified set for β is

$$\mathcal{B} \;=\; \big\{\beta \in \mathbb{R} :\; \Gamma_L(\beta) \;\le\; \Gamma_U(\beta)\big\}.$$

Next, observe that $\Gamma_L(\beta) \le \Gamma_U(\beta)$ holds if and only if

$$L(x) - \beta x \;\le\; U(x') - \beta x' \quad \text{for all pairs } (x, x') \text{ in the support of } X.$$

Equivalently,

$$\beta\,(x' - x) \;\le\; U(x') - L(x) \qquad \forall\, x, x'.$$

For each ordered pair (x, x') this yields a half-line restriction:

$$\begin{cases} \beta \;\le\; \dfrac{U(x') - L(x)}{x' - x}, & \text{if } x' > x, \\[2ex] \beta \;\ge\; \dfrac{U(x') - L(x)}{x' - x}, & \text{if } x' < x. \end{cases}$$

Taking the tightest such restrictions gives the lower and upper boundaries

$$\beta_L \;=\; \sup_{x' < x} \frac{U(x') - L(x)}{x' - x}, \qquad \beta_U \;=\; \inf_{x' > x} \frac{U(x') - L(x)}{x' - x}. \tag{49}$$

With the conventions $\sup \varnothing = -\infty$ and $\inf \varnothing = +\infty$, (49) also covers one-sided unbounded cases. The identified set is then $\mathcal{B} = [\beta_L, \beta_U]$ (and empty if $\beta_L > \beta_U$).

Shifting method then provides the same argument as for Case 1, which leads to the same results, but now applying Equation (49) yields,

$$\begin{aligned} \beta_L = \max_s \beta_L^{(s)} \;&=\; \max_s \sup_{x' < x} \frac{U^{(s)}(x') - L^{(s)}(x)}{x' - x}, \\ \beta_U = \min_s \beta_U^{(s)} \;&=\; \min_s \inf_{x' > x} \frac{U^{(s)}(x') - L^{(s)}(x)}{x' - x}. \end{aligned} \tag{50}$$

A4.3. Case 3: Discretization on Both Sides

In this setup both outcome and regressor are latent: $(Z, Y) \in \mathbb{R}^2$ with the DGP,

$$Y \;=\; \alpha + \beta Z + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid Z] = 0.$$

For Z, let the known cutpoints be

$$c_0^Z < c_1^Z < \cdots < c_{M_Z}^Z, \qquad C_m^Z = [c_{m-1}^Z, c_m^Z), \qquad \ell_m^Z := c_{m-1}^Z, \quad u_m^Z := c_m^Z,$$

and the observed value $Z^* = v_m^Z$ iff $Z \in \mathcal{C}_m^Z$. For Y, let the known cutpoints be

$$c_0^Y < c_1^Y < \cdots < c_{M_Y}^Y, \qquad \mathcal{C}_\nu^Y = [c_{\nu-1}^Y, c_\nu), \qquad \ell_\nu^Y := c_{\nu-1}, \;\; u_\nu^Y := c_\nu,$$

and the observed value $Y^* = w_\nu$ iff $Y \in D_\nu$, with $\nu = 1, \ldots, M_Y$.

Define the endpoint maps

$$\ell_Z(Z^*) = \sum_{m=1}^{M_Z} \ell_m^Z \, \mathbf{1}\{Z^* = v_m\}, \quad u_Z(Z^*) = \sum_{m=1}^{M_Z} u_m^Z \, \mathbf{1}\{Z^* = v_m\},$$

$$\ell_Y(Y^*) = \sum_{\nu=1}^{M_Y} \ell_\nu^Y \, \mathbf{1}\{Y^* = w_\nu\}, \quad u_Y(Y^*) = \sum_{\nu=1}^{M_Y} u_\nu^Y \, \mathbf{1}\{Y^* = w_\nu\}.$$

By construction,

$$\ell_Z(Z^*) \le Z < u_Z(Z^*) \quad \text{and} \quad \ell_Y(Y^*) \le Y < u_Y(Y^*) \qquad \text{a.s.}$$

Conditional bounds we can observe.
For each regressor bin m,

$$p_{\nu|m} := \mathbb{P}(Y^* = w_\nu \mid Z^* = v_m) \quad \text{is observed from data,}$$

and the conditional mean of Y given $Z^* = v_m$ is bounded by

$$L_Y(m) \; := \; \sum_{\nu=1}^{M_Y} p_{\nu|m} \, \ell_\nu^Y \;\; \le \;\; \mu_m \; := \; \mathbb{E}[Y \mid Z^* = v_m] \;\; \le \;\; U_Y(m) \; := \; \sum_{\nu=1}^{M_Y} p_{\nu|m} \, u_\nu^Y. \tag{51}$$

Likewise, because $Z \in [\ell_m^Z, u_m^Z)$ a.s. on $\{Z^* = v_m\}$, we have

$$\eta_m \; := \; \mathbb{E}[Z \mid Z^* = v_m] \; \in \; [\, \ell_m^Z, \; u_m^Z \,]. \tag{52}$$

Structural link across these means.
Using $\mathbb{E}[\varepsilon \mid Z] = 0$ and iterated expectations,

$$\mu_m = \mathbb{E}[Y \mid Z^* = v_m] = \mathbb{E}[\alpha + \beta Z + \varepsilon \mid Z^* = v_m] = \alpha + \beta \, \mathbb{E}[Z \mid Z^* = v_m] = \alpha + \beta \, \eta_m.$$

Combine this with (51)–(52): for each m,

$$\exists \, \eta_m \in [\ell_m^Z, u_m^Z] \;\text{ such that }\; \alpha + \beta \, \eta_m \in [\, L_Y(m), \, U_Y(m) \,]. \tag{53}$$

Geometrically, the line $y = \alpha + \beta z$ must intersect the rectangle

$$\mathcal{R}_m \;:=\; [\,\ell_m^Z, u_m^Z\,] \;\times\; [\,L_Y(m), U_Y(m)\,], \qquad m = 1, \dots, M.$$

Sharp identified set for (α, β).
Condition (53) is *sharp*: given only the discretization and $\mathbb{E}[\varepsilon \mid Z] = 0$, the within-rectangle behavior is otherwise unrestricted. It is equivalent, for each m, to the existence of some $z \in [\ell_m^Z, u_m^Z]$ with

$$L_Y(m) \;\leq\; \alpha + \beta z \;\leq\; U_Y(m).$$

For any fixed β, the admissible α values for bin m are therefore

$$\mathcal{A}_m(\beta) = \bigcup_{z \in [\ell_m^Z, u_m^Z]} \big[\, L_Y(m) - \beta z,\; U_Y(m) - \beta z \,\big] = \big[\, \underline{\alpha}_m(\beta),\; \overline{\alpha}_m(\beta) \,\big],$$

where

$$\underline{\alpha}_m(\beta) = \begin{cases} L_Y(m) - \beta\, u_m^Z, & \beta \geq 0, \\ L_Y(m) - \beta\, \ell_m^Z, & \beta < 0, \end{cases} \qquad \overline{\alpha}_m(\beta) = \begin{cases} U_Y(m) - \beta\, \ell_m^Z, & \beta \geq 0, \\ U_Y(m) - \beta\, u_m^Z, & \beta < 0. \end{cases}$$

(These follow from minimizing/maximizing $y - \beta z$ over $y \in [L_Y(m), U_Y(m)]$ and $z \in [\ell_m^Z, u_m^Z]$.) Hence, for a given slope β, the identified set for α is the intersection

$$\alpha \;\in\; \bigcap_{m=1}^{M} \mathcal{A}_m(\beta) \;=\; \Big[\, L(\beta),\; U(\beta) \,\Big], \quad L(\beta) := \max_m \underline{\alpha}_m(\beta), \quad U(\beta) := \min_m \overline{\alpha}_m(\beta).$$

The *projected* sharp identified set for β is therefore

$$\mathcal{B} \;=\; \{\beta \in \mathbb{R} : \; L(\beta) \leq U(\beta)\}.$$

Boundary (endpoint) characterization for β.
Both $L(\beta)$ and $U(\beta)$ are piecewise-linear in β, with a kink at $\beta = 0$. It is convenient to analyze the nonnegative and nonpositive regions separately.

Region $\beta \geq 0$. In this region

$$L^+(\beta) = \max_m \{\, L_Y(m) - \beta\, u_m^Z \,\}, \qquad U^+(\beta) = \min_m \{\, U_Y(m) - \beta\, \ell_m^Z \,\}.$$

Feasibility requires $L^+(\beta) \le U^+(\beta)$, which is equivalent to the family of pairwise inequalities

$$L_Y(j) - \beta u_j^Z \;\le\; U_Y(k) - \beta \ell_k^Z, \qquad \forall\, j,k.$$

Rearranging gives, for each pair (j,k),

$$\beta\,(u_j^Z - \ell_k^Z) \;\ge\; L_Y(j) - U_Y(k).$$

Thus, for $u_j^Z > \ell_k^Z$ we obtain lower-type restrictions $\beta \ge \dfrac{L_Y(j) - U_Y(k)}{u_j^Z - \ell_k^Z}$, while for $u_j^Z < \ell_k^Z$ we get upper-type restrictions $\beta \le \dfrac{L_Y(j) - U_Y(k)}{u_j^Z - \ell_k^Z}$. Define the index sets

$$\mathcal{S}_+ := \{(j,k) :\; u_j^Z > \ell_k^Z\}, \qquad \mathcal{S}_- := \{(j,k) :\; u_j^Z < \ell_k^Z\}.$$

Then the endpoints within $\{\beta \ge 0\}$ are

$$\beta_L^+ \;=\; \sup_{(j,k)\in\mathcal{S}_+} \frac{L_Y(j) - U_Y(k)}{u_j^Z - \ell_k^Z}, \qquad \beta_U^+ \;=\; \inf_{(j,k)\in\mathcal{S}_-} \frac{L_Y(j) - U_Y(k)}{u_j^Z - \ell_k^Z}. \quad (54)$$

(Use the conventions $\sup\varnothing = -\infty$ and $\inf\varnothing = +\infty$.)

Region $\beta \le 0$. In this region

$$L^-(\beta) = \max_m\{\, L_Y(m) - \beta\,\ell_m^Z\,\}, \qquad U^-(\beta) = \min_m\{\, U_Y(m) - \beta\,u_m^Z\,\}.$$

Feasibility $L^-(\beta) \le U^-(\beta)$ is equivalent to

$$L_Y(j) - \beta \ell_j^Z \;\le\; U_Y(k) - \beta u_k^Z \qquad \forall\, j,k,$$

i.e.,

$$\beta\,(\ell_j^Z - u_k^Z) \;\ge\; L_Y(j) - U_Y(k).$$

Thus, for $\ell_j^Z > u_k^Z$ we obtain lower-type restrictions $\beta \ge \dfrac{L_Y(j) - U_Y(k)}{\ell_j^Z - u_k^Z}$, while for $\ell_j^Z < u_k^Z$ we obtain upper-type restrictions $\beta \le \dfrac{L_Y(j) - U_Y(k)}{\ell_j^Z - u_k^Z}$. Define

$$\widetilde{\mathcal{S}}_+ := \{(j,k) :\; \ell_j^Z > u_k^Z\}, \qquad \widetilde{\mathcal{S}}_- := \{(j,k) :\; \ell_j^Z < u_k^Z\}.$$

Then the endpoints within $\{\beta \leq 0\}$ are

$$\beta_L^- \;=\; \sup_{(j,k)\in\widetilde{\mathcal{S}}_+} \frac{L_Y(j) - U_Y(k)}{\ell_j^Z - u_k^Z}, \qquad \beta_U^- \;=\; \inf_{(j,k)\in\widetilde{\mathcal{S}}_-} \frac{L_Y(j) - U_Y(k)}{\ell_j^Z - u_k^Z}. \quad (55)$$

Identified set regions The sharp identified set for β is the union

$$\mathcal{B} \;=\; \big([\beta_L^+, \beta_U^+] \cap [0,\infty)\big) \;\cup\; \big([\beta_L^-, \beta_U^-] \cap (-\infty, 0]\big),$$

and for any $\beta \in \mathcal{B}$, the sharp identified set for α is

$$\alpha \;\in\; \Big[\,L(\beta),\; U(\beta)\,\Big], \qquad L(\beta) = \max_m \underline{\alpha}_m(\beta), \quad U(\beta) = \min_m \overline{\alpha}_m(\beta).$$

We can compute these boundaries as (Y^*, Z^*) are observed, thus

$$p_{\nu|m} = \mathbb{P}(Y^* = w_\nu \mid Z^* = v_m) \;\text{ is identified,}$$

and we can compute,

$$L_Y(m) \;=\; \sum_{\nu=1}^{M_Y} p_{\nu|m}\, \ell_\nu^Y, \qquad U_Y(m) \;=\; \sum_{\nu=1}^{M_Y} p_{\nu|m}\, u_\nu^Y.$$

These are the *sharp* bounds on $\mu_m = \mathbb{E}[Y \mid Z^* = v_m]$ implied solely by outcome discretization. These can be plugged back to Equation (54) or (55).

Finally, the shifting method provides the same argument as for Case 1 and 2, which leads to the same results, but now applying Equation (55). The formula is quite complicated, however the principle is the same that the maximum and minimum are following each other and as S increases the identified region decreases as long as the observed intervals changing.

A4.4. Speed of convergence with shifting

Fix $M \geq 2$. In split $s = 1, \ldots, S$, recall that the cut points are equally spaced and shifted,

$$h \;=\; \frac{c_m^{(s)} - c_{m-1}^{(s)}}{S(M-1)} \;\propto\; \frac{1}{S},$$

and let $\mathcal{B}^{(s)} = [\beta_L^{(s)}, \beta_U^{(s)}]$ denote the sharp identified set from split s for any of the previous cases. Define the intersection of the combined identified set

$$\mathcal{B}^{(S)} \;=\; \bigcap_{s=1}^{S} \mathcal{B}^{(s)} \;=\; \Big[\max_{s\leq S} \beta_L^{(s)},\; \min_{s\leq S} \beta_U^{(s)}\Big], \qquad \Delta_S \;:=\; \beta_U^{(S)} - \beta_L^{(S)}.$$

Assume: (i) the linear model is correctly specified; (ii) the support of the latent regressor is an interval with density bounded away from zero on its interior; (iii) bin probabilities are uniformly bounded away from zero across splits; (iv) outcomes (or the conditional moments used) are bounded or sub-Gaussian. Suppose the total sample of size N is randomly partitioned into S *independent* splits. If there is no sampling error, then

$$\Delta_S \;=\; O(h) \;=\; O\left(\frac{1}{S}\right).$$

Proof. Fix two interior points $a < b$ in the support of Z with separation $\Delta := b - a \geq \delta > 0$. Because cutpoints are equally spaced and the S splits shift them by steps of size $h = \frac{c_m^{(s)} - c_{m-1}^{(s)}}{S(M-1)}$, for any x in the interior there exists a split whose cutpoint lies within h of x. Hence, for S large enough (so $h < \delta/2$), there exist splits and bins (j,k) such that

$$u_j \in [a,\, a+h] \quad \text{and} \quad \ell_k \in [b-h,\, b],$$

with $u_j < \ell_k$ and $\ell_k - u_j \geq \Delta - 2h > 0$.

Consider Case 1 for concreteness; the other cases are analogous because the identified β-bounds always come from secant-type ratios formed by admissible pairs of $(\mu, \text{endpoint})$ inequalities. Let (α_0, β_0) be the true parameters and $z_m^0 := \mathbb{E}[Z \mid Z^* = v_m] \in [\ell_m, u_m]$ the (unknown) conditional means of Z. Then $\mu_m = \mathbb{E}[Y \mid Z^* = v_m] = \alpha_0 + \beta_0 z_m^0$ and, for the pair (j,k) above,

$$\frac{\mu_j - \mu_k}{u_j - \ell_k} = \beta_0 \, \frac{z_j^0 - z_k^0}{u_j - \ell_k} \qquad \text{with} \quad z_j^0 \leq u_j, \;\; z_k^0 \geq \ell_k.$$

Lower bound. Pick instead bins $\tilde{j}, \tilde{k}$ with $\ell_{\tilde{j}} \in [a-h, a]$ and $u_{\tilde{k}} \in [b, b+h]$, so that $u_{\tilde{j}} > \ell_{\tilde{k}}$ and $u_{\tilde{j}} - \ell_{\tilde{k}} \leq \Delta + 2h$. Then

$$\beta_L^{(S)} \;\geq\; \frac{\mu_{\tilde{j}} - \mu_{\tilde{k}}}{u_{\tilde{j}} - \ell_{\tilde{k}}} = \beta_0 \, \frac{z_{\tilde{j}}^0 - z_{\tilde{k}}^0}{u_{\tilde{j}} - \ell_{\tilde{k}}} \;\geq\; \beta_0 \, \frac{\Delta}{\Delta + 2h} \;=\; \beta_0 \Big(1 - \frac{2h}{\Delta}\Big).$$

Upper bound. For (j,k) with $u_j < \ell_k$ as above,

$$\beta_U^{(S)} \;\leq\; \frac{\mu_j - \mu_k}{u_j - \ell_k} = \beta_0 \, \frac{z_j^0 - z_k^0}{u_j - \ell_k} \;\leq\; \beta_0 \, \frac{\Delta}{\Delta - 2h} \;=\; \beta_0 \Big(1 + \frac{2h}{\Delta - 2h}\Big).$$

Combining,

$$0 \;\leq\; \beta_U^{(S)} - \beta_L^{(S)} \;\leq\; \beta_0 \Big(\frac{2h}{\Delta - 2h} + \frac{2h}{\Delta}\Big) \;=\; O(h),$$

uniformly for $h < \delta/2$. Hence $\Delta_S = \beta_U^{(S)} - \beta_L^{(S)} = O(h) = O(1/S)$. $\square$

A5. Magnifying Method

Here, we discuss another split sampling method that also converges in distribution to the unknown distribution, but has different discretization scheme thus different properties. The idea for magnifying method is to magnify specific parts of the underlying variable's domain. The interval size for each split sample depends on the number of split samples (S) and the number of original intervals (M). As the number of split samples increases, the interval widths decrease, which is the main mechanism for uncovering the unknown distribution. Figure A2 shows the main idea of the magnifying method for the case of $M = 3, S = 4$.

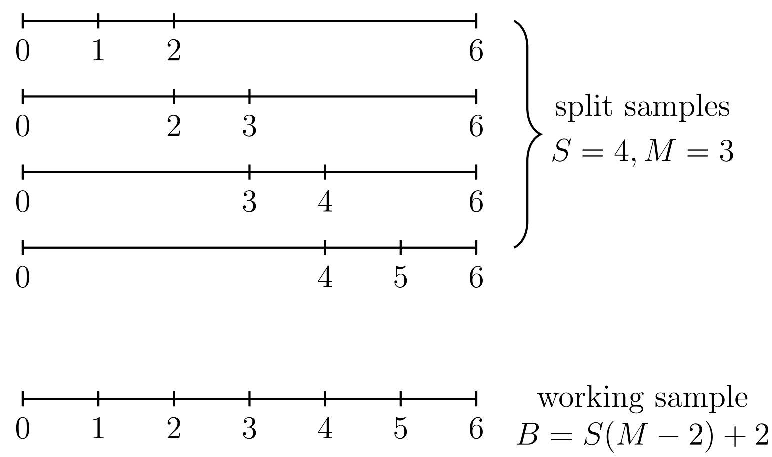

Figure A2: The magnifying method

As seen in Figure A2, there are two types of intervals: i) Around the boundaries of Z, the first and last interval widths are changing and in general they do not decrease as we increase the number of split sizes. Observations that fall into these intervals are called "*non-directly-transferable observations*" or NDTOs. ii) All remaining intervals have widths that are decreasing as S increases. Observations falling into these categories are called "*directly-transferable observations*" or DTOs. These DTOs can then be used to map the original distribution.[5]

[5]Note that, the first and last split samples are slightly different. Their observations from first/last intervals ($c_1^{(1)}$ and $c_M^{(S)}$) are also DTOs, as the interval widths are decreasing

To explore the properties of the magnifying method, let us establish the connection between the number of magnified intervals in the working sample (B), and the number of split samples (S) and intervals (M),

$$B = S(M-2) + 2\,. \tag{56}$$

Note that we have 2 split samples (first and last respectively), where we magnify intervals around the boundary of the domain. Here, we capture $M-1$ intervals of equal size, for all the other $S-2$ split samples, there are $M-2$ intervals.

The widths of the intervals in the working sample are given by,

$$h = \frac{a_u - a_l}{S(M-2)+2}\,. \tag{57}$$

Intuitively, the magnifying method works as if we increase the number of split samples, the magnified interval widths go to zero ($h \to 0$ as $S \to \infty$). Note however, to have this property we will fix the upper and lower bounds of the support for the split samples ($a_l = c_0^{WS} = c_0^{(s)}$; $a_u = c_B^{WS} = c_M^{(s)}$, $\forall s$), thus make sure that the discretized domain does not have infinite support. Next, we derive the boundary points for each magnified split sample and show how and under what assumptions the working sample converges in distribution to the underlying unknown distribution.
The boundary points for each split sample can be derived as

$$c_m^{(s)} = \begin{cases} a_l \text{ or } -\infty & \text{if } m = 0, \\ a_l + mh & \text{if } 0 < m < M \text{ and } s = 1, \\ a_l + h\left[(s-2)(M-2) + M + m - 2\right] & \text{if } 0 < m < M \text{ and } s > 1, \\ a_u \text{ or } \infty & \text{if } m = M. \end{cases} \tag{63}$$

The intuition behind this is that on the boundaries of the support, the split samples take the values of the lower and upper bounds. For the first split sample, one needs to shift the boundary points m times. However, for the other split samples, one needs to push by $h(M-1)$ times to shift through the first questionnaire and then $h(M-2)$ to shift through each split sample in

in S similarly to the intervals that are not at the boundary points ($c_m^{(s)}$, $\forall\, 1 < s < S, 1 < m < M$).

Algorithm A6 Magnifying method – creation of the split samples

1: For any given S and M. Set

$$B = S(M-2)+2 \tag{58}$$
$$h = \frac{a_u - a_l}{B} \tag{59}$$
$$s = 1. \tag{60}$$

2: Set $c_0^{(s)} = a_l$ and $c_M^{(s)} = a_u$.

3: If $s = 1$, then set

$$c_1^{(s)} = c_0^{(s)} + h, \tag{61}$$

else set

$$c_1^{(s)} = c_{M-1}^{(s-1)}. \tag{62}$$

4: Set $c_m^{(s)} = c_{m-1}^{(s)} + h$ for $m = 2, \ldots, M-1$.

5: If $s < S$ then $s := s+1$ and goto Step 2.

between $s = 2$ and $s = S-1$, $s-2$ times. Deriving this process algebraically will result in the above expression.[6] Algorithm A6 shows how to create the boundary points for the split samples in the case of the magnifying method.

The working sample's boundary points are given by the unique boundary points from the split samples, which leads to

$$c_b^{WS} = a_l + bh = a_l + b\frac{a_u - a_l}{S(M-2)+2}\,. \tag{64}$$

To show how and why the magnifying method works, let us derive the different interval widths for each split samples' interval, defined by Equation (63). Let $||\mathcal{C}_m^{(s)}|| = c_m^{(s)} - c_{m-1}^{(s)}$ be the m-th interval width, then for the split samples which are in-between the boundaries ($1 < s < S$) and substituting

[6] There is an alternative way to formalize the boundary points, when one starts from a_u. This formalism leads to the same conclusions.

for h, we can write

$$||\mathcal{C}_m^{(s)}|| = \begin{cases} (a_u - a_l)\left(\frac{s(M-2)+2}{S(M-2)+2} + \frac{1-M}{S(M-2)+2}\right) & \text{if } m = 1,\ 1 < s < S, \\ \frac{a_u - a_l}{S(M-2)+2} & \text{if } 1 < m < M,\ 1 < s < S, \\ (a_u - a_l)\left(1 - \frac{s(M-2)+1}{S(M-2)+2}\right) & \text{if } m = M,\ 1 < s < S\,. \end{cases} \tag{65}$$

We can also define the interval widths for the first and last split samples as

$$\begin{aligned} ||\mathcal{C}_m^{(1)}|| &= \begin{cases} \frac{a_u - a_l}{S(M-2)+2} & \text{if } 1 \leq m < M, \\ (a_u - a_l)\left(1 - \frac{M-1}{S(M-2)+2}\right) & \text{if } m = M, \end{cases} \\ ||\mathcal{C}_m^{(S)}|| &= \begin{cases} (a_u - a_l)\left(1 - \frac{M-1}{S(M-2)+2}\right) & \text{if } m = 1, \\ \frac{a_u - a_l}{S(M-2)+2} & \text{if } 1 < m \leq M. \end{cases} \end{aligned} \tag{66}$$

Note that $||\mathcal{C}_m^{(s)}|| \leq ||\mathcal{C}_1^{(s)}||$ and $||\mathcal{C}_m^{(s)}|| \leq ||\mathcal{C}_M^{(s)}||$. Formally, let us define $\zeta := \{\mathcal{C}_m^{(s)} \mid 1 < m < M, 1 < s < S, \mathcal{C}_m^{(1)} \mid 1 \leq m < M, \mathcal{C}_m^{(S)} \mid 1 < m \leq M\}$ as the set of intervals which have the interval width $\frac{a_u - a_l}{S(M-2)+2}$.

An interesting insight that we can write $\Pr\left((z - z^{(s)})^2 | z \in \zeta \leq (z - z^{(s)})^2 \mid z \notin \zeta\right) = 1$, which is true if and only if, $\mathbb{E}\left[Z\right] = \mathbb{E}\left[Z^{(s)}\right], \forall Z$. One example is when Z is uniformly distributed.

Now, let us check the limit in the number of split samples. We end up with the following limiting cases

$$\lim_{S \to \infty} \left(||\mathcal{C}_m^{(s)}||\right) = \begin{cases} 0 & \text{if } 1 \leq m < M,\ 1 < s < S, \\ a_u - a_l & \text{if } m = M,\ 1 < s < S; \end{cases} \tag{67}$$

and for the first and last split sample

$$\begin{aligned} \lim_{S \to \infty} \left(||\mathcal{C}_m^{(1)}||\right) &= \begin{cases} 0 & \text{if } 1 \leq m < M, \\ a_u - a_l & \text{if } m = M, \end{cases} \\ \lim_{S \to \infty} \left(||\mathcal{C}_m^{(S)}||\right) &= \begin{cases} a_u - a_l & \text{if } m = M, \\ 0 & \text{if } 1 < m \leq M. \end{cases} \end{aligned} \tag{68}$$

This formulation takes a_l as the starting point and expresses the boundary points given a_l. However, we can use a_u as the starting point as well to shift the boundary point. This implies that the convergences on the bounds $(||\mathcal{C}_1^{(s)}||, ||\mathcal{C}_M^{(s)}||)$ will change, resulting in those parts not converging to 0 in general.

Now, it is clear that there are two types of observations: The first type is $Z_i^{(s)} \in \zeta$. These observations are the closest to the underlying unknown observations, as these have the feature of $\lim_{S\to\infty} ||\mathcal{C}_m^{(s)}|| = 0$. Moreover, these observations have the same interval width as the working sample's intervals and each of them can be directly linked to a certain working sample interval by design. Formally, $\exists\, \mathcal{C}_m^{(s)} \cong \mathcal{C}_b^{WS}$ such that $c_m^{(s)} = c_b^{WS}$, $c_{m-1}^{(s)} = c_{b-1}^{WS}$. We call these values "*directly transferable observations*", as we can directly transfer and use them in the working sample. These observations are denoted by $Z_i^{DTO} := Z_i^{(s)} \in \zeta,\ \forall s$, and the related random variable by Z^{DTO}.

The second type of observation is all others for which none of the above is true. We call them "*non–directly transferable observations*". Algorithm A7 describes how to construct the working sample when using only the directly transferable observations.

Algorithm A7 Magnifying method - creation of the "DTO" working sample

---

1: Set $m = 1, s = 1$ and $Z_i^{DTO} = \emptyset$.

2: If $\mathcal{C}_m^{(s)} \in \zeta$, add observations from interval $\mathcal{C}_m^{(s)}$ to the working sample:

$$Z_i^{DTO} := \left\{ Z_i^{DTO}, \bigcup_{j=1}^{N} \left(Z_j^{(s)} \in \mathcal{C}_m^{(s)} \mid \mathcal{C}_m^{(s)} \in \zeta \right) \right\} \tag{69}$$

3: If $s < S$, then $s := s + 1$ and go to Step 2.

4: If $s = S$, then $s := 1$ and set $m = m + 1$ and go to Step 2.

As a next step let us derive the probability that a *directly transferable observation* lies in a given interval of the working sample. Based on Equation (11) from the main text,

$$\Pr\left(Z \in \mathcal{C}_b^{WS}\right) = \Pr(Z \in \mathcal{S}_s) \int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z\,. \tag{70}$$

Note, here we can use the fact that individual i being assigned to a split sample s is independent of i choosing the interval with class value $Z_m^{(s)}$.

An important requirement of the magnifying method is that we want to ensure that in each interval in the working sample, there are directly transferable observations. For each split sample, the expected number of directly transferable observations is

$$\begin{aligned}\mathbb{E}(N_b^{WS}) &= \mathbb{E}\left(\sum_{i=1}^{N} \mathbf{1}_{\{Z_i \in \mathcal{C}_b^{WS}\}}\right) \\ &= N \Pr(Z \in \mathcal{S}_s) \int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z.\end{aligned} \tag{71}$$

Following from Equation (71), consider the following assumptions:

Assumption M1. *Let Z be a continuous random variable with probability density function $f_Z(z)$ with S, N and $\mathcal{C}_m^{(s)}$ follow the definitions above. We require that all split samples will have non-zero respondents,* $\Pr(Z \in \mathcal{S}_s) > 0$.

Assumption M1 ensures utilization of all split samples, i.e., each split sample has non-zero respondents. Similarly, for the shifting case, we need assumption 1.a), which ensures that the number of respondents is always higher than the number of split samples, and assumption 1.b), that imposes a mild assumption on the underlying distribution. (The support of the random variable is not disjoint, thus $\int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z > 0$.) These assumptions allow us to establish proposition 1, which establishes convergence in distribution.

Proposition 1. *Under Assumptions 1a, 1b from the main text and M1,*

1. $$\mathbb{E}(N_b^{WS}) > 0. \tag{72}$$

2. $$\Pr\left(\sum_{i=1}^{b} N_b^{WS} > 0\right) \to 1. \tag{73}$$

3. $$\Pr\left(Z_{DTO}^{WS} < a\right) = \Pr\left(Z < a\right) \text{ for any } a \in [a_l, a_u]. \tag{74}$$

Proof:
The probabilities of the unobserved variable to fall into class $\mathcal{C}_b^{WS}$,

$$\Pr\left(Z \in \mathcal{C}_b^{WS}\right) = \Pr(Z \in \mathcal{S}_s) \int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z\,, \tag{75}$$

where, $\mathcal{S}_s$ is the set for split sample s, and we used the fact that individual i being assigned to a split sample s is independent of i. This is satisfied if the discretization schemes are randomly assigned to observations. To ensure that in each interval from the working sample, there are directly transferable observations, let us write

$$\begin{aligned}\mathbb{E}(N_b^{WS}) &= \mathbb{E}\left(\sum_{i=1}^{N}\mathbf{1}_{\{Z_i\in\mathcal{C}_b^{WS}\}}\right)\\ &= N\Pr(Z\in\mathcal{S}_s)\int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z.\end{aligned} \tag{76}$$

We can reformulate Equation (76) by considering the number of observations up to a certain boundary point, rather than the number of observations in a particular class. That is

$$\Pr\left(\mathbb{E}\left[\sum_{i=1}^{b} N_i^{WS}\right] > 0\right) \to 1\,. \tag{77}$$

This gives the possibility to replace $\int_{c_{b-1}^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z$ with $\int_{c_0^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z$. Since this is a CDF, and hence a non-decreasing function, it effectively shows that each interval has non-empty observations:

$$\begin{aligned}\mathbb{E}\left(\sum_{i=1}^{b} N_i^{WS}\right) =&\mathbb{E}\left(\sum_{i=1}^{N}\mathbf{1}_{\{Z_i<c_b^{WS}\}}\right)\\ =&N\Pr(Z\in\mathcal{S}_s)\int_{c_0^{WS}}^{c_b^{WS}} f_Z(z)\mathrm{d}z.\end{aligned} \tag{78}$$

Next, we need to show that this is an increasing function in $\mathcal{C}_b^{WS}$. As $N\to\infty$, under the assumption that $\Pr(Z\in\mathcal{S}_s)=1/S$ and $S/N\to d$ with $d\in(0,1)$ – which is satisfied when $S=dN$,

$$\begin{aligned}\lim_{n\to\infty}\mathbb{E}\left(\sum_{i=1}^{b} N_i^{WS}\right) =&N\Pr(Z_i<\mathcal{C}_b^{WS})\\ =&\frac{1}{d}\int_{\mathcal{C}_0^{WS}}^{\mathcal{C}_b^{WS}} f_Z(z)dz.\end{aligned} \tag{79}$$

Note that the derivative with respect to $\mathcal{C}_b^{WS}$ is $\frac{1}{d} f_Z\left(\mathcal{C}_b^{WS}\right) > 0$, so the expected number of observations in each class is not 0. This completes our proof in the univariate case. We leave the proof for multivariate cases to future research.

A5.1. Derivation for Estimation – using NDTOs

Let us consider the placement of the *non-directly transferable observations*. We have seen that these observations belong to intervals, where the interval widths do not converge to zero. One way to proceed is to remove them completely so that they do not appear in the working sample. In practice, it seems that too many could fall into this category, resulting in a large efficiency loss. Another approach is to use the information available for these observations namely, the known boundary points for these values. Then we could use all the *directly transferable observations* from the working sample to calculate specific conditional averages for all *non-directly transferable observations* and replace them with those values. Let us denote a new variable $Z_{i,ALL}^{WS}$ that represents all the directly transferable observations and the replaced values for non-directly transferable observations.

For simplicity let us consider the case, when discretization happens with the explanatory variable. In this case, we simply need to calculate the conditional expectation that the underlying variable falls into the interval where the NDTOs are. The other two cases follow the same principle but use different conditioning.

Let us formalize the non-directly transferable observations as $Z_i^{(s)} \in \mathcal{C}_\chi$, where

$$\mathcal{C}_\chi := \bigcup_{s,m} \mathcal{C}_m^{(s)} \bigcap_b \mathcal{C}_b^{WS} = \zeta^\complement$$

is the set for non-directly transferable observations from all split samples, with $\chi = 1, \ldots, 2(S-1)$. We can then replace $Z_i^{(s)} \in \mathcal{C}_\chi$ with $\hat{\nu}_\chi$, which denotes the sample conditional averages

$$\hat{\nu}_\chi = \left(\sum_{i=1}^{N} \mathbf{1}_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}} \right)^{-1} \sum_{i=1}^{N} \mathbf{1}_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}} Z_i^{DTO}. \tag{80}$$

Let us introduce Z_i^{NDTO} as the variable which contains all the replaced values with $\hat{\nu}_\chi$, $\forall Z_i^{(s)} \in \mathcal{C}_\chi$. This way we can create a new working sample as

$Z_i^{ALL} := \{Z_i^{DTO}, Z_i^{NDTO}\}$, which contains information from both types of observations.

Under the WLLN and the same assumptions needed for the magnifying method, it is straightforward to show, $\hat{\nu}_\chi \to \mathbb{E}(Z|Z \in \mathcal{C}_\chi)$, as $N, S \to \infty$. Algorithm A8 shows how to replace NDTO values with the appropriate conditional expectation estimators.

Algorithm A8 The magnifying method - creation of "ALL" working sample

1: Let, $Z_i^{ALL} := \{Z_i^{DTO}\}$.
2: Set, $m = 1, s = 1$.
3: If $\mathcal{C}_m^{(s)} \in \mathcal{C}_\chi$, then calculate $\hat{\nu}_\chi$ and expand the working sample as,

$$Z_i^{ALL} := \left\{ Z_i^{ALL}, \bigcup_{j=1}^{N} \hat{\nu}_\chi \mid \left(Z_j^{(s)} \in \mathcal{C}_m^{(s)} \mid \mathcal{C}_m^{(s)} \in \mathcal{C}_\chi \right) \right\}. \tag{81}$$

4: If $s < S$, then $s := s + 1$ and go to Step 3.
5: If $s = S$, then $s := 1$ and set $m = m + 1$ and go to Step 3.

To obtain the standard errors, one can think of $\hat{\nu}_\chi$ as an LS estimator, regressing $\mathbf{1}_{\{Z_{i,DTO}^{WS} \in \mathcal{C}_\chi\}}$ on Z_i^{DTO}. Here $\mathbf{1}_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}}$ is a vector of indicator variables, created by $2(S-1)$ indicator functions: It takes the value of one for the directly transferable observations, which are within $\mathcal{C}_\chi$.[7] We can now write the following:

$$Z_i^{DTO} = \boldsymbol{\nu}_\chi \mathbf{1}_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}} + \eta_i, \tag{82}$$

where $\boldsymbol{\nu}_\chi$ stands for the vector of $\nu_\chi, \forall \chi$. The LS estimator of $\boldsymbol{\nu}_\chi$ is

$$\hat{\boldsymbol{\nu}}_\chi = \left(\mathbf{1}'_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}} \mathbf{1}_{\{Z_i^{DTO} \in \mathcal{C}_\chi\}} \right)^{-1} \mathbf{1}'_{\{z_i^{DTO} \in \mathcal{C}_\chi\}} Z_i^{DTO}, \tag{83}$$

and under the standard LS assumptions, we can write

$$\sqrt{N^{DTO}} \left(\hat{\boldsymbol{\nu}}_\chi - \boldsymbol{\nu}_\chi \right) \overset{\text{a}}{\sim} \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Omega}_\chi \right), \tag{84}$$

where $\boldsymbol{\nu}_\chi = \mathbb{E}(Z|Z \in \mathcal{C}_\chi), \forall \chi$. The variance of the OLS estimator is

$$\boldsymbol{\Omega}_\chi = \mathrm{V}\left(\eta_i\right) \left(\mathbf{1}'_{\{x_{i,DTO}^{WS} \in \mathcal{C}_\chi\}} \mathbf{1}_{\{x_{i,DTO}^{WS} \in \mathcal{C}_\chi\}} \right)^{-1}. \tag{85}$$

[7] The indicator variables are not independent of each other, while the non-transferable observation intervals ($\mathcal{C}_\chi$) overlap each other.

Using this result, we may decide whether to replace or not the NDTOs.

A6. Estimation with Discretized Regressors[8]

In this section, first, we analyze the bias and consistency of the OLS estimator of β in the univariate case, when the explanatory variable is discretized. We investigate properties when $N \to \infty$ and when $M \to \infty$. The last implies we observe each value directly resulting in a consistent OLS estimator as outlined in Section 4 of the main text. These exercises are helpful to see how these results generalize in the multiple linear regression case discussed in Section A6.4. Last here, we investigate the bias in the panel set up in Section A6.6.

Recall the data-generating process is assumed to be

$$Y_i = X_i'\beta + u_i \,, \tag{86}$$

with the linear regression model using the discretized version of X_i namely,

$$Y_i = X_i^{*'}\beta^* + u_i \,. \tag{87}$$

It is also assumed that there is a known support $[a_l, a_u]$ for X_i with known boundaries ($\mathcal{C}_m$), and let v_m from Equation (1) in the main text be any value X_i^* take, typically the mid-point.

Let N_m be the number of observations in each class $\mathcal{C}_m$, that is $N_m = \sum_{i=1}^N \mathbf{1}_{\{X_i \in \mathcal{C}_m\}}$, where $\mathbf{1}_{\{X \in \mathcal{C}_m\}}$ denotes the indicator function. When X has a cumulative distribution (cdf) $F_X(\cdot)$,

$$\begin{aligned} \mathbb{E}(N_m) &= \mathbb{E}\left(\sum_{i=1}^N \mathbf{1}_{\{X_i \in \mathcal{C}_m\}}\right) \\ &= N \int_{\mathcal{C}_m} f_X(x)\,\mathrm{d}x \\ &= N \Pr(c_{m-1} < X = x \leq c_m), \end{aligned} \tag{88}$$

using the independence assumption. Note, when X has a uniform distribution, we have $\mathbb{E}(N_m) = N/M$ for all $m = 1, \ldots, M$.

[8] We acknowledge the work of Balázs Kertész from this section on the expected value of $\hat{\boldsymbol{\beta}}^*_{\boldsymbol{OLS}}$ and on N and M (in-)consistency results.

The OLS estimator can expanded as,

$$
\begin{aligned}
\hat{\beta}^*_{OLS} &= \left(X^{*'}X^*\right)^{-1}(X^{*'}Y) \\
&= \frac{v_1\left(\sum_{i=1}^{N_1} Y_i\right) + v_2\left(\sum_{i=N_1+1}^{N_1+N_2} Y_i\right) + \cdots + v_M\left(\sum_{i=N-N_M+1}^{N_M} Y_i\right)}{N_1 v_1^2 + N_2 v_2^2 + \cdots + N_M v_M^2} \\
&= \frac{v_1\left(\sum_{i=1}^{N_1} \beta X_i + u_i\right) + \cdots + v_M\left(\sum_{i=N-N_M+1}^{N_M} \beta X_i + u_i\right)}{N_1 v_1^2 + \cdots + N_M v_M^2} \\
&= \frac{v_1\left[\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_1\}}(\beta X_i + u_i)\right] + \cdots + v_M\left[\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_M\}}(\beta X_i + u_i)\right]}{N_1 v_1^2 + \cdots + N_M v_1^2} \\
&= \frac{\sum_{m=1}^{M} v_m\left[\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}}(\beta X_i + u_i)\right]}{\sum_{m=1}^{M} N_m v_m^2}.
\end{aligned}
$$

Using the expression above, we can get the following general formula for the expected value of the OLS estimator:

$$
\begin{aligned}
\mathbb{E}\left(\hat{\beta}^*_{OLS}\right) &= \mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m\left[\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}}\left(\beta(X_i^* + \xi_i) + u_i\right)\right]}{\sum_{m=1}^{M} N_m v_m^2}\right\} \\
&= \mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m\left[\beta\left(\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} X_i^* + \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} \xi_i\right) + \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} u_i\right]}{\sum_{m=1}^{M} N_m v_m^2}\right\} \\
&= \beta\mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} X_i^*}{\sum_{m=1}^{M} N_m v_m^2}\right\} + \beta\mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} \xi_i}{\sum_{m=1}^{M} N_m v_m^2}\right\} \\
&\qquad + \mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} u_i}{\sum_{m=1}^{M} N_m v_m^2}\right\} \\
&= \beta + \beta\mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} \xi_i}{\sum_{m=1}^{M} N_m v_m^2}\right\} \\
&= \beta + \beta\mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m N_m v^m}{\sum_{m=1}^{M} N_m v_m^2}\right\}, \qquad (89)
\end{aligned}
$$

where the discretization error $\xi_i = X_i - X_i^*$ for each observation by setting the possible answer values at X_i^*. The derivation above is based on the

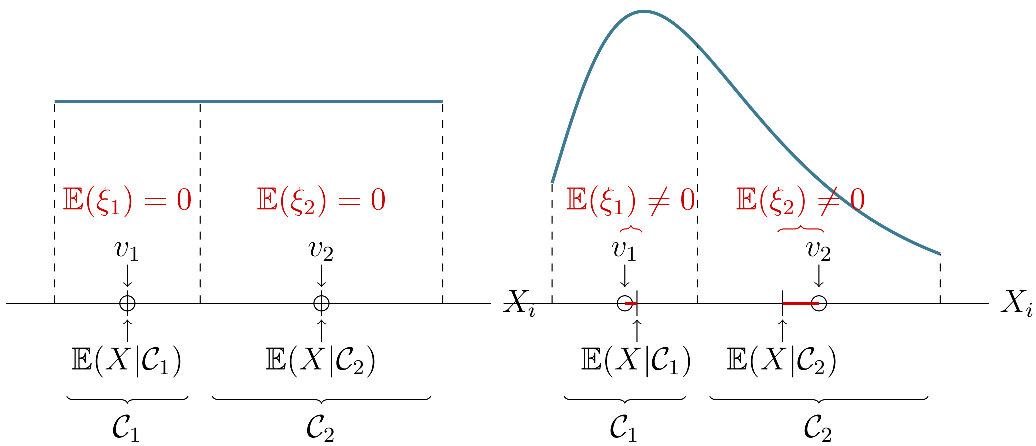

Figure A3: The difference between uniform (left panel) and general distributions (right panel)

disturbance term u_i being independent of regressor X_i and $\mathbb{E}(u_i) = 0$ for all $i = 1, \ldots, N$. The last inference uses the fact that the errors ξ_i have the same conditional distribution over the class $\mathcal{C}_m$, $\upsilon^m \stackrel{\text{d}}{=} \xi_i|\mathcal{C}_m$ for all $m = 1, \ldots, M$ and $i = 1, \ldots, N$. Importantly, the second term in Equation (89) does not vanish in general, since $\upsilon^m|\mathcal{C}_m$ is not independent of $N_m|\mathcal{C}_m$, $\upsilon^m|\mathcal{C}_m \not\perp\!\!\!\perp N_m|\mathcal{C}_m$ nor $\mathbb{E}(\xi_i|\mathcal{C}_m) = \mathbb{E}(\upsilon^m) = 0$ (see Figure A3, right panel for illustrative explanation). The former issue can be eliminated by conditioning on the underlying distribution of X_i. Conditional on the distribution X_i and the class $\mathcal{C}_m$, the number of observations in the class and assuming that the errors are independent of each other, $N_m|X_i, \mathcal{C}_m \perp\!\!\!\perp \upsilon^m|X_i, \mathcal{C}_m$, but knowing the underlying distribution makes the problem trivial. Nonetheless, because of both issues, the "naive" OLS estimator is biased.

Note that the uniform distribution, however, turns out to be a special case. Let us assume that $X_i \sim U(a_l, a_u)$ for all $i = 1, \ldots, N$, then both of the above disappear (see the left panel in Figure A3) if we are using the class midpoints. The first problem is resolved, because, in the case of the uniform distribution, both the number of observations N_m in each class $\mathcal{C}_m$ and the error term υ^m are independent of the regressor's X_i distribution, while the second problem does not appear trivially, since now the class midpoints are proper estimates of the regressor's X_i expected value in the class $\mathcal{C}_m$. From

Equation (89), we obtain that

$$\mathbb{E}\left(\hat{\beta}^{*}_{OLS}\right) = \beta + \beta\mathbb{E}\left\{\frac{\sum_{m=1}^{M} v_m N_m \upsilon^m}{\sum_{m=1}^{M} N_m v_m^2}\right\} = \beta, \tag{90}$$

where υ^m is a uniformly distributed random variable with zero expected value, $\mathbb{E}(\upsilon^m) = 0$ for all $m = 1, \dots, M$. Hence, in the case of uniform distribution, unlike for other distributions, the OLS is unbiased.

A6.1. N (in)consistency

Next, the large sample properties of the estimator are conisdered. First, assume that $\operatorname{plim}_{N\to\infty} \sum_{i=1}^{N} (\mathbf{1}_{\{X_i \in \mathcal{C}_m\}} u_i) = 0$, in other words, that the class set selection is independent of the disturbance terms, and also that with sample size N the number of classes M is fixed. Then

$$\begin{aligned}
\operatorname*{plim}_{N\to\infty} \hat{\beta}^{*}_{OLS} &= \operatorname*{plim}_{N\to\infty} \frac{\sum_{m=1}^{M} v_m \left[\sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} (\beta X_i + u_i)\right]}{\sum_{m=1}^{M} N_m v_m^2} \\
&= \frac{\sum_{m=1}^{M} v_m \left[\operatorname{plim}_{N\to\infty} \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} (\beta X_i + u_i)\right]}{\sum_{m=1}^{M} v_m^2 \operatorname{plim}_{N\to\infty} N_m} \\
&= \frac{\sum_{m=1}^{M} v_m \left[\operatorname{plim}_{N\to\infty} \beta \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} X_i\right]}{\sum_{m=1}^{M} v_m^2 \operatorname{plim}_{N\to\infty} N_m} \\
&= \beta \; \frac{\sum_{m=1}^{M} v_m \left[\operatorname{plim}_{N\to\infty} \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} X_i\right]}{\sum_{m=1}^{M} v_m^2 \operatorname{plim}_{N\to\infty} N_m}.
\end{aligned} \tag{91}$$

Define $X^m = \sum_{i=1}^{N} \mathbf{1}_{\{X_i \in \mathcal{C}_m\}} X_i$, then X^m sums the truncated version of the original random variables X_i on the class $\mathcal{C}_m$, $X_m \overset{\mathrm{d}}{=} X_i | \mathcal{C}_m$, for all $m = 1, \dots, M$, therefore its asymptotic distribution can be calculated by applying the Lindeberg-Levy Central Limit Theorem,

$$X^m / N_m \overset{\mathrm{a}}{\sim} N\big(\mathbb{E}(X_m), \mathrm{V}(X_m)/N_m\big). \tag{92}$$

The $\hat{\beta}^{*}_{OLS}$ estimator is consistent if and only if the probability limit in Equation (91) equals β. To give a condition for consistency, first, we rewrite the previous Equation (91) in terms of the error terms ξ_i,

$$\operatorname*{plim}_{N\to\infty}\left(\hat{\beta}^*_{OLS}-\beta\right)=\frac{\beta\left(\sum_{m=1}^{M}v_m\left[\operatorname{plim}_{N\to\infty}\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}X_i\right]-\sum_{m=1}^{M}v_m^2\operatorname{plim}_{N\to\infty}N_m\right)}{\sum_{m=1}^{M}v_m^2\operatorname{plim}_{N\to\infty}N_m}$$
$$=\frac{\beta\sum_{m=1}^{M}v_m\left[\operatorname{plim}_{N\to\infty}\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}(X_i-X_i^*)\right]}{\sum_{m=1}^{M}v_m^2\operatorname{plim}_{N\to\infty}N_m}$$
$$=\frac{\beta\sum_{m=1}^{M}v_m\left[\operatorname{plim}_{N\to\infty}\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}\xi_i\right]}{\sum_{m=1}^{M}v_m^2\operatorname{plim}_{N\to\infty}N_m},\tag{93}$$

where the asymptotic distribution of the sum of errors in class $\mathcal{C}_m$, $\xi^m=\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}\xi_i$, $m=1,\dots,M$, can be given by

$$\xi^m/N_m\overset{\mathrm{d}}{=}X^m/N_m-v_m\overset{\mathrm{a}}{\sim}N\big(\mathbb{E}(X^m)-v_m,\mathrm{V}(X^m)/N_m\big).\tag{94}$$

Then

$$\operatorname*{plim}_{N\to\infty}\left(\hat{\beta}^*_{OLS}-\beta\right)=\frac{\operatorname{plim}_{N\to\infty}\beta\sum_{m=1}^{M}v_m\xi^m}{\operatorname{plim}_{N\to\infty}\sum_{m=1}^{M}v_m^2N_m}$$
$$=\frac{\operatorname{plim}_{N\to\infty}O(N)\beta\sum_{m=1}^{M}v_m\xi^m/N_m}{\operatorname{plim}_{N\to\infty}O(N)\sum_{m=1}^{M}v_m^2}$$
$$=\frac{\beta\sum_{m=1}^{M}v_m\operatorname{plim}_{N\to\infty}\xi^m/N_m}{\sum_{m=1}^{M}v_m^2}O(N)$$
$$=\frac{\beta\sum_{m=1}^{M}v_m\left\{\mathbb{E}(z_m)-v_m\right\}}{\sum_{m=1}^{M}v_m^2}O(N).\tag{95}$$

The last step in the above derivation can simply be obtained from the definition of the plim operator, i.e., for any $\varepsilon>0$ given. Therefore, to obtain the (in)consistency of the OLS estimator $\hat{\beta}^*_{OLS}$ in the number of observations N, we only need to calculate the expected value of the truncated random variable X_m, $m=1,\dots,M$ and check whether the expression (95) equals 0 to satisfy a sufficient condition. So

$$\operatorname*{plim}_{N\to\infty}\xi^m=\mathbb{E}(X_m)-X_m\tag{96}$$
$$\iff\lim_{N\to\infty}\Pr\left(\left|\xi^m-\{\mathbb{E}(X_m)-X_m\}\right|>\varepsilon\right)\tag{97}$$
$$=\lim_{N\to\infty}F_{\xi^m}\left(-\varepsilon+\mathbb{E}(X_m)-X_m\right)\left[1-F_{\xi^m}\left(\varepsilon+\mathbb{E}(X_m)-X_m\right)\right]=0.\tag{98}$$

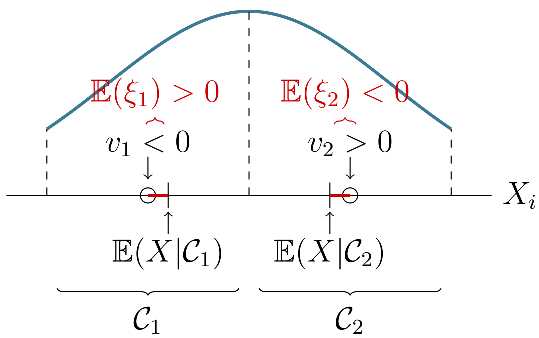

Figure A4: The estimator is inconsistent even in case of symmetric distributions (see Equation (95)).

The convergence holds because, for any given $\delta > 0$, there is a threshold N_0 for which the term in the limit becomes less than δ. This can be seen from $F_{\xi^m}(\cdot)$ being close to a degenerate distribution above a threshold number of observations N_0, or intuitively since the variance of the sequence of random variables ξ^m collapses in N, its probability limit equals its expected value.

Let us apply these results to the uniform distribution. In this case, there is no consistency issue because the class midpoints coincide with the expected value of the truncated uniform random variable in each class, making the expression (95) zero, hence the OLS estimator is consistent.

Note that the consistency of the OLS estimator is not guaranteed even in the case of symmetric distributions and symmetric class boundaries. After appropriate transformations (e.g., demeaning), it can be seen that the sign of the differences between the expectation of the truncated random variables X_m and the class midpoints is opposite to the sign of the class midpoints on either side of the distribution, which implies negative overall asymptotic bias in N (see Figure A4).

In the case of a (truncated) normal variable, for example, we need to substitute the expected value of the truncated normal random variable X_m for each $m = 1, \ldots, M$ in the consistency formula (95). As a result, the difference between the expectation and the class midpoints, in general, is not zero for all m, hence the formula cannot be made arbitrarily small. Therefore, the OLS estimator becomes inconsistent in N.

So far we have focused on the estimation of β in Equation (87). But how about γ? It can be shown that the bias and inconsistency presented above are contagious. Estimation of all parameters of a model is going to be biased and inconsistent unless the measurement error and X are orthogonal (independent), which is quite unlikely in practice. This is important to emphasize: a single interval-type variable in a model is going to infect the estimation of all variables of the model.

A6.2. M Consistency

Let us see next the case when N is fixed but $M \to \infty$. Now, we may have some intervals that do not contain any observations, while others still do. Omitting, however, empty intervals does not cause any bias because of our i.i.d. assumption. Furthermore, while we increase the number of intervals, the size of the intervals itself is likely to shrink and become so narrow that only one observation can fall into each. In the limit, we are going to hit the observations with the interval boundaries. To see that, we derive the consistency formula in the number of intervals M assuming that $\operatorname{plim}_{M\to\infty} \sum_{\{m:\mathcal{C}_m \neq \emptyset, m=1,\ldots,M\}} v_m u_{i_m} = 0$, or with re-indexation $\operatorname{plim}_{M\to\infty} \sum_{i=1}^{N} v_{m_i} u_i = \sum_{i=1}^{N} z_i u_i = 0$, which should hold in the sample and is a stronger assumption than the usual $\operatorname{plim}_{N\to\infty}\sum_{i=1}^{N} X_i u_i = 0$:

$$
\begin{aligned}
\operatorname*{plim}_{M\to\infty}\left(\hat{\beta}^{*}_{OLS}-\beta\right) &= \operatorname*{plim}_{M\to\infty}\frac{\sum_{m=1}^{M} v_m\left[\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}(\beta X_i+u_i)\right]}{\sum_{m=1}^{M} N_m v_m^2}-\beta \\
&= \operatorname*{plim}_{M\to\infty}\frac{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} v_m\left[\sum_{i=1}^{N}\mathbf{1}_{\{X_i\in\mathcal{C}_m\}}(\beta X_i+u_i)\right]}{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} N_m v_m^2}-\beta \\
&= \operatorname*{plim}_{M\to\infty}\frac{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} v_m(\beta X_{i_m}+u_{i_m})}{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} v_m^2}-\beta \\
&= \operatorname*{plim}_{M\to\infty}\beta\left\{\frac{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} v_m X_{i_m}}{\sum_{\{m:\mathcal{C}_m\neq\emptyset,m=1,\ldots,M\}} v_m^2}-1\right\} \\
&= \operatorname*{plim}_{M\to\infty}\beta\left\{\frac{\sum_{i=1}^{N} v_{m_i}X_i}{\sum_{i=1}^{N} v_{m_i}^2}-1\right\} \\
&= \beta\left\{\frac{\sum_{i=1}^{N}\operatorname{plim}_{M\to\infty} v_{m_i}X_i}{\sum_{i=1}^{N}\operatorname{plim}_{M\to\infty} v_{m_i}^2}-1\right\} \\
&= \beta\left\{\frac{\sum_{i=1}^{N} X_iX_i}{\sum_{i=1}^{N} {X_i}^2}-1\right\} \\
&= 0,
\end{aligned}
\tag{99}
$$

where the index $i_m \in \{1,\ldots,N\}$ denotes observation i in class m (at the beginning there might be several observations that belong to the same class m), and index $m_i \in \{1,\ldots,M\}$ denotes the class m that contains observation i (at the and of the derivation one class m includes only one observation i). Note that the derivation does not depend on the distribution of the explanatory variable X, so consistency in the number of classes M holds in general. Let us also note, however, that this convergence in M is slow. Also, as $M\to\infty$, the class sizes go to zero, and the smaller the class sizes the smaller the bias.

A6.3. Some Further Remarks

The above results hold for much simpler cases as well. If instead of model (87) we just take the simple sample average of X, $\bar{X}=\sum_i X_i/N$, then $\bar{X}^*=\sum_i X_i^*/N$ is going to be a biased and inconsistent estimator of $\bar{X}$.

The measurement error due to discretized variables, however, not only induces a correlation between the error terms and the observed variables, but it also induces a non-zero expected value for the disturbance terms of the regression in (87). Consider a simple example where there is an unobserved variable X_i with an observed discretized version:

$$X_i^* = \begin{cases} v_1 & \text{if } c_0 \leq X_i < c_1, \\ v_2 & \text{if } c_1 \leq X_i < c_2, \end{cases} \tag{100}$$

and

$$Y_i = X_i\beta + \varepsilon_i. \tag{101}$$

Using the discretized variable means:

$$Y_i = X_i^*\beta + (X_i - X_i^*)\beta + u_i \tag{102}$$

and

$$\begin{aligned} \mathbb{E}\left[X_i - X_i^*\right] =& \mathbb{E}(X_i) - \mathbb{E}(X_i^*) \\ =& \mathbb{E}(X_i) - \mathbb{E}\left[X_1 \mathbf{1}(c_0 \leq X_i < c_1) + X_2 \mathbf{1}(c_1 \leq X_i < c_2)\right] \\ =& \mathbb{E}(X_i) - v_1 \Pr(c_0 \leq X_i < c_1) - X_2 \Pr(c_1 \leq X_i < c_2). \end{aligned} \tag{103}$$

The last line above is not zero in general. Thus, it would induce a bias in the estimator if the regression did not include an intercept. This result generalizes naturally to variables with multiple class values.

A6.4. Estimation in a Multivariate Linear Regression

Let us generalise the problem and re-write it in matrix form. Consider the following linear regression model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}\,, \tag{104}$$

where $\mathbf{X}$ and $\mathbf{W}$ are $N \times K$ and $N \times J$ data matrices of the explanatory variables, $\mathbf{y}$ is a $N \times 1$ vector containing the data of the dependent variable, $\boldsymbol{\varepsilon}$ is a $N \times 1$ vector of disturbance terms, and finally $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are $K \times 1$ and $J \times 1$ parameter vectors.

$\mathbf{X}$ is not observed, only its discretized version $\mathbf{X}^*$ is. Define the $MK \times K$ matrix as

$$\mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & \mathbf{0} & \dots & \dots \\ \mathbf{0} & \mathbf{V}_2 & \mathbf{0} & \mathbf{0} \\ \vdots & \dots & \ddots & \vdots \\ \dots & \dots & \mathbf{0} & \mathbf{V}_K \end{bmatrix}, \tag{105}$$

where $\mathbf{V}_i = (v_{i1}, \ldots, v_{iM})'$ contains the values for variable i. Let $\mathbf{E} = \{\mathbf{e}_{ki}\}$, where $k = 1, \ldots K$ and $i = 1, \ldots, N$ such that

$$\mathbf{e}_{ki} = \begin{bmatrix} \mathbf{1}(c_{k0} \leq x_{ki} < c_{k1}) \\ \mathbf{1}(c_{k1} \leq x_{ki} < c_{k2}) \\ \vdots \\ \mathbf{1}(c_{kM-1} \leq x_{ki} < c_{kM}) \end{bmatrix}, \tag{106}$$

where x_{ki} denotes the value of the i^{th} observation from the explanatory variable $\mathbf{x}_k$.

This implies $\mathbf{E}$ is a $MK \times N$ matrix since each entry $\mathbf{e}_{ki}$ is a $M \times 1$ vector. Following the definition of X_i^* in the paper, we can rewrite $\mathbf{X}^* = \mathbf{E}'\mathbf{V}$.

A6.5. The OLS Estimator

From Equation (104), consider the regression based on the observed data:

$$\mathbf{y} = \mathbf{X}^*\boldsymbol{\beta} + \mathbf{W}\boldsymbol{\gamma} + (\mathbf{X} - \mathbf{X}^*)\,\boldsymbol{\beta} + \boldsymbol{\varepsilon}\,, \tag{107}$$

then the OLS estimator for $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^{*\prime}\mathbf{M}_{\mathbf{X}}\mathbf{X}^*)^{-1}\,\mathbf{X}^{*\prime}\mathbf{M}_{\mathbf{X}}\mathbf{y}\,, \tag{108}$$

where $\mathbf{M}_{\mathbf{W}} = \mathbf{I} - \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$ defines the usual residual maker. The standard derivation shows that

$$\hat{\boldsymbol{\beta}} = (\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{E}'\mathbf{V})^{-1}\,\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{X}\boldsymbol{\beta} + (\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{E}'\mathbf{V})^{-1}\,\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\boldsymbol{\varepsilon}. \tag{109}$$

This implies OLS is unbiased if and only if $(\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{E}'\mathbf{V})^{-1}\,\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{X} = \mathbf{I}$. This allows us to investigate the bias analytically by examining the elements in $\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{E}'\mathbf{V}$ and $\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{X}$.

To simplify the analysis, we assume for the time being the following:

$$\mathbf{M}_{\mathbf{W}}\mathbf{X} = \mathbf{X} \tag{110}$$

$$\mathbf{M}_{\mathbf{W}}\mathbf{X}^* = \mathbf{X}^*. \tag{111}$$

In other words, we assume independence between $\mathbf{X}$ and $\mathbf{W}$, as well as its discretized version. This may appear to be a strong assumption but it does allow us to see what is happening somewhat better. We relax this at a later stage.

The OLS estimator in this case becomes:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{V}'\mathbf{E}\mathbf{E}'\mathbf{V}\right)^{-1}\mathbf{V}'\mathbf{E}\mathbf{X}\boldsymbol{\beta} + \left(\mathbf{V}'\mathbf{E}\mathbf{E}'\mathbf{V}\right)^{-1}\mathbf{V}'\mathbf{E}\boldsymbol{\varepsilon}. \tag{112}$$

The OLS is unbiased if $\left(\mathbf{V}'\mathbf{E}\mathbf{E}'\mathbf{V}\right)^{-1}\mathbf{V}'\mathbf{E}\mathbf{X} = \mathbf{I}$. Note that $\mathbf{V}'$ and $\mathbf{E}$ are of size $K \times MK$ and $MK \times N$, respectively. This means $\mathbf{V}'\mathbf{E}\mathbf{E}'\mathbf{V}$ are invertible as long as $N > K$, which is a standard assumption in classical regression analysis. Let us consider a typical element in $\mathbf{V}'\mathbf{E}\mathbf{E}'\mathbf{V}$ first. Since $\mathbf{V}$ is non-stochastic as it contains only all the pre-defined interval values, it is sufficient to examine $\mathbf{E}\mathbf{E}'$:

$$\mathbf{E}\mathbf{E}' = \begin{bmatrix} \mathbf{e}_{11} & \dots & \mathbf{e}_{1i} & \dots & \mathbf{e}_{1N} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \mathbf{e}_{k1} & \dots & \mathbf{e}_{ki} & \dots & \mathbf{e}_{kN} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \mathbf{e}_{K1} & \dots & \mathbf{e}_{Ki} & \dots & \mathbf{e}_{KN} \end{bmatrix} \begin{bmatrix} \mathbf{e}'_{11} & \dots & \mathbf{e}'_{k1} & \dots & \mathbf{e}'_{K1} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \mathbf{e}'_{1i} & \dots & \mathbf{e}'_{ki} & \dots & \mathbf{e}'_{ki} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \mathbf{e}'_{1N} & \dots & \mathbf{e}'_{kN} & \dots & \mathbf{e}'_{KN} \end{bmatrix}. \tag{113}$$

Note that each entry in $\mathbf{E}$ is a vector, so $\mathbf{E}\mathbf{E}'$ results in a partition matrix whose elements are the sums of the outer products of $\mathbf{e}_{ki}$ and $\mathbf{e}_{lj}$ for $k, l = 1, \dots, K$ and $i, j = 1, \dots, N$. Specifically, let $\mathbf{q}_{kl}$ be a typical block element in $\mathbf{E}\mathbf{E}'$, then

$$\mathbf{q}_{kl} = \sum_{i=1}^{N} \mathbf{e}_{ki}\mathbf{e}'_{li}. \tag{114}$$

Let $\mathbf{1}_m^{ki} = \mathbf{1}\left(c_{km-1} \le z_{ki} < c_{km}\right)$, then the (m, n) element in $\mathbf{q}_{kl}$, q_{mn} is $\sum\limits_{i=1}^{N} \mathbf{1}_m^{ki}\mathbf{1}_n^{li}$ for $m, n = 1, \dots, M$. Thus, $\mathbb{E}\left(\mathbf{E}\mathbf{E}'\right)$ exists if $\mathbb{E}\left(\mathbf{1}_m^{ki}\mathbf{1}_n^{li}\right)$ exists,

$$\mathbb{E}\left(\mathbf{1}_m^{ki}\mathbf{1}_n^{li}\right) = \int\limits_{\Omega} f(x_k, x_l)dx_k dx_l\,, \tag{115}$$

where $f(x_k, x_l)$ denotes the joint distribution of x_k and x_l and $\Omega = [c_{km-1}, c_{km}] \times [c_{ln-1}, c_{ln}]$ defines the region for integration. Thus, $N^{-1}b_{mn}$ should converge into Equation (115) under the usual WLLN.

Following a similar method, let a_{kl} be the (k, l) element in $\mathbf{V}'\mathbf{E}\mathbf{X}$, then

$$a_{kl} = \sum_{i=1}^{N}\sum_{m=1}^{M} v_{km}\mathbf{1}_m^{ki}x_{li}. \tag{116}$$

Now,

$$
\begin{aligned}
\mathbb{E}\left[\sum_{m=1}^{M} v_{km}\mathbf{1}_{m}^{ki}x_{li}\right] &= \sum_{m=1}^{M} v_{km}\mathbb{E}\left[\mathbf{1}_{m}^{ki}x_{li}\right] \\
&= \sum_{m=1}^{M} v_{km}\int_{\Omega_1} x_l f(x_k, x_l)dx_k dx_l\,,
\end{aligned}
\tag{117}
$$

where $\Omega_1 = [c_{km-1}, c_{km}] \times \Omega_{\mathbf{X}}$ with $\Omega_{\mathbf{X}}$ denotes the sample space of x_k and x_l. Thus, $N^{-1}a_{kl}$ converge into Equation (117) under the usual WLLN.

In the case when Equations (110) and (111) do not hold, the analysis becomes more tedious algebraically, but it does not affect the result that OLS is biased. Recall Equation (109), and let ω_{ij} be the (i, j) element in $\mathbf{M_W}$ for $i = 1, \ldots, N$ and $j = 1, \ldots, J$, then following the same argument as above, $\mathbf{EM_W E'}$ can be expressed as a $M \times M$ block partition matrix with each entry a $K \times K$ matrix. The typical (m, n) element in the (k, l) block is

$$
g_{kl} = \sum_{j=1}^{N}\sum_{i=1}^{N}\omega_{ij}\mathbf{1}_{m}^{ki}\mathbf{1}_{n}^{li} \tag{118}
$$

with its expected value being

$$
\sum_{i=1}^{N}\sum_{j=1}^{N}\int_{\Omega}\omega_{ij} f\left(x_k, x_l, \mathbf{W}\right)dw_k dw_k d\mathbf{W}, \tag{119}
$$

where

$$
\mathbf{W} = \left(w_1, \ldots, w_J\right),\; d\mathbf{W} = \prod_{i=1}^{J} dw_i \text{ and } \Omega = [c_{km-1}, c_{km}] \times [c_{ln-1}, c_{ln}] \times \Omega_{\mathbf{W}}, \tag{120}
$$

where $\Omega_{\mathbf{W}}$ denotes the sample space of $\mathbf{W}$. Note that ω_{ij} is a nonlinear function of $\mathbf{W}$, and so the condition of existence for Equation (119) is complicated. However, under the assumption that the integral in Equation (119) exits, then $N^{-1}g_{kl}$ should converge to Equation (119) under the usual WLLN. It is also worth noting that $\mathbb{E}\left[\mathbf{M_W X}\right] = \mathbb{E}\left[\mathbf{M_W}\right]\mathbb{E}\left[\mathbf{X}\right] = \mathbb{E}\left[\mathbf{X}\right]$ and $\mathbb{E}\left[\mathbf{M_W X}^*\right] = \mathbb{E}\left[\mathbf{M_W}\right]\mathbb{E}\left[\mathbf{X}^*\right] = \mathbb{E}\left[\mathbf{X}^*\right]$ under the assumption of independence, which reduces Equation (119) to Equation (115).

Again, following the same derivation as above, a typical element in $\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{W}}\mathbf{X}$ is

$$h_{kl} = \sum_{m=1}^{M} \sum_{i=1}^{N} x_{km} \mathbf{1}_m^{ki} u_{li}, \tag{121}$$

where $u_{li} = \sum_{v=1}^{N} \omega_{iv} X_{lv}$. Note that u_{li} is the i^{th} residual of the regression of X_l on $\mathbf{W}$. The expected value of h_{kl} can be expressed as

$$\sum_{m=1}^{M} v_{km} \int_{\Omega_m} u_l f(x_k, x_l, \mathbf{W}) dx_k dx_l d\mathbf{W}, \tag{122}$$

where u_l denotes the random variable corresponding to the i^{th} column of $\mathbf{M}_{\mathbf{W}}\mathbf{X}$ and $\Omega_m = [c_{km-1}, c_{km}] \times \Omega_{\mathbf{X}} \times \Omega_{\mathbf{X}}$ with $\Omega_{\mathbf{X}}$ denotes the sample space of $\mathbf{W}$. Note that $u_l = w_l$ under the assumption of independence, which reduces Equation (122) to Equation (117).

A6.6. Extension to Panel Data

So far, we have dealt with cross-sectional data. Next, let us see what changes if we have panel data at hand. We can extend our DGP based on Equation (104), to

$$Y_{it} = X_{it}'\beta + \varepsilon_{it}, \tag{123}$$

where $X_{it} \sim f_{X_i}(a_l, a_u)$ denotes an individual distribution with mean μ_i for $i = 1, \ldots, N$. Here we need to assume that $f_{X_i}(\cdot)$ is stationary, so the distribution may change over individual i but not over time, t.

Now, the most important problem is identification. If the interval for an individual does not change over the time periods covered, the individual effects in the panel and the parameter associated with the class variable cannot be identified separately. The Within transformation would wipe out the interval variable as well. When the interval does change over time, but not much, then we are facing weak identification, i.e., in fact very little information is available for identification, so the parameter estimates are going to be highly unreliable. This is a likely scenario when M is small, for example, $M = 3$ or $M = 5$.

The bias of the panel data Within estimator can easily be shown. Let us re-write Equation (104) in a panel data context without further control variables $\mathbf{W}$. Then

$$\mathbf{y} = \mathbf{D}_N \boldsymbol{\alpha} + \mathbf{X}^* \boldsymbol{\beta} + \left[(\mathbf{X} - \mathbf{X}^*) \boldsymbol{\beta} + \boldsymbol{\varepsilon} \right], \tag{124}$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N)'$ and $\mathbf{D}_N$ is a $NT \times N$ zero-one matrix that appropriately selects the corresponding fixed effect elements to form $\boldsymbol{\alpha}$. The Within estimator is

$$\hat{\boldsymbol{\beta}}^*_X = (\mathbf{X}^{*\prime} \mathbf{M}_{\mathbf{D}_N} \mathbf{X}^*)^{-1} \mathbf{X}^{*\prime} \mathbf{M}_{\mathbf{D}_N} \mathbf{y}, \tag{125}$$

or equivalently

$$\hat{\boldsymbol{\beta}}^*_X = (\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{D}_N}\mathbf{E}'\mathbf{V})^{-1}\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{D}_N}\mathbf{X}\boldsymbol{\beta} + (\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{D}_N}\mathbf{E}'\mathbf{V})^{-1}\mathbf{V}'\mathbf{E}\mathbf{M}_{\mathbf{D}_N}\boldsymbol{\varepsilon}, \tag{126}$$

where

$$\mathbf{M}_{\mathbf{D}_N} \mathbf{y} = \mathbf{M}_{\mathbf{D}_N} \mathbf{X}^* \boldsymbol{\beta} + \mathbf{M}_{\mathbf{D}_N} [(\mathbf{X} - \mathbf{X}^*)\boldsymbol{\beta} + \boldsymbol{\varepsilon}]. \tag{127}$$

The Within estimator is biased as $\mathbb{E}(\hat{\boldsymbol{\beta}}^*_X) \neq \boldsymbol{\beta}$, because $\mathbf{M}_{\mathbf{D}_N}\mathbf{E}'\mathbf{V} = \mathbf{M}_{\mathbf{D}_N}\mathbf{X}^* \neq \mathbf{M}_{\mathbf{D}_N}\mathbf{X}$.

A7. Split Sampling and the Perception Effect

Next, let us discuss a further phenomenon that we call the *perception effect*. The perception effect is relevant if the discretization happens through surveys. There is much evidence in the behavioral literature that the answers to a question may depend on the way the question is asked (see, e.g., Diamond and Hausman, 1994, Haisley et al., 2008 and Fox and Rottenstreich, 2003).[9] Note, that this is present regardless of whether split sampling has been performed or not. However, with split sampling, there is a way to tackle this issue, much akin to the approach a similar problem has been dealt with in the panel data literature.

A7.1. Outcome variable

To sketch out the idea in the univariate case, let us define the *perception effect* B_s for split sample s, as

$$Z_i^{**} = \begin{cases} v_1^{(s)} & \text{if} \quad c_0^{(s)} < Z_i + B_s < c_1^{(s)} \\ \vdots & \vdots \\ v_m^{(s)} & \text{if} \quad c_{m-1}^{(s)} < Z_i + B_s < c_M^{(s)} . \end{cases} \tag{128}$$

[9]Comments by Botond Kőszegi on this section are highly appreciated.

Let Z_i^* be the same quantity, but $B_s = 0\,,\ \forall s$, thus no perception effect. $\tilde{Z}_i^*$ and $\tilde{Z}_i^{**}$ denote the replaced observations in the working sample that derived from Z_i^* and Z_i^{**}, respectively. Following the construction of the working sample,

$$\tilde{Z}_i^{**} = \tilde{Z}_i^* + B_s\,. \tag{129}$$

For example, in the case of discretization happening with the outcome variable, one can use a similar approach as outlined in Section 4.2 in the main text, and use the re-defined equation,

$$\tilde{Y}_i^{**} = \tilde{Y}_i^* + B_s = \beta \tilde{X}_i + u_i\,. \tag{130}$$

Re-write the above in matrix notation,

$$\tilde{\mathbf{y}}^* + \mathbf{TB} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \mathbf{u}, \tag{131}$$

where $\mathbf{B} = (B_1, \ldots, B_S)'$ and $\mathbf{T}$ is a $(N \times S)$ zero-one matrix that extracts the appropriate elements from $\mathbf{B}$.

Note, the estimator for $\mathbb{E}\left(\mathbf{y}|\mathbf{X} \in \mathcal{D}_\mathbf{l}\right)$ needs to be adjusted for the above to hold if identification of $\boldsymbol{\beta}$ is based on the conditional expectation. The main challenge is to keep track of the perception effect. This means we need to identify each split sample and observation when estimating the conditional averages. Specifically,

$$N_l^{-1} \sum_{\mathbf{X}\in\mathcal{D}_\mathbf{l}, \mathbf{X}\in\mathbf{s}} \tilde{\mathbf{y}}^{**} - \mathbb{E}\left(\mathbf{y}|\mathbf{X} \in \mathcal{D}_\mathbf{l}\right) + B_s = o_p(1)\ , \tag{132}$$

where N_l is the number of observations in the partitioned interval $\mathcal{D}_\mathbf{l}$.

Now, the estimation of $\boldsymbol{\beta}$ can be done in the spirit of a fixed effect estimator. Define the usual residual maker, $\mathbf{M_T} = \mathbf{I}_N - \mathbf{T}\left(\mathbf{T}'\mathbf{T}\right)^{-1}\mathbf{T}'$, then

$$\hat{\boldsymbol{\beta}} = \left(\tilde{\mathbf{X}}'\mathbf{M_T}\tilde{\mathbf{X}}\right)^{-1} \tilde{\mathbf{X}}'\mathbf{M_T}\tilde{\mathbf{y}}^{**} \tag{133}$$

is a consistent estimator of $\boldsymbol{\beta}$ given the results presented in this paper and similar arguments for the consistency of the standard fixed effect estimator in the panel data literature (see, e.g., Matyas 2024).

A7.2. Explanatory variable

Let us continue with the case when the explanatory variable is discretized. The discretization of X_i, is the same as defined in Equation (128), with B_s

perception effect for split sample s. Let $\tilde{X}_i^{**} = \tilde{X}_i^* + B_s$ denote the observations in the working sample that derived from X_i^* and X_i^{**}, respectively. The model with perception effects is,

$$Y_i = \beta \tilde{X}_i^{**} + u_i = \beta \tilde{Z}_i^* + \beta B_s + u_i \,. \tag{134}$$

Extension to multiple linear regressions, requires specifying $\mathbf{B_X}$ the perception effect matrix with $S \times K$ dimensions, that allows for different effects for each split sample s and variable k in $\mathbf{X}$. Re-writing in matrix form yields

$$\mathbf{y} = \tilde{\mathbf{X}}^* \boldsymbol{\beta} + \mathbf{D}\mathbf{B_X}\boldsymbol{\beta} + \mathbf{u}, \tag{135}$$

where $\mathbf{D}$ is a $N \times S$ zero-one matrix that extracts the appropriate elements from $\mathbf{B_X}$.

We need to modify the replacement estimator $\boldsymbol{\kappa}$ for the above to hold. We need to keep track of the perception effects, that is from which split sample each observation comes from when estimating the conditional averages. This implies,

$$\hat{\boldsymbol{\kappa}}_s = \left(\mathbf{1}'_{\{\mathbf{X} \in \mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}, \mathbf{X} \in \mathbf{s}\}} \mathbf{1}_{\{\mathbf{X} \in \mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}, \mathbf{X} \in \mathbf{s}\}} \right)^{-1} \mathbf{1}'_{\{\mathbf{X} \in \mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}, \mathbf{X} \in \mathbf{s}\}} \tilde{\mathbf{X}}^{**} \,. \tag{136}$$

As $S, N \to \infty$

$$\hat{\boldsymbol{\kappa}}_s = \operatorname{vec}\left(\mathbb{E}\left[\mathbf{X} | \mathbf{X} \in \mathcal{C}_{\mathbf{m}}^{(\mathbf{s})}) + \mathbf{B_X} \right] \right) + o_p(1) \,, \tag{137}$$

where $\operatorname{vec}(\cdot)$ vectorize the conditional expectations similarly as in Appendix B.1. Note that to identify $\mathbf{B_X}$, we require variation along s and k, thus individual i shall face different split sample discretization for different variables. This is a mild condition and can be satisfied if the survey is constructed accordingly.

Now, the estimation of $\boldsymbol{\beta}$ can be done in the spirit of a fixed effect estimator. Define the usual residual maker, $\mathbf{M_D} = \mathbf{I}_N - \mathbf{D}\left(\mathbf{D}'\mathbf{D}\right)^{-1}\mathbf{D}'$, then

$$\hat{\boldsymbol{\beta}} = \left(\tilde{\mathbf{X}}^{*\prime} \mathbf{M_D} \tilde{\mathbf{X}}^* \right)^{-1} \tilde{\mathbf{X}}^{*\prime} \mathbf{M_D} \mathbf{y} \tag{138}$$

is a consistent estimator of $\boldsymbol{\beta}$ following similar arguments.

Perhaps a more interesting question is the presence of perception effects over different m. In principle, this can also be incorporated by replacing B_s with B_{sm} for $s = 1, \ldots, S$ and $m = 1, \ldots, M$. Therefore, this particular setup does not just allow for perception effects due to different split samples, but also, it provides a framework to investigate different types of perception effects. This would certainly be an interesting avenue for future research in this area.

A7.3. Both variables

Similarly, as before, let us define the observed discretized variables with perception effects in the univariate case,

$$\tilde{Y}_i^{**} = \tilde{Y}_i^{*} + B_Y \qquad \tilde{X}_i^{**} = \tilde{X}_i^{*} + B_X\,. \tag{139}$$

The model without further controls is,

$$\tilde{Y}_i^{**} = \beta \tilde{X}_i^{**} + u_i\,, \tag{140}$$

that is equivalent to

$$\begin{aligned} \tilde{Y}_i^{*} + B_Y &= \beta \tilde{X}_i^{*} + \beta B_X + u_i \\ \tilde{Y}_i^{*} &= \beta \tilde{X}_i^{*} + \beta B_X - B_Y + u_i\,. \end{aligned} \tag{141}$$

Equation (141) is interesting, as it allows for different perception effects in both Y_i and X_i variables in the univariate case. However, in such a setup, the parameters can not be identified in general. To show this, let us consider the matrix formulation,

$$\begin{aligned} \tilde{\mathbf{y}}^{*} + \mathbf{T}\mathbf{B}_Y &= \tilde{\mathbf{X}}^{*}\boldsymbol{\beta} + \mathbf{D}\mathbf{B}_{\mathbf{X}}\boldsymbol{\beta} + \mathbf{u} \\ \tilde{\mathbf{y}}_1^{*} &= \tilde{\mathbf{X}}^{*}\boldsymbol{\beta} + \mathbf{D}\mathbf{B}_{\mathbf{X}}\boldsymbol{\beta} - \mathbf{T}\mathbf{B}_Y + \mathbf{u}\,, \end{aligned} \tag{142}$$

where $\mathbf{B}_Y = (B_{Y,1}, \ldots, B_{Y,S_Y})'$. A unique solution for $\boldsymbol{\beta}$ exists if and only if $\mathbf{T}$ is orthogonal to $\mathbf{D}$. If so, one can use the corresponding residual maker $\mathbf{M}_{\mathbf{TD}} = \mathbf{M}_{\mathbf{T}}\mathbf{M}_{\mathbf{D}}$, that yields in $\hat{\beta} = \left(\tilde{\mathbf{X}}^{*\prime}\mathbf{M}_{\mathbf{TD}}\tilde{\mathbf{X}}^{*}\right)^{-1}\tilde{\mathbf{X}}^{*\prime}\mathbf{M}_{\mathbf{TD}}\tilde{\mathbf{y}}^{*}$. Note, however, that this assumption requires a careful survey design. Lastly, also note that when constructing $\hat{\boldsymbol{\psi}}_Y$ and $\hat{\boldsymbol{\psi}}_X$, one needs to keep track of the split sample as well to ensure convergence.

A7.4. Testing for perception effect

It is theoretically possible to test the impacts of the perception effects on the estimator. Since $\hat{\boldsymbol{\beta}}$ as defined in Equation (133) is consistent regardless of the presence of perception effects. As,

$$\tilde{\beta} = \left(\tilde{\mathbf{X}}'\tilde{\mathbf{X}}\right)^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{y}}^{**} \tag{143}$$

is consistent only in the absence of the perception effects or if the effects are uncorrelated with $\mathbf{X}$, then under the usual regularity conditions, the test statistic is

$$\left(\hat{\beta}-\tilde{\beta}\right)'\left[\mathrm{Var}\left(\hat{\beta}-\tilde{\beta}\right)\right]^{-1}\left(\hat{\beta}-\tilde{\beta}\right)\overset{a}{\sim}\chi^2(K). \tag{144}$$

The exact regularity conditions and the construction of the test statistic would depend on the nature of the perception effect. For example, the case where $\mathbf{B}$ is fixed would be different from the case where $\mathbf{B}$ is a random vector. It would also appear that some assumptions on $\mathbf{B}$ are required to compute the test statistics. This also is another interesting avenue for future research.

A8. Extension to Nonlinear Models

A possible extension is the use of nonlinear models with one or more variables discretized. Let us consider the following general model:

$$\mathbf{y}=g(\mathbf{X},\mathbf{W};\boldsymbol{\beta},\boldsymbol{\gamma})+\mathbf{u}, \tag{145}$$

where $g(\cdot)$ denotes a known continuous function.

The identification procedure is similar to the one in Section 4 of the main text, with the exception that $g(\mathbb{E}[\cdot;\boldsymbol{\beta}])\neq\boldsymbol{\beta}g(\mathbb{E}[\cdot])$ in general. As shown with *split sampling* $\mathbf{Z}^{\dagger}\xrightarrow{p}\mathbf{Z}$. If so, with discretized explanatory variable(s) by continuous mapping theorem $g(\mathbf{X}^{\dagger};\boldsymbol{\beta})\xrightarrow{p}g(\mathbf{X};\boldsymbol{\beta})$. Similar argument applies when discretization happens on the left hand side ($\mathbf{y}^*$) or on both sides ($\mathbf{y}^*,\mathbf{X}^*$).

As for the estimation, let $\hat{\boldsymbol{\beta}}(\mathbf{y},\mathbf{X},\mathbf{W})$ denote a consistent estimator of $\boldsymbol{\beta}$ with $\rho(\mathbf{X},\mathbf{W})=\sqrt{N}\left[\hat{\boldsymbol{\beta}}(\mathbf{y},\mathbf{X},\mathbf{W})-\boldsymbol{\beta}\right]$ such that $\rho(\mathbf{y},\mathbf{X},\mathbf{W})\xrightarrow{d}\mathbf{f}(\mathbf{0},\boldsymbol{\Omega})$. Under the assumptions made earlier, for all cases, the nonlinear quantities should converge too. When the regressors are discretized, we need $\mathbf{X}^{\dagger}\xrightarrow{d}\mathbf{X}\implies\rho(\mathbf{y},\mathbf{X}^{\dagger},\mathbf{W})\xrightarrow{d}\rho(\mathbf{y},\mathbf{X},\mathbf{W})$. In case of discretized outcome, $\mathbf{y}^{\dagger}\xrightarrow{d}\mathbf{y}\implies\rho(\mathbf{y}^{\dagger},\mathbf{X},\mathbf{W})\xrightarrow{d}\rho(\mathbf{y},\mathbf{X},\mathbf{W})$; while for both variables $(\mathbf{y}^{\dagger},\mathbf{X}^{\dagger})\xrightarrow{d}(\mathbf{y},\mathbf{X})\implies\rho(\mathbf{y}^{\dagger},\mathbf{X}^{\dagger},\mathbf{W})\xrightarrow{d}\rho(\mathbf{y},\mathbf{X},\mathbf{W})$. By the continuous mapping theorem under appropriate regularity conditions these convergences do hold. The fine technical details of these conditions, however, could be an interesting subject of future research.

A9. Data Privacy and Split Sampling

A9.1. k-anonymity

A helpful way to interpret the privacy-preserving nature of our discretization strategy is by conceptualizing it as a form of k-anonymity. In k-anonymity, privacy is achieved by replacing exact values of quasi-identifiers with coarser categories so that each released record becomes indistinguishable from at least $k-1$ others. Our approach implements a similar form of protection through interval-based generalization: each sensitive value is mapped to a predefined discretization interval $\mathcal{C}_m$, thereby ensuring that an individual can be associated with only a range of possible values rather than a unique realization. The fixed discretization intervals hide identity by design, and the width of these intervals governs the familiar privacy–accuracy trade-off. Unlike classical k-anonymity, however, the split-sampling framework employs multiple overlapping discretization schemes, so that each observation is represented through several distinct generalizations. This multiplicity preserves the anonymity benefits of coarsening while enabling consistent recovery of the underlying distribution via the convergence results in Proposition 1. In this way, the method preserves the protective logic of k-anonymity while overcoming its typical drawback of severe loss of statistical information. Note that, however, the shifting method does not imply automatically that the k-anonymity privacy condition is satisfied: $|E_j| \geq k$, where E_j is the equivalence class (e.g., around boundary classes); even though it is likely with small S and M. Therefore, one may need to adjust and develop a specific discretization scheme to preserve privacy in this way and ensure convergence.

A9.2. Differential privacy

Our method also has the potential to be extended to the large literature on differential privacy (DP) to protect data privacy. DP has received much attention in the last decade and has become the industry standard.[10] DP quantifies the notion of privacy for downstream tasks and aims to protect the most extreme observations as well. (see, e.g., Dwork 2006, Jordan and

[10]It has been used by many technology companies, including Google Erlingsson et al. (2014), Microsoft (Ding et al., 2017), LinkedIn (Kenthapadi and Tran, 2018), or Facebook. The United States Census Bureau employed DP in 2020 to safeguard individual confidentiality in the U.S. decennial census.

Mitchell 2015, or Bi and Shen 2023). There are two large branches of differential privacy; the first is the "standard" or "central" DP, where data owners can only publish randomized statistics. Our paper relates to the second branch, which is called "local" DP, where data owners do not trust the "central server" or are not allowed/want to share the data in its original form. Local DP has multiple variants[11] and the major issue is that such data privatization may alter the analysis (estimation) results vis-a-vis the original data. As this problem is not new, Duchi et al. (2018) and Cai et al. (2021) work out this trade-off between privacy and statistical accuracy in different setups, such as the mean estimator or least-squares parameters in linear regression. In the econometric context, Bi and Shen (2023) proposes a DP discretization that maintains statistical accuracy for a pre-specified model, while allowing for data privacy. Our approach to some extent is close to theirs in the sense of maintaining (asymptotic) consistency; however, this is the only link it shares with the differential privacy literature. For example, instead of adding noise (Laplace or any other) to the data, we employ discretization to "distort" or rather "dilute" information.[12]

In differential privacy, there is a well-established notion for data protection called ε-differential privacy (Dwork 2006, Dwork et al. 2006). The idea is the following: one changes a single observation in a dataset and wants to see how it impacts the publicly provided information. ε-differential privacy captures this notion as a ratio of probabilities. Let $\mathcal{Z}$ be an arbitrary sample taken from $f_Z(z; a_l, a_u)$, and $\mathcal{M}(\cdot)$ the privatization mechanism, which privatizes/discretizes $\mathcal{Z}$ into $\mathcal{Z}^*$ for public release. Consider two neighboring realizations of $\mathcal{Z}$; z, and z$'$ that are different in just one observation. In our case it means to drop one observation from z to get z$'$.[13] Consequently,

[11] Dwork et al. (2014) uses a Laplace mechanism, Rohde and Steinberger (2020) uses local differential privacy, or Avella-Medina (2021) proposes parametric estimation, just to list a few variants.

[12] Compared to the novel method introduced by Bi and Shen (2023), in which they require specifying the model before the analysis, our discretization process is model-agnostic in the sense that it can be used with different model specifications. Furthermore, Bi and Shen (2023) achieves asymptotic convergence through splitting the sample into two parts and providing only one part to the end user. The other part is then used to transform the original data, add noise, and convert back to a representative distribution of the original data, resulting in an efficiency loss in the number of observations. Our method does not require such sample splitting.

[13] Differential privacy may also imply adding different noise to one observation.

ε-differential privacy requires that the ratio of the probability of any privatized value given one sample to the probability of it given the other sample is upper bound by e^{ε}, that is:

$$\sup_{\mathrm{z},\mathrm{z}'} \sup_{\mathcal{C}_m} \frac{\Pr\left[\mathcal{M}(\mathcal{Z}) \in \mathcal{C}_m | \mathcal{Z} = \mathrm{z}\right]}{\Pr\left[\mathcal{M}(\mathcal{Z}) \in \mathcal{C}_m | \mathcal{Z} = \mathrm{z}'\right]} \leq e^{\epsilon}, \tag{146}$$

where $\varepsilon \geq 0$ is a privacy factor. Note that the $\Pr[\cdot]$ refers to the randomness implied in the assignment mechanism $\mathcal{M}(\cdot)$. If ε is small, we say privacy protection is high. Typically, the literature set ε to 1 or 2. When the numerator and denominator are zero, the convention is to define them as 0. In the case when the denominator is zero, but the numerator is not, we face information leakage. To address this case, the literature uses the "approximate differential privacy", given by:

$$\sup_{\mathrm{z}} \sup_{\mathcal{C}_m} \Pr\left[\mathcal{M}(\mathcal{Z}) \in \mathcal{C}_m | \mathcal{Z} = \mathrm{z}\right] \leq e^{\epsilon} \sup_{\mathrm{z}'} \sup_{\mathcal{C}_m} \left(\Pr\left[\mathcal{M}(\mathcal{Z}) \in \mathcal{C}_m | \mathcal{Z} = \mathrm{z}'\right]\right) + \delta, \tag{147}$$

where δ gives the probability of information leakage, typically a small value.

Remark 1: The use of **known fixed** discretization intervals $\mathcal{C}_m$ allows to hide identity, however it causes the denominator in Equation (146) to be zero in some cases, enforcing large δ in approximate DP. This can be resolved if $\mathcal{C}_m$ is not known, but **stochastic** ensuring to satisfy approximate DP. Then the stochastic nature of $\mathcal{C}_m$ will govern the values of ε and δ. The exact details are also an interesting venue for future research.

Remark 2: In our setup, the discretization intervals' size (M) mitigates the trade-off between privacy and accuracy. To illustrate this, the size of the discretization interval is $||\mathcal{C}_m|| = c_m - c_{m-1}$. One end of the spectrum is an infinite number of interval; $M \to \infty \implies ||\mathcal{C}_m|| \to 0$, so we observe each sensitive value in its original form without providing any privacy. On the other hand, if we use only one interval, $M = 1 \implies ||\mathcal{C}_m|| = a_u - a_l$, which means Z_i^* takes only one value.

A10. Further Monte Carlo evidence

The Monte Carlo simulations are extended in five different ways. The basic setup is the same as in Section 5.1 of the main text, and each time only one parameter is changed compared to the basic setup. First, we investigate the effect of sample size on our shifting method, and how the magnitude of the bias changes when we use $N = 1,000$. As a second exercise, we investigate how the bias changes if the generated distributions are symmetric. In the third exercise, we check how the bias changes if instead of $M = 5$ we only use $M = 3$ intervals representing 'low-mid-high' categories. In last exercise, we show some results on how the bias vanishes as we increase N and S, and the inconsistency of the alternative(s). All the following tables show the Monte Carlo average bias (or distortion if you prefer) of $\hat{\beta}$ from $\beta = 0.5$. In parenthesis, we report the Monte Carlo standard deviation of the estimated parameter.

A10.1. Explanatory variable

First, we investigate the case, when X_i is discretized, hence we only observe X_i^*.

A10.1.1. Moderate sample size

For moderate sample size set $N = 1,000$. Table A1 shows the results which are similar to the results with $N = 10,000$ as reported in the paper.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression	-0.0251	-0.0100	-0.0170	0.0003	0.0009	-0.0414
	(0.0175)	(0.0143)	(0.0159)	(0.0126)	(0.033)	(0.0229)
Shifting ($S = 10$)	-0.0002	0.0001	-0.0005	0.0002	0.0008	-0.0004
	(0.0179)	(0.0140)	(0.0155)	(0.0120)	(0.0287)	(0.0228)

Table A1: Monte Carlo average bias and standard deviation with moderate sample size, $N = 1,000$, when discretization happens to the explanatory variable

Shifting method always outperforms the alternatives, except in the case of uniform and exponential, where there is no bias or only small.

A10.1.2. Symmetric boundaries

Next, we investigate symmetric boundary cases. We set the domain of the explanatory variable to $a_l = -2, a_u = 2$ and keep ε_i generated in the same way. For the log-normal, exponential, and weibull cases, we truncate at 3 and subtract 1 from the generated distribution.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression	-0.0312	-0.0228	-0.0051	-0.0169	-0.0015	-0.0172
	(0.0067)	(0.0062)	(0.0092)	(0.006)	(0.0285)	(0.0174)
Shifting ($S = 10$)	0.0001	-0.0001	0.0002	0.0000	0.0006	0.0002
	(0.0066)	(0.0062)	(0.0087)	(0.0059)	(0.0242)	(0.0157)

Table A2: Monte Carlo average bias and standard deviation with symmetric boundary points: $a_l = -2, a_u = 2$, when discretization happens to the explanatory variable

In this case the bias is even more severe for the mid-point regression than in the asymmetric case. This relates to the distance of the midpoints and the actual excepted values within the intervals. Shifting performs in this setting well and the bias vanishes.

A10.1.3. Number of intervals (M)

Another question is how the number of intervals (M) affects the bias. In this exercise, we investigated the $M = 3$ case, where interval defines (known) low-mid-high ranges. In general, the bias increases for the methods, however, it shows up in a larger standard deviation.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression	-0.0252	-0.0101	-0.0174	0.0002	0.0005	-0.0422
	(0.0057)	(0.0046)	(0.0051)	(0.0040)	(0.0102)	(0.0073)
Shifting ($S = 10$)	0.0002	0.0001	0.0000	0.0002	0.0005	0.0002
	(0.0056)	(0.0044)	(0.0049)	(0.0038)	(0.009)	(0.0072)

Table A3: Monte Carlo average bias and standard deviation with small number of interval options, $M = 3$, when discretization happens to the explanatory variable

A10.1.4. Convergence in N

Tables A4, A5, and A6 show the (asymptotic) reduction in the bias with the split sampling method under different distributions. We use normal distribution for ε_i in Table A4, whereas log-normal and Weibull for Tables A6 and A5 to show that the convergence results are robust to different distributional specifications. In general, the shifting method decreases the bias towards zero, but one needs larger N and S as well. Mid-point regression remains biased regardless of N.

Standard Normal distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		-0.0262 (0.0727)	-0.0260 (0.0221)	-0.0257 (0.0070)	-0.0255 (0.0022)
Shifting	$S = 3$	-0.123 (0.0726)	-0.0051 (0.0230)	-0.0041 (0.0073)	-0.0038 (0.0023)
	$S = 5$	-0.0073 (0.0718)	-0.0032 (0.0226)	-0.0021 (0.0073)	-0.0018 (0.0023)
	$S = 10$	-0.0059 (0.0723)	-0.0018 (0.0225)	-0.0011 (0.0072)	-0.0010 (0.0023)
	$S = 25$	-0.0047 (0.0722)	-0.0015 (0.0224)	-0.0009 (0.0072)	-0.0006 (0.0022)
	$S = 50$	-0.0040 (0.0719)	-0.0015 (0.0222)	-0.0007 (0.0072)	-0.0005 (0.0022)
	$S = 100$	-0.0038 (0.0717)	-0.0012 (0.0225)	-0.0006 (0.0071)	-0.0003 (0.0022)

Table A4: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the explanatory variable – normal distribution

Log-Normal distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		-0.0186 (0.0627)	-0.0182 (0.0198)	-0.0179 (0.0063)	-0.0176 (0.0020)
Shifting	$S = 3$	-0.0075 (0.0632)	-0.0049 (0.0199)	-0.0045 (0.0064)	-0.0044 (0.0020)
	$S = 5$	-0.0070 (0.0638)	-0.0037 (0.0200)	-0.0030 (0.0064)	-0.0026 (0.0020)
	$S = 10$	-0.0051 (0.0626)	-0.0026 (0.0197)	-0.0020 (0.0062)	-0.0019 (0.0020)
	$S = 25$	-0.0024 (0.0630)	-0.0025 (0.0197)	-0.0018 (0.0063)	-0.0015 (0.0020)
	$S = 50$	-0.0033 (0.0621)	-0.0018 (0.0197)	-0.0014 (0.0063)	-0.0012 (0.0020)
	$S = 100$	-0.0028 (0.0623)	-0.0015 (0.0196)	-0.0012 (0.0062)	-0.0010 (0.0020)

Table A5: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the explanatory variable – log-normal distribution

Weibull distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		-0.0433 (0.0929)	-0.0436 (0.0283)	-0.0424 (0.0091)	-0.0423 (0.0029)
Shifting	$S = 3$	-0.0171 (0.0945)	-0.0109 (0.0297)	-0.0097 (0.0095)	-0.0087 (0.0029)
	$S = 5$	-0.0113 (0.0947)	-0.0063 (0.0293)	-0.0050 (0.0094)	-0.0047 (0.0030)
	$S = 10$	-0.0083 (0.0937)	-0.0043 (0.0295)	-0.0031 (0.0093)	-0.0031 (0.0030)
	$S = 25$	-0.0069 (0.0933)	-0.0037 (0.0290)	-0.0024 (0.0092)	-0.0022 (0.0029)
	$S = 50$	-0.0064 (0.0937)	-0.0031 (0.0290)	-0.0021 (0.0091)	-0.0017 (0.0029)
	$S = 100$	-0.0053 (0.0933)	-0.0026 (0.0288)	-0.0018 (0.0092)	-0.0014 (0.0029)

Table A6: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the explanatory variable – Weibull distribution

A10.2. Outcome variable

In this subsection, we provide further evidence on the bias reduction, when discretization happens on the left-hand side, thus we observe discretized outcome variable Y_i^*.

Notes: For all the upcoming Tables, for *"Set identification"* we estimate the lower and upper boundaries for the identified set-region. First, we subtract the true parameter from the boundaries and then report it, therefore it should give a (close) interval around zero. For ordered choice models, we report the "distortion" from the true β. Ordered probit and logit models' maximum likelihood parameters do not aim to recover the true β parameter, therefore it is not appropriate to call it bias in general.

A10.2.1. Moderate sample size

First, we investigate the magnitude of the bias, when the sample size is moderate, namely $N = 1,000$. Table A7 shows the results which are similar to the results with $N = 10,000$ as reported in the paper.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Set identification†	$[-1.1, 1.15]$ (0.06),(0.07)	$[-1.09, 1.15]$ (0.08),(0.08)	$[-1.09, 1.16]$ (0.07),(0.07)	$[-1.07, 1.17]$ (0.09),(0.09)	$[-1.06, 1.18]$ (0.08),(0.09)	$[-1.09, 1.15]$ (0.05),(0.06)
Ordered probit*	0.1978 (0.0810)	0.0690 (0.0797)	0.2138 (0.0827)	0.0181 (0.0763)	0.0965 (0.0795)	0.4484 (0.0908)
Ordered logit*	0.6523 (0.1479)	0.3828 (0.1431)	0.6967 (0.1561)	0.2419 (0.1364)	0.4309 (0.1455)	1.2109 (0.1682)
Interval regression	0.0254 (0.0618)	0.0329 (0.0784)	0.0398 (0.0694)	0.0512 (0.0882)	0.0638 (0.0825)	0.0396 (0.0505)
Midpoint regression	0.0209 (0.0643)	0.0293 (0.0786)	0.0310 (0.0733)	0.0453 (0.0895)	0.2029 (0.0426)	0.0275 (0.0526)
Shifting ($S = 10$)	-0.0043 (0.0611)	-0.0021 (0.0758)	-0.0036 (0.0685)	-0.0014 (0.0869)	-0.0019 (0.0389)	-0.0019 (0.0475)

Table A7: Monte Carlo average bias and standard deviation with moderate sample size, $N = 1,000$, when discretization happens to the outcome variable

Shifting method always outperforms the alternatives.

A10.2.2. Symmetric boundaries

Next, we investigate symmetric boundary cases. We set the domain of the outcome variable to $a_l = -3, a_u = 3$ and keep X_i generated in the same way. ε_i is generated/truncated such that its lower and upper bound is -2 and 2. In the normal, logistic, and uniform cases, it means the lower and upper bounds are -2 and 2. For the log-normal, exponential, and weibull cases, we truncate at 4 and subtract 2 from the generated distribution.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Set identification†	$[-1.11, 1.13]$ (0.02),(0.02)	$[-1.15, 1.10]$ (0.02),(0.02)	$[-1.09, 1.16]$ (0.02),(0.02)	$[-1.07, 1.17]$ (0.03),(0.03)	$[-1.06, 1.19]$ (0.03),(0.03)	$[-1.09, 1.15]$ (0.02),(0.02)
Ordered probit*	0.0890 (0.0252)	0.0029 (0.0243)	0.2085 (0.0262)	0.0158 (0.0234)	0.0986 (0.0241)	0.4461 (0.0295)
Ordered logit*	0.4513 (0.0446)	0.3198 (0.0427)	0.6862 (0.0499)	0.2379 (0.0422)	0.4338 (0.044)	1.2085 (0.0546)
Interval regression	0.0085 (0.022)	-0.0267 (0.0234)	0.0371 (0.0221)	0.0491 (0.0271)	0.0663 (0.0249)	0.0397 (0.0166)
Midpoint regression	0.0070 (0.0211)	0.0240 (0.0242)	0.0362 (0.0216)	0.0490 (0.0273)	0.2077 (0.0128)	0.0314 (0.0157)
Shifting ($S = 10$)	-0.0001 (0.0199)	0.0004 (0.0232)	-0.0001 (0.0204)	-0.0007 (0.0262)	-0.0015 (0.0115)	-0.0001 (0.0140)

Table A8: Monte Carlo average bias and standard deviation with symmetric boundary points: $a_l = -3, a_u = 3$, when discretization happens to the outcome variable

As we expected the maximum likelihood methods, have a closer fit to the assumed distribution the distortion is somewhat smaller in the case of ordered probit model.[14] This is the case with the normal and logistic distributions for the disturbance term. However, the distortion remains of the same magnitude for all the other misspecified cases. The shifting method outperforms all other methods.

A10.2.3. Number of intervals (M)

We investigated the $M = 3$ case, where interval defines (known) low-mid-high ranges. In general, the bias increases for the methods. Interesting exceptions are interval regression and midpoint regression, where the results become more volatile: in some cases, they give better results, while in others even worse. The shifting method gives fairly accurate estimates.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Set identification†	$[-1.83, 1.90]$ (0.03),(0.03)	$[-1.85, 1.88]$ (0.03),(0.03)	$[-1.85, 1.89]$ (0.03),(0.03)	$[-1.87, 1.87]$ (0.03),(0.03)	$[-1.89, 1.85]$ (0.03),(0.03)	$[-1.81, 1.93]$ (0.02),(0.02)
Ordered probit*	0.1062 (0.0272)	-0.0220 (0.0266)	0.0197 (0.0278)	-0.1028 (0.0250)	-0.0752 (0.0253)	0.2347 (0.0302)
Ordered logit*	0.5193 (0.0462)	0.3220 (0.0457)	0.3916 (0.0472)	0.1700 (0.0423)	0.2169 (0.0428)	0.7246 (0.0509)
Interval regression	0.0124 (0.0224)	0.0124 (0.0281)	0.0122 (0.0268)	-0.0044 (0.0306)	-0.0243 (0.0280)	-0.0061 (0.0200)
Midpoint regression	0.0336 (0.0233)	0.0168 (0.0274)	0.0229 (0.0267)	-0.0011 (0.0307)	-0.2026 (0.0170)	0.0647 (0.0216)
Shifting ($S = 10$)	-0.0274 (0.0237)	-0.0114 (0.0256)	-0.0009 (0.0226)	-0.0027 (0.0277)	0.0011 (0.0135)	-0.0008 (0.0151)

Table A9: Monte Carlo average bias and standard deviation with small number of interval options, $M = 3$, when discretization happens to the outcome variable

Also note that with the shifting method, the average bias is within 1 standard deviation, which is not true for the other methods, especially when the underlying distribution is exponential or weibull.

A10.2.4. Convergence in N

Tables A10, A11, and A12 show the (asymptotic) reduction of the bias with the split sampling method under different distributions. We use normal distribution for ε_i in Table A10, whereas log-normal and Weibull for Tables

[14] Note that ordered probit and logit uses different scaling (depending on the assumed distribution), which results in different parameter estimates. In our case it means ordered logit has higher average distortions than ordered probit, but this is only a matter of scaling. One can map one to the other with the scaling factor, $\hat{\beta}^{ML}_{probit} \approx \hat{\beta}^{ML}_{logit} \times 0.25/0.3989$. This is why we use the term distortion rather than bias for these methods.

A11 and A12 to show that the convergence results are robust to different distributional specifications. The results suggest, as we increase the number of observations the bias vanishes for the shifting method. Also if we increase the number of split samples the bias tends to decrease. One important exception is with few number of observations: if $N = 100$, then if the ratio of S/N increases after 0.1, the bias increases. This is not a big surprise as with $S = 50$, only two observations are assigned to each split sample. It is also important to highlight the other methods' bias remains the same as we increase the number of observations, therefore they give inconsistent estimates.

Normal distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		0.0176 (0.1896)	0.0257 (0.0635)	0.0251 (0.0195)	0.0251 (0.0061)
Shifting	$S = 3$	-0.0039 (0.1966)	-0.0019 (0.0635)	0.0014 (0.0197)	-0.0008 (0.0062)
	$S = 5$	0.0054 (0.1902)	-0.0016 (0.0614)	-0.0007 (0.0189)	-0.0005 (0.0060)
	$S = 10$	-0.0093 (0.1905)	-0.0067 (0.0605)	-0.0025 (0.0190)	-0.0006 (0.0059)
	$S = 25$	0.0050 (0.1829)	0.0052 (0.0602)	0.0008 (0.0185)	-0.0001 (0.0057)
	$S = 50$	-0.0136 0.1853	-0.0027 (0.0587)	-0.0011 (0.0185)	-0.0004 (0.0058)
	$S = 100$	-0.0533 0.1802	-0.0006 (0.0596)	-0.0002 (0.0183)	-0.0002 (0.0057)

Table A10: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the outcome variable – normal distribution

Log-Normal distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		0.0342 (0.2186)	0.0346 (0.0702)	0.0358 (0.0222)	0.0362 (0.0068)
Shifting	$S = 3$	0.0135 (0.2200)	-0.0054 (0.0711)	0.0001 (0.0222)	-0.0007 (0.0068)
	$S = 5$	0.0106 (0.2127)	-0.0026 (0.0675)	-0.0001 (0.0213)	-0.0005 (0.0066)
	$S = 10$	-0.0025 (0.2176)	-0.0063 (0.0682)	-0.0019 (0.0211)	-0.0008 (0.0066)
	$S = 25$	0.0086 (0.2106)	-0.0036 (0.0682)	-0.0006 (0.0213)	-0.0001 (0.0066)
	$S = 50$	-0.0063 (0.2102)	-0.0034 (0.0684)	-0.0004 (0.0212)	-0.0003 (0.0064)
	$S = 100$	-0.0503 (0.2059)	0.0005 (0.0670)	0.0003 (0.0206)	-0.0002 (0.0065)

Table A11: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the outcome variable – log-normal distribution

Weibull distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		0.0309 (0.1631)	0.0295 (0.0514)	0.0312 (0.0159)	0.0319 (0.0049)
Shifting	$S = 3$	0.0092 (0.1545)	0.0068 (0.0531)	0.0024 (0.0162)	-0.0004 (0.0050)
	$S = 5$	-0.0013 (0.1602)	0.0022 (0.0503)	0.0004 (0.0156)	-0.0002 (0.0048)
	$S = 10$	-0.0127 (0.1575)	-0.0041 (0.0485)	-0.0017 (0.0151)	-0.0004 (0.0048)
	$S = 25$	0.0156 (0.1511)	-0.0009 (0.0482)	-0.0011 (0.0152)	-0.0002 (0.0046)
	$S = 50$	-0.0060 (0.1471)	-0.0008 (0.0471)	0.0000 (0.0146)	-0.0002 (0.0046)
	$S = 100$	-0.0532 0.1431	0.0008 (0.0476)	-0.0002 (0.0146)	-0.0001 (0.0045)

Table A12: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens to the outcome variable – weibull distribution

A10.3. Both sides

In our final simulations, we investigate the properties of bias when both outcome and explanatory variables are discretized.

A10.3.1. Moderate sample size

For moderate sample size we set $N = 1,000$. Table A13 shows the results which are similar to the pattern with $N = 10,000$ as reported in the paper.

		Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression		-0.0856	-0.0797	-0.0759	-0.0647	0.0809	-0.0771
		(0.0571)	(0.0691)	(0.0616)	(0.0790)	(0.0377)	(0.0456)
Shifting ($S = 10$)		0.0139	0.0094	0.0042	0.0064	0.0001	0.0035
		(0.0607)	(0.0768)	(0.0668)	(0.0862)	(0.0364)	(0.0452)

Table A13: Monte Carlo average bias and standard deviation with moderate sample size, $N = 1,000$, when discretization happens on both sides

Shifting method always outperforms the alternative mid-point regression.

A10.3.2. Symmetric boundaries

For the symmetric boundary case, we set the domain of the disturbance term to $a_l = -2, a_u = 2$ and keep X_i generated in the same way. Now the outcome variable's domain is between -3 and 3. For the log-normal, exponential, and Weibull cases, we truncate at 3 and subtract 1 from the generated distribution.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression	-0.0975	-0.0838	-0.0752	-0.0635	0.0797	-0.0759
	(0.0189)	(0.0217)	(0.019)	(0.0243)	(0.0116)	(0.0137)
Shifting ($S = 10$)	0.0088	0.0075	0.0056	0.0057	0.0026	0.0047
	(0.0205)	(0.0237)	(0.02211)	(0.0264)	(0.0118)	(0.0142)

Table A14: Monte Carlo average bias and standard deviation with symmetric boundary points: $a_l = -2, a_u = 2$, when discretization happens on both sides

Results are similar to the reported table in the main paper.

A10.3.3. Number of intervals (M)

We investigated the $M = 3$ case, where interval defines (known) low-mid-high ranges. In general, the bias increases for the methods. Shifting gives closer results to zero bias.

	Normal	Logistic	Log-Normal	Uniform	Exponential	Weibull
Mid-point regression	-0.1998 (0.0177)	-0.2091 (0.0210)	-0.2221 (0.0198)	-0.2143 (0.0235)	-0.3552 (0.0108)	-0.2012 (0.0154)
Shifting ($S = 10$)	-0.1062 (0.0269)	-0.0168 (0.0301)	-0.0123 (0.0257)	0.0142 (0.0304)	-0.0049 (0.0137)	0.0092 (0.0162)

Table A15: Monte Carlo average bias and standard deviation with small number of interval options, $M = 3$, when discretization happens on both sides

A10.3.4. Convergence in N

Tables A16, A17, and A18 show the (asymptotic) reduction in the bias with the split sampling method under different distributions. We use normal distribution for ε_i in Table A16, whereas log-normal and Weibull for Tables A17 and A18 to show that the convergence results are robust to differenet distributional specifications. Here, we see a big reduction with few S, but then the bias is not decreasing as quickly as in the other cases.

Normal distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		-0.1516 (0.1652)	-0.1562 (0.0517)	-0.1547 (0.0162)	-0.1543 (0.0051)
Shifting	$S = 3$	-0.1011 (0.1877)	-0.0694 (0.0633)	-0.0609 (0.0205)	-0.0597 (0.0064)
	$S = 5$	-0.0857 (0.1922)	-0.0543 (0.0606)	-0.0515 (0.0198)	-0.0505 (0.0062)
	$S = 10$	-0.0768 (0.1896)	-0.0539 (0.0594)	-0.0481 (0.0194)	-0.0463 (0.0059)
	$S = 25$	-0.0707 (0.1896)	-0.0506 (0.0593)	-0.0471 (0.0188)	-0.0462 (0.0060)
	$S = 50$	-0.0736 (0.1884)	-0.0520 (0.0570)	-0.0482 (0.0186)	-0.0470 (0.0058)
	$S = 100$	-0.0860 (0.1832)	-0.0520 (0.0564)	-0.0487 (0.0183)	-0.0470 (0.0057)

Table A16: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens on both sides – normal

Log-normal distribution					
		$N = 100$	$N = 1{,}000$	$N = 10{,}000$	$N = 100{,}000$
Midpoint regression		-0.1297 (0.1681)	-0.1373 (0.0520)	-0.1337 (0.0168)	-0.1333 (0.0051)
Shifting	$S = 3$	-0.1275 (0.1924)	-0.1051 (0.0652)	-0.0917 (0.0209)	-0.0891 (0.0064)
	$S = 5$	-0.1126 (0.1911)	-0.0825 (0.0612)	-0.0816 (0.0198)	-0.0807 (0.0063)
	$S = 10$	-0.1068 (0.1916)	-0.0867 (0.0601)	-0.0783 (0.0196)	-0.0764 (0.0060)
	$S = 25$	-0.0931 (0.1895)	-0.0821 (0.0594)	-0.0769 (0.0189)	-0.0750 (0.0060)
	$S = 50$	-0.0932 (0.1842)	-0.0841 (0.0576)	-0.0774 (0.0192)	-0.0751 (0.0058)
	$S = 100$	-0.1014 (0.1834)	-0.0804 (0.0581)	-0.0768 (0.0181)	-0.0752 (0.0057)

Table A17: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens on both sides – log-normal

Weibull distribution					
		$N = 100$	$N = 1,000$	$N = 10,000$	$N = 100,000$
Midpoint regression		-0.0837 (0.1413)	-0.0898 (0.0457)	-0.0883 (0.0147)	-0.0882 (0.0045)
Shifting	$S = 3$	-0.0675 (0.1636)	-0.0365 (0.0519)	-0.0263 (0.0169)	-0.0254 (0.0052)
	$S = 5$	-0.0415 (0.1618)	-0.0179 (0.0492)	-0.0146 (0.0160)	-0.0142 (0.0049)
	$S = 10$	-0.0373 (0.1574)	-0.0171 (0.0487)	-0.0094 (0.0156)	-0.0082 (0.0050)
	$S = 25$	-0.0314 (0.1527)	-0.0122 (0.0468)	-0.0074 (0.0153)	-0.0063 (0.0049)
	$S = 50$	-0.0370 (0.1544)	-0.0119 (0.0471)	-0.0075 (0.0155)	-0.0060 (0.0047)
	$S = 100$	-0.0725 (0.1513)	-0.0099 (0.0458)	-0.0072 (0.0148)	-0.0060 (0.0047)

Table A18: Bias reduction for split sampling methods: different sample sizes and number of split samples, when discretization happens on both sides – weibull

A11. Gender Wage Gap: Further Details

To demonstrate how our method works in practice, we used a dataset where we can measure the difference between the parameter estimated on a non-discretized variable and the parameter(s) using some discretized version of the data. The Australian Tax Office's (ATO) individual sample files record income and some basic socio-economic variables for a 2% sample of the whole population.[15] To contrast this non-discretized income variable with our approach, we use different discretization processes. We employ a simple equally distanced discretization method and a specific method, which is used in the Household, Income, and Labour Dynamics in Australia (HILDA) Survey.[16] HILDA is an annual survey and dataset, which is well known and widely used in Australia for economic research.[17] This way we can estimate parameters on the complete sample and the parameter estimates on "what if it is observed through a discretization process".

A11.1. Data

The Australian Tax Office (ATO) dataset is a confidentialised 2% sample of the whole population. It records individual income tax returns for various income for separate years. We use data from 2016-17, which contains overall 277,202 records. Our outcome variable is yearly earned wage in the Australian dollar. Our parameter of interest is the coefficient for the gender variable. Further variables are,

- Expected age for each age group
    - We have calculated the expected age conditional on the age groups. This is necessary, while the ATO dataset only uses age groups: $0-20, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70+$. To circumvent this discretization process, we are using the Australian Bureau of Statistics on demographic statistics[18], which contains the number of males and female for each age. Based on this we calculate the conditional expected values for the year 2016-17, conditioning on gender.

[15] For details see ATO's website: https://www.ato.gov.au/about-ato/research-and-statistics/in-detail/taxation-statistics/taxation-statistics-previous-editions/taxation-statistics-2016-17 .

[16] https://melbourneinstitute.unimelb.edu.au/hilda.

[17] see:https://melbourneinstitute.unimelb.edu.au/hilda/publications.

[18] File,31010$DO002_2$01906, available at https://www.abs.gov.au/AUSSTATS/abs@.nsf/DetailsPage/3101.0Jun%202019?OpenDocument

- occupation code (as a series of dummies)
- spouse (dummy)
- region (dummies)
- lodgment method (dummy - via tax agent or self-prepared return)
- private health insurance (PHI) indicator (dummy)

We restrict our sample to the working population (older than 25 and younger than 65 years old) and we remove individuals whose wage is lower than the 2017 minimum wage (weekly minimum wage was 627.7 AUD) and whose wage is more than 225.000AUD (that is the top 1%). Table A19 shows basic descriptive statistics of the variables used in our sample.

A11.2. Discretization process and model results

The discretization process influences the magnitude of the bias. We have fixed the lower bound to zero and the upper bound to $225,000$ for the wages. We use two different discretization processes:

- $M = 10$ with equal distances for mid-point regression and shifting as this was the closer in our reported main result in Table 1 from the main text.
- HILDA's household questionnaire, which uses $M = 12+1$ categories in 2017[19]: $1 - 9.999$, $10.000 - 19.999$, $20.000 - 29.999$, $30.000 - 39.999$, $40.000 - 49.999$, $50.000 - 59.999$, $60.000 - 79.999$, $80.000 - 99.999$, $100.000 - 124.999$, $125.000 - 149.999$, $150.000 - 199.999$ and 200.000 or more. Three extra options are added: negative or zero refused, and don't know.

We need to note that HILDA is aiming for total income and not for wages/salaries, thus we discretize the income jointly with the wages. For the shifting method, we use equal distances and check for $S = 10$ during the modeling.

Our outcome variable is yearly wage in Australian dollar and our parameter of interest is the coefficient for the gender dummy. We compare the conditional average outcome based on gender, using three different linear models, all of them estimated by OLS:

[19] https://melbourneinstitute.unimelb.edu.au/__data/assets/pdf_file/0005/2409674/HouseholdQuestionnaireW17M.pdf.

	Mean	Median	Std	Min	Max
wage	77,808	68,864	35,369	34,981	22,4951
total income	83,069	71,337	55,609	-13,5989	3,001,331
exp. age	46.77	47.02	10.78	27.00	61.95
Positive income	Total income> 0 0.9998			Total income≤ 0 0.0002	
Gender	Male 0.5598			Female 0.4402	
Spouse	No 0.3867			Yes 0.6133	
LM	Tax agent 0.7304			Self preparer 0.2696	
PHI	No 0.3562			Yes 0.6438	
Occ. codes*	0 0.0005	1 0.1537	2 0.2759	3 0.1320	4 0.0774
	5 0.1392	6 0.0485	7 0.0681	8 0.0714	9 0.0333
Region. codes†	New South Wales				
	Capital 0.1971	Other Urban 0.0412	R.H.U. 0.0388	R.L.U. 0.0169	Rural 0.0250
	Queensland				
	Capital 0.0970	Other Urban 0.0522	R.H.U. 0.0131	R.L.U. 0.0081	Rural 0.0290
	Tasmania				
	Capital 0.0055	Other Urban 0.0026	R.H.U. 0.0026	R.L.U. 0.0018	Rural 0.0061
	Victoria				
	Capital 0.1830	Other Urban 0.0128	R.H.U. 0.0193	R.L.U. 0.0120	Rural 0.0249
	Western Australia				
	Capital 0.0716	Other Urban 0.0030	R.H.U. 0.0159	R.L.U. 0.0066	Rural 0.0125
	ACT	South Australia			
	0.0218	Capital 0.0472	R.H.U. 0.0060	R.L.U. 0.0043	Rural 0.0079
	Northern Territory			Overseas/ invalid	
	Capital 0.0057	R.H.U. 0.0033	R.L.U. 0.0026	0.0026	

** Occupation codes: 0 - Occupation not listed/ Occupation not specified, 1 - Managers, 2 - Professionals, 3 - Technicians and Trades Workers, 4 - Community and Personal Service Workers, 5 - Clerical and Administrative Workers, 6 - Sales workers, 7 - Machinery operators and drivers, 8 - Labourers, 9 - Consultants, apprentices, and type not specified or not listed*

†: R.H.U: regional high urbanization, R.L.U: regional low urbanization.

Table A19: Summary table of variables used

- Model 1: $y_i = \alpha + \beta \times gender_i + \epsilon_i$.
- Model 2: $y_i = \alpha + \beta \times gender_i + \gamma_1 age_i + \gamma_2 age_i^2 + \varepsilon_i$.
- Model 3: adding further controls for design 2, occupation code (dummies), having a spouse (dummy), region (dummies), and whether having private health insurance (dummy).
- Model 4: Model 3 and interaction terms of occupation and age.

Discretization	Model 1	Model 2	Model 3	Model 4
Non-discretized	-0.2027 (0.0023)	-0.2025 (0.0023)	-0.2277 (0.0023)	-0.2261 (0.0023)
HILDA	-0.2012 0.0023	-0.2010 0.0023	-0.2258 0.0023	-0.2242 0.0023
Mid-point regression	-0.2124 (0.0025)	-0.2122 (0.0024)	-0.2381 (0.0025)	-0.2364 (0.0025)
Shifting method (S=10)	-0.2040 0.0024	-0.2038 0.0024	-0.2296 0.0024	-0.2280 0.0024

Non-discretized row uses the actual wage data. HILDA uses its special 13-category discretization outlined above. Mid-point regression and shifting method use $M = 10$ intervals. Shifting uses $S = 10$ split samples. Standard errors are in parenthesis.

Table A20: Estimated $\hat{\beta}$ parameters for the gender dummy with different model specifications

Table A20 shows that the shifting method gives close and statistically the same parameter estimates relative the non-discretized parameter value(s). HILDA performs similarly well, while mid-point regression gives statistically different parameter values at 5% in almost all cases.

Directly observed/True		0.0000		
		Mid-point	Set-identification[†]	Shifting ($S = 10$)
$M = 3$	bias	-0.0486	[-7.8128, 5.5318]	0.0047
	bias (%)	21.49%		-2.08%
$M = 5$	bias	-0.0374	[-2.7532, 1.5553]	-0.0087
	bias (%)	16.54%		3.84%
$M = 10$	bias	-0.0103	[-0.3791, 0.3422]	-0.0019
	bias (%)	4.55%		0.84%
N	125,995			

[†] Set identification gives the lower and upper boundaries for the valid parameter set. We adjusted it by the true parameter values, hence if it conntains zero, then it also contains the original parameter.

Table A21: Bias of the estimates for gender wage gap based on different discretizations of yearly wages

	$M = 3$	$M = 5$	$M = 10$
$S = 3$	-0.2306	-0.2365	-0.2284
	(0.0031)	(0.0028)	(0.0024)
$S = 5$	-0.2255	-0.2339	-0.2295
	(0.0030)	(0.0027)	(0.0024)
$S = 10$	-0.2214	-0.2348	-0.2280
	(0.0030)	(0.0027)	(0.0024)
$S = 25$	-0.2233	-0.2355	-0.2292
	(0.0030)	(0.0027)	(0.0024)
$S = 50$	-0.2226	-0.2344	-0.2273
	(0.0030)	(0.0027)	(0.0024)
$S = 100$	-0.2207	-0.2334	-0.2290
	(0.0030)	(0.0027)	(0.0024)

Table A22: Point estimate approximation with shifting method with different S and M

	$M = 3$	$M = 5$	$M = 10$
$S = 3$	[-4.5460, 2.8385]	[-1.6672, 0.7846]	[-0.5905, 0.1190]
	[(0.0237), (0.0199)]	[(0.0126), (0.0093)]	[(0.0030), (0.0029)]
$S = 5$	[-4.0849, 2.4915]	[-1.4564, 0.6791]	[-0.5910, 0.1173]
	[(0.0226), (0.0180)]	[(0.0108), (0.0072)]	[(0.0030), (0.0029)]
$S = 10$	[-3.7205, 2.2694]	[-1.3842, 0.6540]	[-0.5896, 0.1193]
	[(0.0205), (0.0170)]	[(0.0101), (0.0077)]	[(0.0030), (0.0029)]
$S = 25$	[-3.5748, 2.1657]	[-1.3321, 0.6245]	[-0.5906, 0.1176]
	[(0.0204), (0.0163)]	[(0.0091), (0.0062)]	[(0.0029), (0.0029)]
$S = 50$	[-3.5100, 2.1352]	[-1.3054, 0.6172]	[-0.5885, 0.1195]
	[(0.0198), (0.0162)]	[(0.0091), (0.0063)]	[(0.0030), (0.0029)]
$S = 100$	[-3.4886, 2.1232]	[-1.3027, 0.6132]	[-0.5906, 0.1180]
	[(0.0199), (0.0163)]	[(0.0088), (0.0061)]	[(0.0029), (0.0029)]

Table A23: Set-identification estimates with using split samples from shifting method with different S and M

A12. Notation Tables

See next pages, due to the size of the tables.

Indexes	
$i = 1, \ldots, N$	unit of observation
$m = 1, \ldots, M$	intervals
$s = 1, \ldots, S$	split samples
$b = 1, \ldots, B$	intervals in the resulting working sample
$l = 1, \ldots, S$	alternative split sample index to prove Proposition 1
Scalars	
N	number of observations
$N^{(s)}$	number of observations in split sample s
M	number of intervals (known and fixed)
S	number of split samples
B	number of intervals in the working sample
a_l	lower boundary for the pdf of Z
a_u	upper boundary for the pdf of Z
c_m	interval boundary point
$c_m^{(s)}$	interval boundary point for split sample s
c_b^{WS}	interval boundary point for b in the working sample
v_m	assigned value to the discretized interval $\mathcal{C}_m$
$v_m^{(s)}$	assigned value to the discretized interval $\mathcal{C}_m^{(s)}$
β	parameter of interest (scalar or vector)
σ_u^2	variance of $\mathbf{u}$
h	size of the shift
a	known scalar for illustrative purposes
b	known scalar for illustrative purposes
c	scalar that is $c \in (0, 1)$, used for convergence assumption
Random Variables	
Z	unobserved random variable
Z_i	potential realization of Z, which are unobserved
Z_i^*	observed discretized realization of Z
$Z_i^{(s)}$	observed discretized realization of Z in split sample s
$Z^\dagger$	synthetic variable created from $Z_i^{(s)}$
Y	random variable for outcome in regression model
X	random variable for explanatory variable(s) in regression model
Y^*	discretized version of the random variable for outcome in regression model
X^*	discretized version of the random variable for explanatory variable(s) in regression model
W	other non-discretized random variables for explanatory variables in regression model
u	idiosyncratic term from unobserved regression model

Table A24: Table of notations I

Sets	
$\mathcal{C}_m$	interval set for m with $\mathcal{C}_m = [c_{m-1}, c_m)$
$\mathcal{C}_m^{(s)}$	interval set for m and for split sample s with $\mathcal{C}_m^{(s)} = [c_{m-1}^{(s)}, c_m^{(s)})$
$\mathcal{C}_b^{WS}$	interval set in the working sample for b with $\mathcal{C}_b^{WS} = [c_{b-1}^{WS}, c_b^{WS})$
$\mathcal{B}$	parameter space for β, that is $\mathcal{B} \subset \mathbb{R}$
$\mathcal{S}_s$	set for observables/random variable that falls into split sample s
Functions	
$f_Z(\cdot)$	probability density function (pdf) for Z
$F_Z(\cdot)$	cumulative density function (cdf) for Z
$F_{Z^\dagger}(\cdot)$	cumulative density function (cdf) for $Z^\dagger$
$\kappa(s, \mathbf{m})$	maps the conditional expectations of $\mathbf{X}$ given grid $\mathcal{C}_\mathbf{m}^{(\mathbf{s})}$
$\pi(s, m, \mathbf{l})$	maps the conditional expectations of $\mathbf{y}$ for a given interval $\mathbf{y} \in \mathcal{C}_m^{(s)}$ and partition $\mathbf{X} \in \mathcal{D}_\mathbf{l}$
Multivariate case	
$p = 1, \ldots, P$	index for number of discretized variable
P	number of discretized variables
$\mathbf{Z}$	$\mathbf{z} = (\mathbf{z}_1, \ldots, \mathbf{z}_p, \ldots, \mathbf{z}_P)$ matrix of discretized variables
$\mathbf{f_Z}(\mathbf{z}; \mathbf{a}_l, \mathbf{a}_u)$	multivariate pdf for $\mathbf{Z}$ with lower and upper boundaries.
$\mathbb{A}^P$	boundaries for $\mathbf{Z}$ with $\mathbb{A}^P = [a_{1,l}, a_{1,u}] \times \ldots \times [a_{P,l}, a_{P,u}] \subseteq \mathbb{R}^P$
$\mathbf{Z}^*$	discretized version of $\mathbf{Z}$
$\mathcal{C}_\mathbf{m}$	multi-dimensional grid for discretization with $\mathcal{C}_\mathbf{m} = [\mathbf{c_{1,m-1}}, \mathbf{c_{1,m}}) \times \ldots \times [\mathbf{c_{P,m-1}}, \mathbf{c_{P,m}})$
$\mathbf{v}_m$	assigned values for each grid
h_p	size of the shift in dimension p
$\mathbf{y}$	$\mathbf{y} = (Y_1, \ldots, Y_N)'$ a $(N \times 1)$ vector for outcome
$\mathbf{y}^*$	discretized version of $\mathbf{y}$
$\mathbf{y}^\dagger$	synthetic random variable created from $\mathbf{y}^*$
$\ddot{\mathbf{y}}$	replaced version of $\mathbf{y}$ with $\hat{\mathbb{E}}\,[\mathbf{y} \vert \mathbf{X}^* \in \mathcal{C}_\mathbf{m}]$
$\tilde{\mathbf{y}}$	replaced version of $\mathbf{y}^*$ with $\sum_l \sum_m \hat{\mathbb{E}}[\mathbf{y} \vert \mathbf{y}^* \in \mathcal{C}_m, \mathbf{X} \in \mathcal{D}_\mathbf{l}] \Pr[\mathbf{y}^* \in \mathcal{C}_\mathbf{m} \vert \mathbf{X} \in \mathcal{D}_\mathbf{l}]$, using $\hat{\boldsymbol{\pi}}$ and sample analogues for $\Pr[\mathbf{y}^* \in \mathcal{C}_m \vert \mathbf{X} \in \mathcal{D}_\mathbf{l}]$
$\mathbf{X}$	$\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_k, \ldots, \mathbf{x}_K)$ with dimensions $(N \times K)$ where $\mathbf{x}_k = (\mathbf{x}_{k,1}, \ldots, \mathbf{x}_{k,N})'$ are covariates
$\mathbf{X}^*$	discretized version of $\mathbf{X}$
$\mathbf{X}^\dagger$	synthetic random variable created from $\mathbf{X}^*$
$\ddot{\mathbf{X}}$	replaced version of $\mathbf{X}^*$ with the corresponding conditional expectations via $\hat{\boldsymbol{\kappa}}$
$\tilde{\mathbf{X}}$	conditional sample means for $\mathbf{X}$ in partition $\mathcal{D}_\mathbf{l}$
$\mathbf{W}$	controls, $\mathbf{W} = (\mathbf{w}_1, \ldots, \mathbf{w}_j, \ldots, \mathbf{w}_J)$ i.e., a $(N \times J)$ matrix with $\mathbf{w}_j = (\mathbf{w}_{j,1}, \ldots, \mathbf{x}_{j,N})'$
$\ddot{\mathbf{W}}$	replaced versions of $\mathbf{W}$ with $\ddot{\mathbf{W}} = \hat{\mathbb{E}}\,[\mathbf{W} \vert \mathbf{X}^* \in \mathcal{C}_\mathbf{m}]$
$\mathbf{M}$	residual maker matrix, that residualizes with respect to the variable in the subscript
$\boldsymbol{\beta}$	$\boldsymbol{\beta}$ is a $(K \times 1)$ parameter vector of interest: $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)'$.
$\boldsymbol{\gamma}$	parameter vector for controls $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_J)'$
$\mathbf{u}$	idiosyncratic term in regression models
$\mathcal{D}_\mathbf{l}$	mutually exclusive partitioning based on the domain of $\mathbf{X}$

Table A25: Table of notations II

References

Avella-Medina, M., 2021. Privacy-preserving parametric inference: a case for robust statistics. Journal of the American Statistical Association 116, 969–983. Doi: https://doi.org/10.1080/01621459.2019.1700130.

Bi, X., Shen, X., 2023. Distribution-invariant differential privacy. Journal of Econometrics 235, 444–453. Doi: https://doi.org/10.1016/j.jeconom.2022.05.004.

Cai, T.T., Wang, Y., Zhang, L., 2021. The cost of privacy: Optimal rates of convergence for parameter estimation with differential privacy. The Annals of Statistics 49, pp. 2825–2850.

Diamond, P.A., Hausman, J.A., 1994. Contingent valuation: Is some number better than no number? American Economic Review 8, 45–64.

Ding, B., Kulkarni, J., Yekhanin, S., 2017. Collecting telemetry data privately, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, Curran Associates Inc., Red Hook, NY, USA. p. 3574–3583.

Duchi, J.C., Jordan, M.I., Wainwright, M.J., 2018. Minimax optimal procedures for locally private estimation. Journal of the American Statistical Association 113, 182–201. doi:10.1080/01621459.2017.1389735.

Dwork, C., 2006. Differential privacy, in: Bugliesi, M., Preneel, B., Sassone, V., Wegener, I. (Eds.), Automata, Languages and Programming, Springer Berlin Heidelberg, Berlin, Heidelberg. pp. 1–12. Doi: https://doi.org/10.1007/11787006_1.

Dwork, C., McSherry, F., Nissim, K., Smith, A., 2006. Calibrating noise to sensitivity in private data analysis, in: Proceedings of the Third Theory of Cryptography Conference, Springer. pp. 265–284. Doi: https://doi.org/10.1007/11681878_14.

Dwork, C., Roth, A., et al., 2014. The algorithmic foundations of differential privacy. Foundations and Trends in Theoretical Computer Science 9, 211–407. Doi: http://dx.doi.org/10.1561/0400000042.

Erlingsson, Ú., Pihur, V., Korolova, A., 2014. Rappor: Randomized aggregatable privacy-preserving ordinal response, in: Proceedings of the 2014 ACM SIGSAC conference on computer and communications security, pp. 1054–1067.

Fox, C.R., Rottenstreich, Y., 2003. Partition priming in judgment under uncertainty. Psychological Science 14, 195–200.

Haisley, E., Mostafa, R., Loewenstein, G., 2008. Subjective relative income and lottery ticket purchases. Journal of Behavioral Decision Making 21, 283–295. Doi: https://doi.org/10.1002/bdm.588.

Jordan, M.I., Mitchell, T.M., 2015. Machine learning: Trends, perspectives, and prospects. Science 349, 255–260. Doi: https://doi.org/10.1126/science.aaa8415.

Kenthapadi, K., Tran, T.T., 2018. Pripearl: A framework for privacy-preserving analytics and reporting at linkedin, in: Proceedings of the 27th ACM International Conference on Information and Knowledge Management, pp. 2183–2191.

Matyas, L. (Ed.), 2024. The econometrics of multi-dimensional panels. Springer Nature.

Rohde, A., Steinberger, L., 2020. Geometrizing rates of convergence under local differential privacy constraints. The Annals of Statistics 48, 2646–2670. Doi: https://doi.org/10.1214/19-AOS1901.